

| LOC | OBJECT CODE | ADDR1 | ADDR2 | STMT                                                                               |
|-----|-------------|-------|-------|------------------------------------------------------------------------------------|
|     |             |       |       | 2 *****                                                                            |
|     |             |       |       | 3 *                                                                                |
|     |             |       |       | 4 *     Zvector E7 instruction tests for VRR-a encoded:                            |
|     |             |       |       | 5 *                                                                                |
|     |             |       |       | 6 *     E754 VGEM    - Vector Generate Element Masks                               |
|     |             |       |       | 7 *                                                                                |
|     |             |       |       | 8 *             James Wekel March 2026                                             |
|     |             |       |       | 9 *****                                                                            |
|     |             |       |       |                                                                                    |
|     |             |       |       | 11 *****                                                                           |
|     |             |       |       | 12 *                                                                               |
|     |             |       |       | 13 *             basic instruction tests                                           |
|     |             |       |       | 14 *                                                                               |
|     |             |       |       | 15 *****                                                                           |
|     |             |       |       | 16 *     This program tests proper functioning of the z/arch E7 VRR-a              |
|     |             |       |       | 17 *     Vector Generate Element Masks instruction.                                |
|     |             |       |       | 18 *     Exceptions are not tested.                                                |
|     |             |       |       | 19 *                                                                               |
|     |             |       |       | 20 *     PLEASE NOTE that the tests are very SIMPLE TESTS designed to catch        |
|     |             |       |       | 21 *     obvious coding errors.   None of the tests are thorough.   They are       |
|     |             |       |       | 22 *     NOT designed to test all aspects of any of the instructions.              |
|     |             |       |       | 23 *                                                                               |
|     |             |       |       | 24 *****                                                                           |
|     |             |       |       | 25 *                                                                               |
|     |             |       |       | 26 *     *Testcase zvector-e7-32-VGEM                                              |
|     |             |       |       | 27 *     *                                                                         |
|     |             |       |       | 28 *     *     Zvector E7 instruction tests for VRR-a encoded:                     |
|     |             |       |       | 29 *     *                                                                         |
|     |             |       |       | 30 *     *     E754 VGEM    - Vector Generate Element Masks                        |
|     |             |       |       | 31 *     *                                                                         |
|     |             |       |       | 32 *     *     # -----                                                             |
|     |             |       |       | 33 *     *     #     This tests only the basic function of the instruction.        |
|     |             |       |       | 34 *     *     #     Exceptions are NOT tested.                                    |
|     |             |       |       | 35 *     *     # -----                                                             |
|     |             |       |       | 36 *     *                                                                         |
|     |             |       |       | 37 *     mainsize        2                                                         |
|     |             |       |       | 38 *     numcpu         1                                                          |
|     |             |       |       | 39 *     sysclear                                                                  |
|     |             |       |       | 40 *     archlvl        z/Arch                                                     |
|     |             |       |       | 41 *     *                                                                         |
|     |             |       |       | 42 *     loadcore        "\$(testpath)/zvector-e7-32-VGEM.core" 0x0                |
|     |             |       |       | 43 *     *                                                                         |
|     |             |       |       | 44 *     diag8cmd        enable        # (needed for messages to Hercules console) |
|     |             |       |       | 45 *     runtest         5             #                                           |
|     |             |       |       | 46 *     diag8cmd        disable       # (reset back to default)                   |
|     |             |       |       | 47 *     *                                                                         |
|     |             |       |       | 48 *     *Done                                                                     |
|     |             |       |       | 49 *     *                                                                         |
|     |             |       |       | 50 *****                                                                           |

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|-----|-------------|-------|-------|-------------------------------------------------------------------|
|     |             |       |       | 52 *****                                                          |
|     |             |       |       | 53 * FCHECK Macro - Is a Facility Bit set?                        |
|     |             |       |       | 54 *                                                              |
|     |             |       |       | 55 * If the facility bit is NOT set, an message is issued and     |
|     |             |       |       | 56 * the test is skipped.                                         |
|     |             |       |       | 57 *                                                              |
|     |             |       |       | 58 * Fcheck uses R0, R1 and R2                                    |
|     |             |       |       | 59 *                                                              |
|     |             |       |       | 60 * eg. FCHECK 134,'vector-packed-decimal'                       |
|     |             |       |       | 61 *****                                                          |
|     |             |       |       | 62 MACRO                                                          |
|     |             |       |       | 63 FCHECK &BITNO,&NOTSETMSG                                       |
|     |             |       |       | 64 .* &BITNO : facility bit number to check                       |
|     |             |       |       | 65 .* &NOTSETMSG : 'facility name'                                |
|     |             |       |       | 66 LCLA &FBBYTE Facility bit in Byte                              |
|     |             |       |       | 67 LCLA &FBBIT Facility bit within Byte                           |
|     |             |       |       | 68                                                                |
|     |             |       |       | 69 LCLA &L(8)                                                     |
|     |             |       |       | 70 &L(1) SetA 128,64,32,16,8,4,2,1 bit positions within byte      |
|     |             |       |       | 71                                                                |
|     |             |       |       | 72 &FBBYTE SETA &BITNO/8                                          |
|     |             |       |       | 73 &FBBIT SETA &L((&BITNO-(&FBBYTE*8))+1)                         |
|     |             |       |       | 74 .* MNOTE 0,'checking Bit=&BITNO: FBBYTE=&FBBYTE, FBBIT=&FBBIT' |
|     |             |       |       | 75                                                                |
|     |             |       |       | 76 B X&SYSNDX                                                     |
|     |             |       |       | 77 * Fcheck data area                                             |
|     |             |       |       | 78 * skip messgae                                                 |
|     |             |       |       | 79 SKT&SYSNDX DC C' Skipping tests: '                             |
|     |             |       |       | 80 DC C&NOTSETMSG                                                 |
|     |             |       |       | 81 DC C' (bit &BITNO) is not installed.'                          |
|     |             |       |       | 82 SKL&SYSNDX EQU *-SKT&SYSNDX                                    |
|     |             |       |       | 83 * facility bits                                                |
|     |             |       |       | 84 DS FD gap                                                      |
|     |             |       |       | 85 FB&SYSNDX DS 4FD                                               |
|     |             |       |       | 86 DS FD gap                                                      |
|     |             |       |       | 87 *                                                              |
|     |             |       |       | 88 X&SYSNDX EQU *                                                 |
|     |             |       |       | 89 LA R0,((X&SYSNDX-FB&SYSNDX)/8)-1                               |
|     |             |       |       | 90 STFLE FB&SYSNDX get facility bits                              |
|     |             |       |       | 91                                                                |
|     |             |       |       | 92 XGR R0,R0                                                      |
|     |             |       |       | 93 IC R0,FB&SYSNDX+&FBBYTE get fbit byte                          |
|     |             |       |       | 94 N R0,=F'&FBBIT' is bit set?                                    |
|     |             |       |       | 95 BNZ XC&SYSNDX                                                  |
|     |             |       |       | 96 *                                                              |
|     |             |       |       | 97 * facility bit not set, issue message and exit                 |
|     |             |       |       | 98 *                                                              |
|     |             |       |       | 99 LA R0,SKL&SYSNDX message length                                |
|     |             |       |       | 100 LA R1,SKT&SYSNDX message address                              |
|     |             |       |       | 101 BAL R2,MSG                                                    |
|     |             |       |       | 102                                                               |
|     |             |       |       | 103 B EOJ                                                         |
|     |             |       |       | 104 XC&SYSNDX EQU *                                               |
|     |             |       |       | 105 MEND                                                          |



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|----------|-------------------|----------|----------|------|------------------------------------------------------------|
|          |                   |          |          | 132  | *****                                                      |
|          |                   |          |          | 133  | * Low core PSWs                                            |
|          |                   |          |          | 134  | *****                                                      |
| 00000000 |                   | 00000000 | 000032D3 | 135  | ZVE7TST START 0                                            |
|          |                   | 00000000 |          | 136  | USING ZVE7TST,R0 Low core addressability                   |
|          |                   | 00000140 | 00000000 | 137  |                                                            |
|          |                   |          |          | 138  | SVOLDPSW EQU ZVE7TST+X'140' z/Arch Supervisor call old PSW |
| 00000000 |                   | 00000000 | 000001A0 | 140  | ORG ZVE7TST+X'1A0' z/Architecture RESTART PSW              |
| 000001A0 | 00000001 80000000 |          |          | 141  | DC X'0000000180000000'                                     |
| 000001A8 | 00000000 00000200 |          |          | 142  | DC AD(BEGIN)                                               |
| 000001B0 |                   | 000001B0 | 000001D0 | 144  | ORG ZVE7TST+X'1D0' z/Architecture PROGRAM CHECK PSW        |
| 000001D0 | 00020001 80000000 |          |          | 145  | DC X'0002000180000000'                                     |
| 000001D8 | 00000000 0000DEAD |          |          | 146  | DC AD(X'DEAD')                                             |
| 000001E0 |                   | 000001E0 | 00000200 | 148  | ORG ZVE7TST+X'200' Start of actual test program...         |
|          |                   |          |          | 150  | *****                                                      |
|          |                   |          |          | 151  | * The actual "ZVE7TST" program itself...                   |
|          |                   |          |          | 152  | *****                                                      |
|          |                   |          |          | 153  | *                                                          |
|          |                   |          |          | 154  | * Architecture Mode: z/Arch                                |
|          |                   |          |          | 155  | * Register Usage:                                          |
|          |                   |          |          | 156  | *                                                          |
|          |                   |          |          | 157  | * R0 (work)                                                |
|          |                   |          |          | 158  | * R1-4 (work)                                              |
|          |                   |          |          | 159  | * R5 Testing control table - current test base             |
|          |                   |          |          | 160  | * R6-R7 (work)                                             |
|          |                   |          |          | 161  | * R8 First base register                                   |
|          |                   |          |          | 162  | * R9 Second base register                                  |
|          |                   |          |          | 163  | * R10 Third base register                                  |
|          |                   |          |          | 164  | * R11 E7TEST call return                                   |
|          |                   |          |          | 165  | * R12 E7TESTS register                                     |
|          |                   |          |          | 166  | * R13 (work)                                               |
|          |                   |          |          | 167  | * R14 Subroutine call                                      |
|          |                   |          |          | 168  | * R15 Secondary Subroutine call or work                    |
|          |                   |          |          | 169  | *                                                          |
|          |                   |          |          | 170  | *****                                                      |
| 00000200 |                   | 00000200 |          | 172  | USING BEGIN,R8 FIRST Base Register                         |
| 00000200 |                   | 00001200 |          | 173  | USING BEGIN+4096,R9 SECOND Base Register                   |
| 00000200 |                   | 00002200 |          | 174  | USING BEGIN+8192,R10 THIRD Base Register                   |
| 00000200 | 0580              |          |          | 176  | BEGIN BALR R8,0 Initalize FIRST base register              |
| 00000202 | 0680              |          |          | 177  | BCTR R8,0 Initalize FIRST base register                    |
| 00000204 | 0680              |          |          | 178  | BCTR R8,0 Initalize FIRST base register                    |
| 00000206 | 4190 8800         |          | 00000800 | 180  | LA R9,2048(,R8) Initalize SECOND base register             |
| 0000020A | 4190 9800         |          | 00000800 | 181  | LA R9,2048(,R9) Initalize SECOND base register             |
|          |                   |          |          | 182  |                                                            |















| LOC      | OBJECT CODE       | ADDR1    | ADDR2    | STMT |                                                                     |
|----------|-------------------|----------|----------|------|---------------------------------------------------------------------|
|          |                   |          |          | 356  | *****                                                               |
|          |                   |          |          | 357  | *            Normal completion or Abnormal termination PSWs         |
|          |                   |          |          | 358  | *****                                                               |
| 00000460 | 00020001 80000000 |          |          | 360  | E0JPSW    DC    0D'0',X'0002000180000000',AD(0)                     |
| 00000470 | B2B2 8260         |          | 00000460 | 362  | E0J            LPSWE E0JPSW            Normal completion            |
| 00000478 | 00020001 80000000 |          |          | 364  | FAILPSW    DC    0D'0',X'0002000180000000',AD(X'BAD')               |
| 00000488 | B2B2 8278         |          | 00000478 | 366  | FAILTEST    LPSWE FAILPSW            Abnormal termination           |
|          |                   |          |          | 368  | *****                                                               |
|          |                   |          |          | 369  | *            Working Storage                                        |
|          |                   |          |          | 370  | *****                                                               |
| 0000048C | 00000000          |          |          | 372  | CTLR0       DS    F            CR0                                  |
| 00000490 | 00000000          |          |          | 373  | DS    F                                                             |
| 00000494 |                   |          |          | 375  | LTORG ,            Literals pool                                    |
| 00000494 | 00000002          |          |          | 376  | =F'2'                                                               |
| 00000498 | 000031BC          |          |          | 377  | =A(E7TESTS)                                                         |
| 0000049C | 00000001          |          |          | 378  | =F'1'                                                               |
| 000004A0 | 0000              |          |          | 379  | =H'0'                                                               |
| 000004A2 | 005F              |          |          | 380  | =AL2(L'MSGMSG)                                                      |
|          |                   |          |          | 381  |                                                                     |
|          |                   |          |          | 382  | *            some constants                                         |
|          |                   |          |          | 383  |                                                                     |
|          |                   | 00000400 | 00000001 | 384  | K            EQU    1024            One KB                          |
|          |                   | 00001000 | 00000001 | 385  | PAGE        EQU    (4*K)            Size of one page                |
|          |                   | 00010000 | 00000001 | 386  | K64        EQU    (64*K)            64 KB                           |
|          |                   | 00100000 | 00000001 | 387  | MB         EQU    (K*K)            1 MB                             |
|          |                   |          |          | 388  |                                                                     |
|          |                   | AABBCCDD | 00000001 | 389  | REG2PATT    EQU    X'AABBCCDD'            Polluted Register pattern |
|          |                   | 000000DD | 00000001 | 390  | REG2LOW    EQU            X'DD'            (last byte above)        |





| LOC      | OBJECT   | CODE     | ADDR1    | ADDR2    | STMT                                                                      |
|----------|----------|----------|----------|----------|---------------------------------------------------------------------------|
|          |          |          |          |          | 433 *****                                                                 |
|          |          |          |          |          | 434 *            E7TEST DSECT                                             |
|          |          |          |          |          | 435 *****                                                                 |
|          |          |          |          |          | 437 E7TEST    DSECT ,                                                     |
| 00000000 | 00000000 |          |          |          | 438 TSUB      DC    A(0)            pointer to test                       |
| 00000004 | 0000     |          |          |          | 439 TNUM      DC    H'00'            Test Number                          |
| 00000006 | 00       |          |          |          | 440            DC    X'00'                                                |
| 00000007 | 00       |          |          |          | 441 M3        DC    HL1'00'            m4 used                            |
|          |          |          |          |          | 442                                                                       |
| 00000008 | 40404040 | 40404040 |          |          | 443 OPNAME    DC    CL8' '            E6 name                             |
| 00000010 | 00000000 |          |          |          | 444 V2ADDR    DC    A(0)            address of v2 source                  |
| 00000014 | 00000000 |          |          |          | 445 RELEN     DC    A(0)            RESULT LENGTH                         |
| 00000018 | 00000000 |          |          |          | 446 READDR    DC    A(0)            result (expected) address             |
| 00000020 | 00000000 | 00000000 |          |          | 447            DS    FD            gap                                    |
| 00000028 | 00000000 | 00000000 |          |          | 448 V10OUTPUT DS    XL16            V1 Output                             |
| 00000038 | 00000000 | 00000000 |          |          | 449            DS    FD            gap                                    |
|          |          |          |          |          | 450                                                                       |
|          |          |          |          |          | 451 *            test routine will be here (from VRR-a macro)             |
|          |          |          |          |          | 452 *                                                                     |
|          |          |          |          |          | 453 *            followed by                                              |
|          |          |          |          |          | 454 *            EXPECTED RESULT                                          |
|          |          |          |          |          |                                                                           |
| 000010B4 |          |          | 00000000 | 000032D3 | 456 ZVE7TST   CSECT ,                                                     |
|          |          |          |          |          | 457            DS    0F                                                   |
|          |          |          |          |          | 459 *****                                                                 |
|          |          |          |          |          | 460 *            Macros to help build test tables                         |
|          |          |          |          |          | 461 *****                                                                 |
|          |          |          |          |          |                                                                           |
|          |          |          |          |          | 463 *                                                                     |
|          |          |          |          |          | 464 *            macro to generate individual test                        |
|          |          |          |          |          | 465 *                                                                     |
|          |          |          |          |          | 466            MACRO                                                      |
|          |          |          |          |          | 467            VRR_A &INST,&M3                                            |
|          |          |          |          |          | 468 .*                            &INST    - VRR-a instruction under test |
|          |          |          |          |          | 469 .*                            &M3      - m3 field                     |
|          |          |          |          |          | 470                                                                       |
|          |          |          |          |          | 471            GBLA   &TNUM                                               |
|          |          |          |          |          | 472 &TNUM      SETA   &TNUM+1                                             |
|          |          |          |          |          | 473                                                                       |
|          |          |          |          |          | 474            DS    0FD                                                  |
|          |          |          |          |          | 475            USING *,R5            base for test data and test routine  |
|          |          |          |          |          | 476                                                                       |
|          |          |          |          |          | 477 T&TNUM    DC    A(X&TNUM)            address of test routine          |
|          |          |          |          |          | 478            DC    H'&TNUM'            test number                      |
|          |          |          |          |          | 479            DC    X'00'                                                |
|          |          |          |          |          | 480            DC    HL1'&M3'            M3                               |
|          |          |          |          |          | 481            DC    CL8'&INST'            instruction name               |
|          |          |          |          |          | 482            DC    A(RE&TNUM+16)            address of v2 source        |
|          |          |          |          |          | 483            DC    A(16)            result length                       |



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|----------|-------------------|----------|----------|-------------------------------------------------|-------------------------------------|
|          |                   |          |          | 527 *****                                       |                                     |
|          |                   |          |          | 528 * E7 VRR-a tests                            |                                     |
|          |                   |          |          | 529 *****                                       |                                     |
|          |                   |          |          | 530 PRINT DATA                                  |                                     |
|          |                   |          |          | 531 *                                           |                                     |
|          |                   |          |          | 532 * E754 VGEM - Vector Generate Element Masks |                                     |
|          |                   |          |          | 533 *                                           |                                     |
|          |                   |          |          | 534 * VRR-a instruction, M3                     |                                     |
|          |                   |          |          | 535 * followed by                               |                                     |
|          |                   |          |          | 536 * 16 byte expected result (V1)              |                                     |
|          |                   |          |          | 537 * 16 byte V2 source                         |                                     |
|          |                   |          |          | 538 * -----                                     |                                     |
|          |                   |          |          | 539 * VGEM - Vector Generate Element Masks      |                                     |
|          |                   |          |          | 540 * -----                                     |                                     |
|          |                   |          |          | 541 * Byte M3 = 0                               |                                     |
|          |                   |          |          | 542 VRR_A VGEM, 0                               |                                     |
| 000010B8 |                   |          |          | 543+ DS 0FD                                     |                                     |
| 000010B8 |                   | 000010B8 |          | 544+ USING *, R5                                | base for test data and test routine |
| 000010B8 | 000010F8          |          |          | 545+T1 DC A(X1)                                 | address of test routine             |
| 000010BC | 0001              |          |          | 546+ DC H'1'                                    | test number                         |
| 000010BE | 00                |          |          | 547+ DC X'00'                                   |                                     |
| 000010BF | 00                |          |          | 548+ DC HL1'0'                                  | M3                                  |
| 000010C0 | E5C7C5D4 40404040 |          |          | 549+ DC CL8'VGEM'                               | instruction name                    |
| 000010C8 | 00001124          |          |          | 550+ DC A(RE1+16)                               | address of v2 source                |
| 000010CC | 00000010          |          |          | 551+ DC A(16)                                   | result length                       |
| 000010D0 | 00001114          |          |          | 552+REA1 DC A(RE1)                              | result address                      |
| 000010D8 | 00000000 00000000 |          |          | 553+ DS FD                                      | gap                                 |
| 000010E0 | 00000000 00000000 |          |          | 554+V101 DS XL16                                | V1 output                           |
| 000010E8 | 00000000 00000000 |          |          |                                                 |                                     |
| 000010F0 | 00000000 00000000 |          |          | 555+ DS FD                                      | gap                                 |
|          |                   |          |          | 556+*                                           |                                     |
| 000010F8 |                   |          |          | 557+X1 DS 0F                                    |                                     |
| 000010F8 | E310 5010 0014    |          | 00000010 | 558+ LGF R1,V2ADDR                              | load v2 source                      |
| 000010FE | E721 0000 0006    |          | 00000000 | 559+ VL v2, 0(R1)                               |                                     |
|          |                   |          |          |                                                 |                                     |
| 00001104 |                   |          |          | 562+ DS 0H                                      | E754 VGEM                           |
| 00001104 | E7                |          |          | 563+ DC X'E7'                                   | - Vector Generate Element Masks     |
| 00001105 | 12                |          |          | 564+ DC X'12'                                   | v1, v2                              |
| 00001106 | 0000              |          |          | 565+ DC x'0000'                                 | reserved                            |
| 00001108 | 00                |          |          | 566+ DC HL1'0'                                  | M3, RXB                             |
| 00001109 | 54                |          |          | 567+ DC X'54'                                   |                                     |
|          |                   |          |          |                                                 |                                     |
| 0000110A | E710 5028 000E    |          | 000010E0 | 569+ VST V1,V101                                | save v1 output                      |
| 00001110 | 07FB              |          |          | 570+ BR R11                                     | return                              |
| 00001114 |                   |          |          | 571+RE1 DC 0F                                   | xl16 expected result                |
| 00001114 |                   |          |          | 572+ DROP R5                                    |                                     |
| 00001114 | 00000000 00000000 |          |          | 573 DC XL16'0000000000000000 0000000000000000'  | result                              |
| 0000111C | 00000000 00000000 |          |          |                                                 |                                     |
| 00001124 | 00000000 00000000 |          |          | 574 DC XL16'0000000000000000 0000000000000000'  | v2                                  |
| 0000112C | 00000000 00000000 |          |          |                                                 |                                     |
|          |                   |          |          | 575                                             |                                     |
|          |                   |          |          | 576 VRR_A VGEM, 0                               |                                     |
| 00001138 |                   |          |          | 577+ DS 0FD                                     |                                     |
| 00001138 |                   | 00001138 |          | 578+ USING *, R5                                | base for test data and test routine |
| 00001138 | 00001178          |          |          | 579+T2 DC A(X2)                                 | address of test routine             |
| 0000113C | 0002              |          |          | 580+ DC H'2'                                    | test number                         |



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|----------|-------------------|----------|----------|----------|-------|-----------------------------------------|-------------------------------------|
| 0000113E | 00                |          |          | 581+     | DC    | X'00'                                   |                                     |
| 0000113F | 00                |          |          | 582+     | DC    | HL1'0'                                  | M3                                  |
| 00001140 | E5C7C5D4 40404040 |          |          | 583+     | DC    | CL8'VGEM'                               | instruction name                    |
| 00001148 | 000011A4          |          |          | 584+     | DC    | A(RE2+16)                               | address of v2 source                |
| 0000114C | 00000010          |          |          | 585+     | DC    | A(16)                                   | result length                       |
| 00001150 | 00001194          |          |          | 586+REA2 | DC    | A(RE2)                                  | result address                      |
| 00001158 | 00000000 00000000 |          |          | 587+     | DS    | FD                                      | gap                                 |
| 00001160 | 00000000 00000000 |          |          | 588+V102 | DS    | XL16                                    | V1 output                           |
| 00001168 | 00000000 00000000 |          |          |          |       |                                         |                                     |
| 00001170 | 00000000 00000000 |          |          | 589+     | DS    | FD                                      | gap                                 |
|          |                   |          |          | 590+*    |       |                                         |                                     |
| 00001178 |                   |          |          | 591+X2   | DS    | 0F                                      |                                     |
| 00001178 | E310 5010 0014    |          | 00000010 | 592+     | LGF   | R1,V2ADDR                               | load v2 source                      |
| 0000117E | E721 0000 0006    |          | 00000000 | 593+     | VL    | v2,0(R1)                                |                                     |
| 00001184 |                   |          |          | 596+     | DS    | 0H                                      | E754 VGEM                           |
| 00001184 | E7                |          |          | 597+     | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00001185 | 12                |          |          | 598+     | DC    | X'12'                                   | v1, v2                              |
| 00001186 | 0000              |          |          | 599+     | DC    | x'0000'                                 | reserved                            |
| 00001188 | 00                |          |          | 600+     | DC    | HL1'0'                                  | M3, RXB                             |
| 00001189 | 54                |          |          | 601+     | DC    | X'54'                                   |                                     |
| 0000118A | E710 5028 000E    |          | 00001160 | 603+     | VST   | V1,V102                                 | save v1 output                      |
| 00001190 | 07FB              |          |          | 604+     | BR    | R11                                     | return                              |
| 00001194 |                   |          |          | 605+RE2  | DC    | 0F                                      | xl16 expected result                |
| 00001194 |                   |          |          | 606+     | DROP  | R5                                      |                                     |
| 00001194 | 00000000 00000000 |          |          | 607      | DC    | XL16'0000000000000000 0000000000000000' | result                              |
| 0000119C | 00000000 00000000 |          |          |          |       |                                         |                                     |
| 000011A4 | 0000FFFF FFFFFFFF |          |          | 608      | DC    | XL16'0000FFFFFFFFFFFFFF FFFFFFFF'       | v2                                  |
| 000011AC | FFFFFFFF FFFFFFFF |          |          |          |       |                                         |                                     |
|          |                   |          |          | 609      |       |                                         |                                     |
|          |                   |          |          | 610      |       |                                         |                                     |
| 000011B8 |                   |          |          | 611+     | VRR_A | VGEM,0                                  |                                     |
| 000011B8 |                   | 000011B8 |          | 612+     | DS    | 0FD                                     |                                     |
| 000011B8 | 000011F8          |          |          | 613+T3   | USING | *,R5                                    | base for test data and test routine |
| 000011BC | 0003              |          |          | 614+     | DC    | A(X3)                                   | address of test routine             |
| 000011BE | 00                |          |          | 615+     | DC    | H'3'                                    | test number                         |
| 000011BF | 00                |          |          | 616+     | DC    | X'00'                                   |                                     |
| 000011C0 | E5C7C5D4 40404040 |          |          | 617+     | DC    | HL1'0'                                  | M3                                  |
| 000011C8 | 00001224          |          |          | 618+     | DC    | CL8'VGEM'                               | instruction name                    |
| 000011CC | 00000010          |          |          | 619+     | DC    | A(RE3+16)                               | address of v2 source                |
| 000011D0 | 00001214          |          |          | 620+REA3 | DC    | A(16)                                   | result length                       |
| 000011D8 | 00000000 00000000 |          |          | 621+     | DC    | A(RE3)                                  | result address                      |
| 000011E0 | 00000000 00000000 |          |          | 622+V103 | DS    | FD                                      | gap                                 |
| 000011E8 | 00000000 00000000 |          |          |          | DS    | XL16                                    | V1 output                           |
| 000011F0 | 00000000 00000000 |          |          | 623+     | DS    | FD                                      | gap                                 |
|          |                   |          |          | 624+*    |       |                                         |                                     |
| 000011F8 |                   |          |          | 625+X3   | DS    | 0F                                      |                                     |
| 000011F8 | E310 5010 0014    |          | 00000010 | 626+     | LGF   | R1,V2ADDR                               | load v2 source                      |
| 000011FE | E721 0000 0006    |          | 00000000 | 627+     | VL    | v2,0(R1)                                |                                     |
| 00001204 |                   |          |          | 630+     | DS    | 0H                                      | E754 VGEM                           |
| 00001204 | E7                |          |          | 631+     | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00001205 | 12                |          |          | 632+     | DC    | X'12'                                   | v1, v2                              |
| 00001206 | 0000              |          |          | 633+     | DC    | x'0000'                                 | reserved                            |
| 00001208 | 00                |          |          | 634+     | DC    | HL1'0'                                  | M3, RXB                             |



| LOC      | OBJECT CODE       | ADDR1    | ADDR2    | STMT     |       |                                         |                                     |
|----------|-------------------|----------|----------|----------|-------|-----------------------------------------|-------------------------------------|
| 00001209 | 54                |          |          | 635+     | DC    | X'54'                                   |                                     |
| 0000120A | E710 5028 000E    |          | 000011E0 | 637+     | VST   | V1,V103                                 | save v1 output                      |
| 00001210 | 07FB              |          |          | 638+     | BR    | R11                                     | return                              |
| 00001214 |                   |          |          | 639+RE3  | DC    | 0F                                      | xl16 expected result                |
| 00001214 |                   |          |          | 640+     | DROP  | R5                                      |                                     |
| 00001214 | 00000000 00000000 |          |          | 641      | DC    | XL16'0000000000000000 00000000000000FF' | result                              |
| 0000121C | 00000000 000000FF |          |          |          |       |                                         |                                     |
| 00001224 | 00010000 00000000 |          |          | 642      | DC    | XL16'0001000000000000 0000000000000000' | v2                                  |
| 0000122C | 00000000 00000000 |          |          |          |       |                                         |                                     |
|          |                   |          |          | 643      |       |                                         |                                     |
| 00001238 |                   |          |          | 644      | VRR_A | VGEM, 0                                 |                                     |
| 00001238 |                   | 00001238 |          | 645+     | DS    | 0FD                                     |                                     |
| 00001238 | 00001278          |          |          | 646+     | USING | *,R5                                    | base for test data and test routine |
| 0000123C | 0004              |          |          | 647+T4   | DC    | A(X4)                                   | address of test routine             |
| 0000123E | 00                |          |          | 648+     | DC    | H'4'                                    | test number                         |
| 0000123E | 00                |          |          | 649+     | DC    | X'00'                                   |                                     |
| 0000123F | 00                |          |          | 650+     | DC    | HL1'0'                                  | M3                                  |
| 00001240 | E5C7C5D4 40404040 |          |          | 651+     | DC    | CL8'VGEM'                               | instruction name                    |
| 00001248 | 000012A4          |          |          | 652+     | DC    | A(RE4+16)                               | address of v2 source                |
| 0000124C | 00000010          |          |          | 653+     | DC    | A(16)                                   | result length                       |
| 00001250 | 00001294          |          |          | 654+REA4 | DC    | A(RE4)                                  | result address                      |
| 00001258 | 00000000 00000000 |          |          | 655+     | DS    | FD                                      | gap                                 |
| 00001260 | 00000000 00000000 |          |          | 656+V104 | DS    | XL16                                    | V1 output                           |
| 00001268 | 00000000 00000000 |          |          |          |       |                                         |                                     |
| 00001270 | 00000000 00000000 |          |          | 657+     | DS    | FD                                      | gap                                 |
|          |                   |          |          | 658+*    |       |                                         |                                     |
| 00001278 |                   |          |          | 659+X4   | DS    | 0F                                      |                                     |
| 00001278 | E310 5010 0014    |          | 00000010 | 660+     | LGF   | R1,V2ADDR                               | load v2 source                      |
| 0000127E | E721 0000 0006    |          | 00000000 | 661+     | VL    | v2, 0(R1)                               |                                     |
| 00001284 |                   |          |          | 664+     | DS    | 0H                                      | E754 VGEM                           |
| 00001284 | E7                |          |          | 665+     | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00001285 | 12                |          |          | 666+     | DC    | X'12'                                   | v1, v2                              |
| 00001286 | 0000              |          |          | 667+     | DC    | x'0000'                                 | reserved                            |
| 00001288 | 00                |          |          | 668+     | DC    | HL1'0'                                  | M3, RXB                             |
| 00001289 | 54                |          |          | 669+     | DC    | X'54'                                   |                                     |
| 0000128A | E710 5028 000E    |          | 00001260 | 671+     | VST   | V1,V104                                 | save v1 output                      |
| 00001290 | 07FB              |          |          | 672+     | BR    | R11                                     | return                              |
| 00001294 |                   |          |          | 673+RE4  | DC    | 0F                                      | xl16 expected result                |
| 00001294 |                   |          |          | 674+     | DROP  | R5                                      |                                     |
| 00001294 | 00000000 00000000 |          |          | 675      | DC    | XL16'0000000000000000 000000000000FF00' | result                              |
| 0000129C | 00000000 0000FF00 |          |          |          |       |                                         |                                     |
| 000012A4 | 00020000 00000000 |          |          | 676      | DC    | XL16'0002000000000000 0000000000000000' | v2                                  |
| 000012AC | 00000000 00000000 |          |          |          |       |                                         |                                     |
|          |                   |          |          | 677      |       |                                         |                                     |
| 000012B8 |                   |          |          | 678      | VRR_A | VGEM, 0                                 |                                     |
| 000012B8 |                   | 000012B8 |          | 679+     | DS    | 0FD                                     |                                     |
| 000012B8 | 000012F8          |          |          | 680+     | USING | *,R5                                    | base for test data and test routine |
| 000012BC | 0005              |          |          | 681+T5   | DC    | A(X5)                                   | address of test routine             |
| 000012BE | 00                |          |          | 682+     | DC    | H'5'                                    | test number                         |
| 000012BE | 00                |          |          | 683+     | DC    | X'00'                                   |                                     |
| 000012BF | 00                |          |          | 684+     | DC    | HL1'0'                                  | M3                                  |
| 000012C0 | E5C7C5D4 40404040 |          |          | 685+     | DC    | CL8'VGEM'                               | instruction name                    |
| 000012C8 | 00001324          |          |          | 686+     | DC    | A(RE5+16)                               | address of v2 source                |

| LOC      | OBJECT CODE       | ADDR1    | ADDR2    | STMT     |       |                                         |                                     |
|----------|-------------------|----------|----------|----------|-------|-----------------------------------------|-------------------------------------|
| 000012CC | 00000010          |          |          | 687+     | DC    | A(16)                                   | result length                       |
| 000012D0 | 00001314          |          |          | 688+REA5 | DC    | A(RE5)                                  | result address                      |
| 000012D8 | 00000000 00000000 |          |          | 689+     | DS    | FD                                      | gap                                 |
| 000012E0 | 00000000 00000000 |          |          | 690+V105 | DS    | XL16                                    | V1 output                           |
| 000012E8 | 00000000 00000000 |          |          |          |       |                                         |                                     |
| 000012F0 | 00000000 00000000 |          |          | 691+     | DS    | FD                                      | gap                                 |
|          |                   |          |          | 692+*    |       |                                         |                                     |
| 000012F8 |                   |          |          | 693+X5   | DS    | 0F                                      |                                     |
| 000012F8 | E310 5010 0014    |          | 00000010 | 694+     | LGF   | R1,V2ADDR                               | load v2 source                      |
| 000012FE | E721 0000 0006    |          | 00000000 | 695+     | VL    | v2,0(R1)                                |                                     |
| 00001304 |                   |          |          | 698+     | DS    | 0H                                      | E754 VGEM                           |
| 00001304 | E7                |          |          | 699+     | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00001305 | 12                |          |          | 700+     | DC    | X'12'                                   | v1, v2                              |
| 00001306 | 0000              |          |          | 701+     | DC    | x'0000'                                 | reserved                            |
| 00001308 | 00                |          |          | 702+     | DC    | HL1'0'                                  | M3, RXB                             |
| 00001309 | 54                |          |          | 703+     | DC    | X'54'                                   |                                     |
| 0000130A | E710 5028 000E    |          | 000012E0 | 705+     | VST   | V1,V105                                 | save v1 output                      |
| 00001310 | 07FB              |          |          | 706+     | BR    | R11                                     | return                              |
| 00001314 |                   |          |          | 707+RE5  | DC    | 0F                                      | xl16 expected result                |
| 00001314 |                   |          |          | 708+     | DROP  | R5                                      |                                     |
| 00001314 | 00000000 00000000 |          |          | 709      | DC    | XL16'0000000000000000 000000000000FFFF' | result                              |
| 0000131C | 00000000 0000FFFF |          |          |          |       |                                         |                                     |
| 00001324 | 00030000 00000000 |          |          | 710      | DC    | XL16'0003000000000000 0000000000000000' | v2                                  |
| 0000132C | 00000000 00000000 |          |          |          |       |                                         |                                     |
|          |                   |          |          | 711      |       |                                         |                                     |
|          |                   |          |          | 712      | VRR_A | VGEM,0                                  |                                     |
| 00001338 |                   |          |          | 713+     | DS    | 0FD                                     |                                     |
| 00001338 |                   | 00001338 |          | 714+     | USING | *,R5                                    | base for test data and test routine |
| 00001338 | 00001378          |          |          | 715+T6   | DC    | A(X6)                                   | address of test routine             |
| 0000133C | 0006              |          |          | 716+     | DC    | H'6'                                    | test number                         |
| 0000133E | 00                |          |          | 717+     | DC    | X'00'                                   |                                     |
| 0000133F | 00                |          |          | 718+     | DC    | HL1'0'                                  | M3                                  |
| 00001340 | E5C7C5D4 40404040 |          |          | 719+     | DC    | CL8'VGEM'                               | instruction name                    |
| 00001348 | 000013A4          |          |          | 720+     | DC    | A(RE6+16)                               | address of v2 source                |
| 0000134C | 00000010          |          |          | 721+     | DC    | A(16)                                   | result length                       |
| 00001350 | 00001394          |          |          | 722+REA6 | DC    | A(RE6)                                  | result address                      |
| 00001358 | 00000000 00000000 |          |          | 723+     | DS    | FD                                      | gap                                 |
| 00001360 | 00000000 00000000 |          |          | 724+V106 | DS    | XL16                                    | V1 output                           |
| 00001368 | 00000000 00000000 |          |          |          |       |                                         |                                     |
| 00001370 | 00000000 00000000 |          |          | 725+     | DS    | FD                                      | gap                                 |
|          |                   |          |          | 726+*    |       |                                         |                                     |
| 00001378 |                   |          |          | 727+X6   | DS    | 0F                                      |                                     |
| 00001378 | E310 5010 0014    |          | 00000010 | 728+     | LGF   | R1,V2ADDR                               | load v2 source                      |
| 0000137E | E721 0000 0006    |          | 00000000 | 729+     | VL    | v2,0(R1)                                |                                     |
| 00001384 |                   |          |          | 732+     | DS    | 0H                                      | E754 VGEM                           |
| 00001384 | E7                |          |          | 733+     | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00001385 | 12                |          |          | 734+     | DC    | X'12'                                   | v1, v2                              |
| 00001386 | 0000              |          |          | 735+     | DC    | x'0000'                                 | reserved                            |
| 00001388 | 00                |          |          | 736+     | DC    | HL1'0'                                  | M3, RXB                             |
| 00001389 | 54                |          |          | 737+     | DC    | X'54'                                   |                                     |
| 0000138A | E710 5028 000E    |          | 00001360 | 739+     | VST   | V1,V106                                 | save v1 output                      |
| 00001390 | 07FB              |          |          | 740+     | BR    | R11                                     | return                              |

| LOC      | OBJECT CODE       | ADDR1    | ADDR2    | STMT     |       |                                         |                                     |
|----------|-------------------|----------|----------|----------|-------|-----------------------------------------|-------------------------------------|
| 00001394 |                   |          |          | 741+RE6  | DC    | 0F                                      | xl16 expected result                |
| 00001394 |                   |          |          | 742+     | DROP  | R5                                      |                                     |
| 00001394 | 00000000 00000000 |          |          | 743      | DC    | XL16'0000000000000000 0000000000FF0000' | result                              |
| 0000139C | 00000000 00FF0000 |          |          |          |       |                                         |                                     |
| 000013A4 | 00040000 00000000 |          |          | 744      | DC    | XL16'0004000000000000 0000000000000000' | v2                                  |
| 000013AC | 00000000 00000000 |          |          |          |       |                                         |                                     |
|          |                   |          |          | 745      |       |                                         |                                     |
|          |                   |          |          | 746      | VRR_A | VGEM, 0                                 |                                     |
| 000013B8 |                   |          |          | 747+     | DS    | 0FD                                     |                                     |
| 000013B8 |                   | 000013B8 |          | 748+     | USING | *, R5                                   | base for test data and test routine |
| 000013B8 | 000013F8          |          |          | 749+T7   | DC    | A(X7)                                   | address of test routine             |
| 000013BC | 0007              |          |          | 750+     | DC    | H'7'                                    | test number                         |
| 000013BE | 00                |          |          | 751+     | DC    | X'00'                                   |                                     |
| 000013BF | 00                |          |          | 752+     | DC    | HL1'0'                                  | M3                                  |
| 000013C0 | E5C7C5D4 40404040 |          |          | 753+     | DC    | CL8'VGEM'                               | instruction name                    |
| 000013C8 | 00001424          |          |          | 754+     | DC    | A(RE7+16)                               | address of v2 source                |
| 000013CC | 00000010          |          |          | 755+     | DC    | A(16)                                   | result length                       |
| 000013D0 | 00001414          |          |          | 756+REA7 | DC    | A(RE7)                                  | result address                      |
| 000013D8 | 00000000 00000000 |          |          | 757+     | DS    | FD                                      | gap                                 |
| 000013E0 | 00000000 00000000 |          |          | 758+V107 | DS    | XL16                                    | V1 output                           |
| 000013E8 | 00000000 00000000 |          |          |          |       |                                         |                                     |
| 000013F0 | 00000000 00000000 |          |          | 759+     | DS    | FD                                      | gap                                 |
|          |                   |          |          | 760+*    |       |                                         |                                     |
| 000013F8 |                   |          |          | 761+X7   | DS    | 0F                                      |                                     |
| 000013F8 | E310 5010 0014    |          | 00000010 | 762+     | LGF   | R1, V2ADDR                              | load v2 source                      |
| 000013FE | E721 0000 0006    |          | 00000000 | 763+     | VL    | v2, 0(R1)                               |                                     |
|          |                   |          |          |          |       |                                         |                                     |
| 00001404 |                   |          |          | 766+     | DS    | 0H                                      | E754 VGEM                           |
| 00001404 | E7                |          |          | 767+     | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00001405 | 12                |          |          | 768+     | DC    | X'12'                                   | v1, v2                              |
| 00001406 | 0000              |          |          | 769+     | DC    | x'0000'                                 | reserved                            |
| 00001408 | 00                |          |          | 770+     | DC    | HL1'0'                                  | M3, RXB                             |
| 00001409 | 54                |          |          | 771+     | DC    | X'54'                                   |                                     |
|          |                   |          |          |          |       |                                         |                                     |
| 0000140A | E710 5028 000E    |          | 000013E0 | 773+     | VST   | V1, V107                                | save v1 output                      |
| 00001410 | 07FB              |          |          | 774+     | BR    | R11                                     | return                              |
| 00001414 |                   |          |          | 775+RE7  | DC    | 0F                                      | xl16 expected result                |
| 00001414 |                   |          |          | 776+     | DROP  | R5                                      |                                     |
| 00001414 | 00000000 00000000 |          |          | 777      | DC    | XL16'0000000000000000 0000000000FF00FF' | result                              |
| 0000141C | 00000000 00FF00FF |          |          |          |       |                                         |                                     |
| 00001424 | 00050000 00000000 |          |          | 778      | DC    | XL16'0005000000000000 0000000000000000' | v2                                  |
| 0000142C | 00000000 00000000 |          |          |          |       |                                         |                                     |
|          |                   |          |          | 779      |       |                                         |                                     |
|          |                   |          |          | 780      | VRR_A | VGEM, 0                                 |                                     |
| 00001438 |                   |          |          | 781+     | DS    | 0FD                                     |                                     |
| 00001438 |                   | 00001438 |          | 782+     | USING | *, R5                                   | base for test data and test routine |
| 00001438 | 00001478          |          |          | 783+T8   | DC    | A(X8)                                   | address of test routine             |
| 0000143C | 0008              |          |          | 784+     | DC    | H'8'                                    | test number                         |
| 0000143E | 00                |          |          | 785+     | DC    | X'00'                                   |                                     |
| 0000143F | 00                |          |          | 786+     | DC    | HL1'0'                                  | M3                                  |
| 00001440 | E5C7C5D4 40404040 |          |          | 787+     | DC    | CL8'VGEM'                               | instruction name                    |
| 00001448 | 000014A4          |          |          | 788+     | DC    | A(RE8+16)                               | address of v2 source                |
| 0000144C | 00000010          |          |          | 789+     | DC    | A(16)                                   | result length                       |
| 00001450 | 00001494          |          |          | 790+REA8 | DC    | A(RE8)                                  | result address                      |
| 00001458 | 00000000 00000000 |          |          | 791+     | DS    | FD                                      | gap                                 |
| 00001460 | 00000000 00000000 |          |          | 792+V108 | DS    | XL16                                    | V1 output                           |

| LOC      | OBJECT CODE       | ADDR1    | ADDR2    | STMT     |       |                                         |                                     |
|----------|-------------------|----------|----------|----------|-------|-----------------------------------------|-------------------------------------|
| 00001468 | 00000000 00000000 |          |          |          |       |                                         |                                     |
| 00001470 | 00000000 00000000 |          |          | 793+     | DS    | FD                                      | gap                                 |
|          |                   |          |          | 794+*    |       |                                         |                                     |
| 00001478 |                   |          |          | 795+X8   | DS    | 0F                                      |                                     |
| 00001478 | E310 5010 0014    |          | 00000010 | 796+     | LGF   | R1,V2ADDR                               | load v2 source                      |
| 0000147E | E721 0000 0006    |          | 00000000 | 797+     | VL    | v2,0(R1)                                |                                     |
| 00001484 |                   |          |          | 800+     | DS    | 0H                                      | E754 VGEM                           |
| 00001484 | E7                |          |          | 801+     | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00001485 | 12                |          |          | 802+     | DC    | X'12'                                   | v1, v2                              |
| 00001486 | 0000              |          |          | 803+     | DC    | x'0000'                                 | reserved                            |
| 00001488 | 00                |          |          | 804+     | DC    | HL1'0'                                  | M3, RXB                             |
| 00001489 | 54                |          |          | 805+     | DC    | X'54'                                   |                                     |
| 0000148A | E710 5028 000E    |          | 00001460 | 807+     | VST   | V1,V108                                 | save v1 output                      |
| 00001490 | 07FB              |          |          | 808+     | BR    | R11                                     | return                              |
| 00001494 |                   |          |          | 809+RE8  | DC    | 0F                                      | xl16 expected result                |
| 00001494 |                   |          |          | 810+     | DROP  | R5                                      |                                     |
| 00001494 | 00000000 00000000 |          |          | 811      | DC    | XL16'0000000000000000 0000000000FFFF00' | result                              |
| 0000149C | 00000000 00FFFF00 |          |          |          |       |                                         |                                     |
| 000014A4 | 00060000 00000000 |          |          | 812      | DC    | XL16'0006000000000000 0000000000000000' | v2                                  |
| 000014AC | 00000000 00000000 |          |          |          |       |                                         |                                     |
|          |                   |          |          | 813      |       |                                         |                                     |
| 000014B8 |                   |          |          | 814      | VRR_A | VGEM,0                                  |                                     |
| 000014B8 |                   | 000014B8 |          | 815+     | DS    | 0FD                                     |                                     |
| 000014B8 | 000014F8          |          |          | 816+     | USING | *,R5                                    | base for test data and test routine |
| 000014BC | 0009              |          |          | 817+T9   | DC    | A(X9)                                   | address of test routine             |
| 000014BE | 00                |          |          | 818+     | DC    | H'9'                                    | test number                         |
| 000014BF | 00                |          |          | 819+     | DC    | X'00'                                   |                                     |
| 000014C0 | E5C7C5D4 40404040 |          |          | 820+     | DC    | HL1'0'                                  | M3                                  |
| 000014C8 | 00001524          |          |          | 821+     | DC    | CL8'VGEM'                               | instruction name                    |
| 000014CC | 00000010          |          |          | 822+     | DC    | A(RE9+16)                               | address of v2 source                |
| 000014D0 | 00001514          |          |          | 823+     | DC    | A(16)                                   | result length                       |
| 000014D8 | 00000000 00000000 |          |          | 824+REA9 | DC    | A(RE9)                                  | result address                      |
| 000014E0 | 00000000 00000000 |          |          | 825+     | DS    | FD                                      | gap                                 |
| 000014E8 | 00000000 00000000 |          |          | 826+V109 | DS    | XL16                                    | V1 output                           |
| 000014F0 | 00000000 00000000 |          |          |          |       |                                         |                                     |
|          |                   |          |          | 827+     | DS    | FD                                      | gap                                 |
|          |                   |          |          | 828+*    |       |                                         |                                     |
| 000014F8 |                   |          |          | 829+X9   | DS    | 0F                                      |                                     |
| 000014F8 | E310 5010 0014    |          | 00000010 | 830+     | LGF   | R1,V2ADDR                               | load v2 source                      |
| 000014FE | E721 0000 0006    |          | 00000000 | 831+     | VL    | v2,0(R1)                                |                                     |
| 00001504 |                   |          |          | 834+     | DS    | 0H                                      | E754 VGEM                           |
| 00001504 | E7                |          |          | 835+     | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00001505 | 12                |          |          | 836+     | DC    | X'12'                                   | v1, v2                              |
| 00001506 | 0000              |          |          | 837+     | DC    | x'0000'                                 | reserved                            |
| 00001508 | 00                |          |          | 838+     | DC    | HL1'0'                                  | M3, RXB                             |
| 00001509 | 54                |          |          | 839+     | DC    | X'54'                                   |                                     |
| 0000150A | E710 5028 000E    |          | 000014E0 | 841+     | VST   | V1,V109                                 | save v1 output                      |
| 00001510 | 07FB              |          |          | 842+     | BR    | R11                                     | return                              |
| 00001514 |                   |          |          | 843+RE9  | DC    | 0F                                      | xl16 expected result                |
| 00001514 |                   |          |          | 844+     | DROP  | R5                                      |                                     |
| 00001514 | 00000000 00000000 |          |          | 845      | DC    | XL16'0000000000000000 0000000000FFFFFF' | result                              |
| 0000151C | 00000000 00FFFFFF |          |          |          |       |                                         |                                     |

| LOC      | OBJECT CODE       | ADDR1    | ADDR2    | STMT      |       |                                         |                                     |
|----------|-------------------|----------|----------|-----------|-------|-----------------------------------------|-------------------------------------|
| 00001524 | 00070000 00000000 |          |          | 846       | DC    | XL16'0007000000000000 0000000000000000' | v2                                  |
| 0000152C | 00000000 00000000 |          |          |           |       |                                         |                                     |
|          |                   |          |          | 847       |       |                                         |                                     |
|          |                   |          |          | 848       | VRR_A | VGEM, 0                                 |                                     |
| 00001538 |                   |          |          | 849+      | DS    | 0FD                                     |                                     |
| 00001538 |                   | 00001538 |          | 850+      | USING | *, R5                                   | base for test data and test routine |
| 00001538 | 00001578          |          |          | 851+T10   | DC    | A(X10)                                  | address of test routine             |
| 0000153C | 000A              |          |          | 852+      | DC    | H'10'                                   | test number                         |
| 0000153E | 00                |          |          | 853+      | DC    | X'00'                                   |                                     |
| 0000153F | 00                |          |          | 854+      | DC    | HL1'0'                                  | M3                                  |
| 00001540 | E5C7C5D4 40404040 |          |          | 855+      | DC    | CL8'VGEM'                               | instruction name                    |
| 00001548 | 000015A4          |          |          | 856+      | DC    | A(RE10+16)                              | address of v2 source                |
| 0000154C | 00000010          |          |          | 857+      | DC    | A(16)                                   | result length                       |
| 00001550 | 00001594          |          |          | 858+REA10 | DC    | A(RE10)                                 | result address                      |
| 00001558 | 00000000 00000000 |          |          | 859+      | DS    | FD                                      | gap                                 |
| 00001560 | 00000000 00000000 |          |          | 860+V1010 | DS    | XL16                                    | V1 output                           |
| 00001568 | 00000000 00000000 |          |          |           |       |                                         |                                     |
| 00001570 | 00000000 00000000 |          |          | 861+      | DS    | FD                                      | gap                                 |
|          |                   |          |          | 862+*     |       |                                         |                                     |
| 00001578 |                   |          |          | 863+X10   | DS    | 0F                                      |                                     |
| 00001578 | E310 5010 0014    |          | 00000010 | 864+      | LGF   | R1, V2ADDR                              | load v2 source                      |
| 0000157E | E721 0000 0006    |          | 00000000 | 865+      | VL    | v2, 0(R1)                               |                                     |
|          |                   |          |          |           |       |                                         |                                     |
| 00001584 |                   |          |          | 868+      | DS    | 0H                                      | E754 VGEM                           |
| 00001584 | E7                |          |          | 869+      | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00001585 | 12                |          |          | 870+      | DC    | X'12'                                   | v1, v2                              |
| 00001586 | 0000              |          |          | 871+      | DC    | x'0000'                                 | reserved                            |
| 00001588 | 00                |          |          | 872+      | DC    | HL1'0'                                  | M3, RXB                             |
| 00001589 | 54                |          |          | 873+      | DC    | X'54'                                   |                                     |
|          |                   |          |          |           |       |                                         |                                     |
| 0000158A | E710 5028 000E    |          | 00001560 | 875+      | VST   | V1, V1010                               | save v1 output                      |
| 00001590 | 07FB              |          |          | 876+      | BR    | R11                                     | return                              |
| 00001594 |                   |          |          | 877+RE10  | DC    | 0F                                      | xl16 expected result                |
| 00001594 |                   |          |          | 878+      | DROP  | R5                                      |                                     |
| 00001594 | 00000000 00000000 |          |          | 879       | DC    | XL16'0000000000000000 00000000FF000000' | result                              |
| 0000159C | 00000000 FF000000 |          |          |           |       |                                         |                                     |
| 000015A4 | 00080000 00000000 |          |          | 880       | DC    | XL16'0008000000000000 0000000000000000' | v2                                  |
| 000015AC | 00000000 00000000 |          |          |           |       |                                         |                                     |
|          |                   |          |          | 881       |       |                                         |                                     |
|          |                   |          |          | 882       | VRR_A | VGEM, 0                                 |                                     |
| 000015B8 |                   |          |          | 883+      | DS    | 0FD                                     |                                     |
| 000015B8 |                   | 000015B8 |          | 884+      | USING | *, R5                                   | base for test data and test routine |
| 000015B8 | 000015F8          |          |          | 885+T11   | DC    | A(X11)                                  | address of test routine             |
| 000015BC | 000B              |          |          | 886+      | DC    | H'11'                                   | test number                         |
| 000015BE | 00                |          |          | 887+      | DC    | X'00'                                   |                                     |
| 000015BF | 00                |          |          | 888+      | DC    | HL1'0'                                  | M3                                  |
| 000015C0 | E5C7C5D4 40404040 |          |          | 889+      | DC    | CL8'VGEM'                               | instruction name                    |
| 000015C8 | 00001624          |          |          | 890+      | DC    | A(RE11+16)                              | address of v2 source                |
| 000015CC | 00000010          |          |          | 891+      | DC    | A(16)                                   | result length                       |
| 000015D0 | 00001614          |          |          | 892+REA11 | DC    | A(RE11)                                 | result address                      |
| 000015D8 | 00000000 00000000 |          |          | 893+      | DS    | FD                                      | gap                                 |
| 000015E0 | 00000000 00000000 |          |          | 894+V1011 | DS    | XL16                                    | V1 output                           |
| 000015E8 | 00000000 00000000 |          |          |           |       |                                         |                                     |
| 000015F0 | 00000000 00000000 |          |          | 895+      | DS    | FD                                      | gap                                 |
|          |                   |          |          | 896+*     |       |                                         |                                     |
| 000015F8 |                   |          |          | 897+X11   | DS    | 0F                                      |                                     |



| LOC      | OBJECT CODE       | ADDR1    | ADDR2    | STMT      |       |                                         |                                     |
|----------|-------------------|----------|----------|-----------|-------|-----------------------------------------|-------------------------------------|
| 000015F8 | E310 5010 0014    |          | 00000010 | 898+      | LGF   | R1,V2ADDR                               | load v2 source                      |
| 000015FE | E721 0000 0006    |          | 00000000 | 899+      | VL    | v2,0(R1)                                |                                     |
| 00001604 |                   |          |          | 902+      | DS    | 0H                                      | E754 VGEM                           |
| 00001604 | E7                |          |          | 903+      | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00001605 | 12                |          |          | 904+      | DC    | X'12'                                   | v1, v2                              |
| 00001606 | 0000              |          |          | 905+      | DC    | x'0000'                                 | reserved                            |
| 00001608 | 00                |          |          | 906+      | DC    | HL1'0'                                  | M3, RXB                             |
| 00001609 | 54                |          |          | 907+      | DC    | X'54'                                   |                                     |
| 0000160A | E710 5028 000E    |          | 000015E0 | 909+      | VST   | V1,V1011                                | save v1 output                      |
| 00001610 | 07FB              |          |          | 910+      | BR    | R11                                     | return                              |
| 00001614 |                   |          |          | 911+RE11  | DC    | 0F                                      | xl16 expected result                |
| 00001614 |                   |          |          | 912+      | DROP  | R5                                      |                                     |
| 00001614 | 00000000 00000000 |          |          | 913       | DC    | XL16'0000000000000000 00000000FF0000FF' | result                              |
| 0000161C | 00000000 FF0000FF |          |          |           |       |                                         |                                     |
| 00001624 | 00090000 00000000 |          |          | 914       | DC    | XL16'0009000000000000 0000000000000000' | v2                                  |
| 0000162C | 00000000 00000000 |          |          |           |       |                                         |                                     |
|          |                   |          |          | 915       |       |                                         |                                     |
|          |                   |          |          | 916       | VRR_A | VGEM,0                                  |                                     |
| 00001638 |                   |          |          | 917+      | DS    | 0FD                                     |                                     |
| 00001638 |                   | 00001638 |          | 918+      | USING | *,R5                                    | base for test data and test routine |
| 00001638 | 00001678          |          |          | 919+T12   | DC    | A(X12)                                  | address of test routine             |
| 0000163C | 000C              |          |          | 920+      | DC    | H'12'                                   | test number                         |
| 0000163E | 00                |          |          | 921+      | DC    | X'00'                                   |                                     |
| 0000163F | 00                |          |          | 922+      | DC    | HL1'0'                                  | M3                                  |
| 00001640 | E5C7C5D4 40404040 |          |          | 923+      | DC    | CL8'VGEM'                               | instruction name                    |
| 00001648 | 000016A4          |          |          | 924+      | DC    | A(RE12+16)                              | address of v2 source                |
| 0000164C | 00000010          |          |          | 925+      | DC    | A(16)                                   | result length                       |
| 00001650 | 00001694          |          |          | 926+REA12 | DC    | A(RE12)                                 | result address                      |
| 00001658 | 00000000 00000000 |          |          | 927+      | DS    | FD                                      | gap                                 |
| 00001660 | 00000000 00000000 |          |          | 928+V1012 | DS    | XL16                                    | V1 output                           |
| 00001668 | 00000000 00000000 |          |          |           |       |                                         |                                     |
| 00001670 | 00000000 00000000 |          |          | 929+      | DS    | FD                                      | gap                                 |
|          |                   |          |          | 930+*     |       |                                         |                                     |
| 00001678 |                   |          |          | 931+X12   | DS    | 0F                                      |                                     |
| 00001678 | E310 5010 0014    |          | 00000010 | 932+      | LGF   | R1,V2ADDR                               | load v2 source                      |
| 0000167E | E721 0000 0006    |          | 00000000 | 933+      | VL    | v2,0(R1)                                |                                     |
| 00001684 |                   |          |          | 936+      | DS    | 0H                                      | E754 VGEM                           |
| 00001684 | E7                |          |          | 937+      | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00001685 | 12                |          |          | 938+      | DC    | X'12'                                   | v1, v2                              |
| 00001686 | 0000              |          |          | 939+      | DC    | x'0000'                                 | reserved                            |
| 00001688 | 00                |          |          | 940+      | DC    | HL1'0'                                  | M3, RXB                             |
| 00001689 | 54                |          |          | 941+      | DC    | X'54'                                   |                                     |
| 0000168A | E710 5028 000E    |          | 00001660 | 943+      | VST   | V1,V1012                                | save v1 output                      |
| 00001690 | 07FB              |          |          | 944+      | BR    | R11                                     | return                              |
| 00001694 |                   |          |          | 945+RE12  | DC    | 0F                                      | xl16 expected result                |
| 00001694 |                   |          |          | 946+      | DROP  | R5                                      |                                     |
| 00001694 | 00000000 00000000 |          |          | 947       | DC    | XL16'0000000000000000 00000000FF00FF00' | result                              |
| 0000169C | 00000000 FF00FF00 |          |          |           |       |                                         |                                     |
| 000016A4 | 000A0000 00000000 |          |          | 948       | DC    | XL16'000A000000000000 0000000000000000' | v2                                  |
| 000016AC | 00000000 00000000 |          |          |           |       |                                         |                                     |
|          |                   |          |          | 949       |       |                                         |                                     |
|          |                   |          |          | 950       | VRR_A | VGEM,0                                  |                                     |

| LOC      | OBJECT CODE       | ADDR1    | ADDR2    | STMT      |       |                                         |                                     |
|----------|-------------------|----------|----------|-----------|-------|-----------------------------------------|-------------------------------------|
| 000016B8 |                   |          |          | 951+      | DS    | 0FD                                     |                                     |
| 000016B8 |                   | 000016B8 |          | 952+      | USING | *,R5                                    | base for test data and test routine |
| 000016B8 | 000016F8          |          |          | 953+T13   | DC    | A(X13)                                  | address of test routine             |
| 000016BC | 000D              |          |          | 954+      | DC    | H'13'                                   | test number                         |
| 000016BE | 00                |          |          | 955+      | DC    | X'00'                                   |                                     |
| 000016BF | 00                |          |          | 956+      | DC    | HL1'0'                                  | M3                                  |
| 000016C0 | E5C7C5D4 40404040 |          |          | 957+      | DC    | CL8'VGEM'                               | instruction name                    |
| 000016C8 | 00001724          |          |          | 958+      | DC    | A(RE13+16)                              | address of v2 source                |
| 000016CC | 00000010          |          |          | 959+      | DC    | A(16)                                   | result length                       |
| 000016D0 | 00001714          |          |          | 960+REA13 | DC    | A(RE13)                                 | result address                      |
| 000016D8 | 00000000 00000000 |          |          | 961+      | DS    | FD                                      | gap                                 |
| 000016E0 | 00000000 00000000 |          |          | 962+V1013 | DS    | XL16                                    | V1 output                           |
| 000016E8 | 00000000 00000000 |          |          |           |       |                                         |                                     |
| 000016F0 | 00000000 00000000 |          |          | 963+      | DS    | FD                                      | gap                                 |
|          |                   |          |          | 964+*     |       |                                         |                                     |
| 000016F8 |                   |          |          | 965+X13   | DS    | 0F                                      |                                     |
| 000016F8 | E310 5010 0014    |          | 00000010 | 966+      | LGF   | R1,V2ADDR                               | load v2 source                      |
| 000016FE | E721 0000 0006    |          | 00000000 | 967+      | VL    | v2,0(R1)                                |                                     |
| 00001704 |                   |          |          | 970+      | DS    | 0H                                      | E754 VGEM                           |
| 00001704 | E7                |          |          | 971+      | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00001705 | 12                |          |          | 972+      | DC    | X'12'                                   | v1, v2                              |
| 00001706 | 0000              |          |          | 973+      | DC    | x'0000'                                 | reserved                            |
| 00001708 | 00                |          |          | 974+      | DC    | HL1'0'                                  | M3, RXB                             |
| 00001709 | 54                |          |          | 975+      | DC    | X'54'                                   |                                     |
| 0000170A | E710 5028 000E    |          | 000016E0 | 977+      | VST   | V1,V1013                                | save v1 output                      |
| 00001710 | 07FB              |          |          | 978+      | BR    | R11                                     | return                              |
| 00001714 |                   |          |          | 979+RE13  | DC    | 0F                                      | xl16 expected result                |
| 00001714 |                   |          |          | 980+      | DROP  | R5                                      |                                     |
| 00001714 | 00000000 00000000 |          |          | 981       | DC    | XL16'0000000000000000 00000000FF00FFFF' | result                              |
| 0000171C | 00000000 FF00FFFF |          |          |           |       |                                         |                                     |
| 00001724 | 000B0000 00000000 |          |          | 982       | DC    | XL16'000B000000000000 0000000000000000' | v2                                  |
| 0000172C | 00000000 00000000 |          |          |           |       |                                         |                                     |
|          |                   |          |          | 983       |       |                                         |                                     |
| 00001738 |                   |          |          | 984       | VRR_A | VGEM,0                                  |                                     |
| 00001738 |                   | 00001738 |          | 985+      | DS    | 0FD                                     |                                     |
| 00001738 | 00001778          |          |          | 986+      | USING | *,R5                                    | base for test data and test routine |
| 00001738 | 00001778          |          |          | 987+T14   | DC    | A(X14)                                  | address of test routine             |
| 0000173C | 000E              |          |          | 988+      | DC    | H'14'                                   | test number                         |
| 0000173E | 00                |          |          | 989+      | DC    | X'00'                                   |                                     |
| 0000173F | 00                |          |          | 990+      | DC    | HL1'0'                                  | M3                                  |
| 00001740 | E5C7C5D4 40404040 |          |          | 991+      | DC    | CL8'VGEM'                               | instruction name                    |
| 00001748 | 000017A4          |          |          | 992+      | DC    | A(RE14+16)                              | address of v2 source                |
| 0000174C | 00000010          |          |          | 993+      | DC    | A(16)                                   | result length                       |
| 00001750 | 00001794          |          |          | 994+REA14 | DC    | A(RE14)                                 | result address                      |
| 00001758 | 00000000 00000000 |          |          | 995+      | DS    | FD                                      | gap                                 |
| 00001760 | 00000000 00000000 |          |          | 996+V1014 | DS    | XL16                                    | V1 output                           |
| 00001768 | 00000000 00000000 |          |          |           |       |                                         |                                     |
| 00001770 | 00000000 00000000 |          |          | 997+      | DS    | FD                                      | gap                                 |
|          |                   |          |          | 998+*     |       |                                         |                                     |
| 00001778 |                   |          |          | 999+X14   | DS    | 0F                                      |                                     |
| 00001778 | E310 5010 0014    |          | 00000010 | 1000+     | LGF   | R1,V2ADDR                               | load v2 source                      |
| 0000177E | E721 0000 0006    |          | 00000000 | 1001+     | VL    | v2,0(R1)                                |                                     |
| 00001784 |                   |          |          | 1004+     | DS    | 0H                                      | E754 VGEM                           |

| LOC      | OBJECT CODE       | ADDR1    | ADDR2    | STMT       |       |                                         |                                     |
|----------|-------------------|----------|----------|------------|-------|-----------------------------------------|-------------------------------------|
| 00001784 | E7                |          |          | 1005+      | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00001785 | 12                |          |          | 1006+      | DC    | X'12'                                   | v1, v2                              |
| 00001786 | 0000              |          |          | 1007+      | DC    | x'0000'                                 | reserved                            |
| 00001788 | 00                |          |          | 1008+      | DC    | HL1'0'                                  | M3, RXB                             |
| 00001789 | 54                |          |          | 1009+      | DC    | X'54'                                   |                                     |
| 0000178A | E710 5028 000E    |          | 00001760 | 1011+      | VST   | V1,V1014                                | save v1 output                      |
| 00001790 | 07FB              |          |          | 1012+      | BR    | R11                                     | return                              |
| 00001794 |                   |          |          | 1013+RE14  | DC    | 0F                                      | xl16 expected result                |
| 00001794 |                   |          |          | 1014+      | DROP  | R5                                      |                                     |
| 00001794 | 00000000 00000000 |          |          | 1015       | DC    | XL16'0000000000000000 00000000FFFF0000' | result                              |
| 0000179C | 00000000 FFFF0000 |          |          |            |       |                                         |                                     |
| 000017A4 | 000C0000 00000000 |          |          | 1016       | DC    | XL16'000C000000000000 0000000000000000' | v2                                  |
| 000017AC | 00000000 00000000 |          |          |            |       |                                         |                                     |
|          |                   |          |          | 1017       |       |                                         |                                     |
| 000017B8 |                   |          |          | 1018       | VRR_A | VGEM, 0                                 |                                     |
| 000017B8 |                   | 000017B8 |          | 1019+      | DS    | 0FD                                     |                                     |
| 000017B8 | 000017F8          |          |          | 1020+      | USING | *,R5                                    | base for test data and test routine |
| 000017BC | 000F              |          |          | 1021+T15   | DC    | A(X15)                                  | address of test routine             |
| 000017BE | 00                |          |          | 1022+      | DC    | H'15'                                   | test number                         |
| 000017BF | 00                |          |          | 1023+      | DC    | X'00'                                   |                                     |
| 000017C0 | E5C7C5D4 40404040 |          |          | 1024+      | DC    | HL1'0'                                  | M3                                  |
| 000017C8 | 00001824          |          |          | 1025+      | DC    | CL8'VGEM'                               | instruction name                    |
| 000017CC | 00000010          |          |          | 1026+      | DC    | A(RE15+16)                              | address of v2 source                |
| 000017D0 | 00001814          |          |          | 1027+      | DC    | A(16)                                   | result length                       |
| 000017D8 | 00000000 00000000 |          |          | 1028+REA15 | DC    | A(RE15)                                 | result address                      |
| 000017E0 | 00000000 00000000 |          |          | 1029+      | DS    | FD                                      | gap                                 |
| 000017E8 | 00000000 00000000 |          |          | 1030+V1015 | DS    | XL16                                    | V1 output                           |
| 000017F0 | 00000000 00000000 |          |          | 1031+      | DS    | FD                                      | gap                                 |
|          |                   |          |          | 1032+*     |       |                                         |                                     |
| 000017F8 |                   |          |          | 1033+X15   | DS    | 0F                                      |                                     |
| 000017F8 | E310 5010 0014    |          | 00000010 | 1034+      | LGF   | R1,V2ADDR                               | load v2 source                      |
| 000017FE | E721 0000 0006    |          | 00000000 | 1035+      | VL    | v2,0(R1)                                |                                     |
| 00001804 |                   |          |          | 1038+      | DS    | 0H                                      | E754 VGEM                           |
| 00001804 | E7                |          |          | 1039+      | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00001805 | 12                |          |          | 1040+      | DC    | X'12'                                   | v1, v2                              |
| 00001806 | 0000              |          |          | 1041+      | DC    | x'0000'                                 | reserved                            |
| 00001808 | 00                |          |          | 1042+      | DC    | HL1'0'                                  | M3, RXB                             |
| 00001809 | 54                |          |          | 1043+      | DC    | X'54'                                   |                                     |
| 0000180A | E710 5028 000E    |          | 000017E0 | 1045+      | VST   | V1,V1015                                | save v1 output                      |
| 00001810 | 07FB              |          |          | 1046+      | BR    | R11                                     | return                              |
| 00001814 |                   |          |          | 1047+RE15  | DC    | 0F                                      | xl16 expected result                |
| 00001814 |                   |          |          | 1048+      | DROP  | R5                                      |                                     |
| 00001814 | 00000000 00000000 |          |          | 1049       | DC    | XL16'0000000000000000 00000000FFFF00FF' | result                              |
| 0000181C | 00000000 FFFF00FF |          |          |            |       |                                         |                                     |
| 00001824 | 000D0000 00000000 |          |          | 1050       | DC    | XL16'000D000000000000 0000000000000000' | v2                                  |
| 0000182C | 00000000 00000000 |          |          |            |       |                                         |                                     |
|          |                   |          |          | 1051       |       |                                         |                                     |
| 00001838 |                   |          |          | 1052       | VRR_A | VGEM, 0                                 |                                     |
| 00001838 |                   | 00001838 |          | 1053+      | DS    | 0FD                                     |                                     |
| 00001838 | 00001878          |          |          | 1054+      | USING | *,R5                                    | base for test data and test routine |
| 0000183C | 0010              |          |          | 1055+T16   | DC    | A(X16)                                  | address of test routine             |
|          |                   |          |          | 1056+      | DC    | H'16'                                   | test number                         |



| LOC      | OBJECT CODE        | ADDR1    | ADDR2    | STMT       |       |                                         |                                     |
|----------|--------------------|----------|----------|------------|-------|-----------------------------------------|-------------------------------------|
| 0000183E | 00                 |          |          | 1057+      | DC    | X'00'                                   |                                     |
| 0000183F | 00                 |          |          | 1058+      | DC    | HL1'0'                                  | M3                                  |
| 00001840 | E5C7C5D4 40404040  |          |          | 1059+      | DC    | CL8'VGEM'                               | instruction name                    |
| 00001848 | 000018A4           |          |          | 1060+      | DC    | A(RE16+16)                              | address of v2 source                |
| 0000184C | 00000010           |          |          | 1061+      | DC    | A(16)                                   | result length                       |
| 00001850 | 00001894           |          |          | 1062+REA16 | DC    | A(RE16)                                 | result address                      |
| 00001858 | 00000000 00000000  |          |          | 1063+      | DS    | FD                                      | gap                                 |
| 00001860 | 00000000 00000000  |          |          | 1064+V1016 | DS    | XL16                                    | V1 output                           |
| 00001868 | 00000000 00000000  |          |          |            |       |                                         |                                     |
| 00001870 | 00000000 00000000  |          |          | 1065+      | DS    | FD                                      | gap                                 |
|          |                    |          |          | 1066+*     |       |                                         |                                     |
| 00001878 |                    |          |          | 1067+X16   | DS    | 0F                                      |                                     |
| 00001878 | E310 5010 0014     |          | 00000010 | 1068+      | LGF   | R1,V2ADDR                               | load v2 source                      |
| 0000187E | E721 0000 0006     |          | 00000000 | 1069+      | VL    | v2,0(R1)                                |                                     |
| 00001884 |                    |          |          | 1072+      | DS    | 0H                                      | E754 VGEM                           |
| 00001884 | E7                 |          |          | 1073+      | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00001885 | 12                 |          |          | 1074+      | DC    | X'12'                                   | v1, v2                              |
| 00001886 | 0000               |          |          | 1075+      | DC    | x'0000'                                 | reserved                            |
| 00001888 | 00                 |          |          | 1076+      | DC    | HL1'0'                                  | M3, RXB                             |
| 00001889 | 54                 |          |          | 1077+      | DC    | X'54'                                   |                                     |
| 0000188A | E710 5028 000E     |          | 00001860 | 1079+      | VST   | V1,V1016                                | save v1 output                      |
| 00001890 | 07FB               |          |          | 1080+      | BR    | R11                                     | return                              |
| 00001894 |                    |          |          | 1081+RE16  | DC    | 0F                                      | xl16 expected result                |
| 00001894 |                    |          |          | 1082+      | DROP  | R5                                      |                                     |
| 00001894 | 00000000 00000000  |          |          | 1083       | DC    | XL16'0000000000000000 00000000FFFFFF00' | result                              |
| 0000189C | 00000000 FFFFFFF00 |          |          |            |       |                                         |                                     |
| 000018A4 | 000E0000 00000000  |          |          | 1084       | DC    | XL16'000E000000000000 0000000000000000' | v2                                  |
| 000018AC | 00000000 00000000  |          |          |            |       |                                         |                                     |
|          |                    |          |          | 1085       |       |                                         |                                     |
| 000018B8 |                    |          |          | 1086       | VRR_A | VGEM,0                                  |                                     |
| 000018B8 |                    | 000018B8 |          | 1087+      | DS    | 0FD                                     |                                     |
| 000018B8 | 000018F8           |          |          | 1088+      | USING | *,R5                                    | base for test data and test routine |
| 000018BC | 0011               |          |          | 1089+T17   | DC    | A(X17)                                  | address of test routine             |
| 000018BE | 00                 |          |          | 1090+      | DC    | H'17'                                   | test number                         |
| 000018BF | 00                 |          |          | 1091+      | DC    | X'00'                                   |                                     |
| 000018C0 | E5C7C5D4 40404040  |          |          | 1092+      | DC    | HL1'0'                                  | M3                                  |
| 000018C8 | 00001924           |          |          | 1093+      | DC    | CL8'VGEM'                               | instruction name                    |
| 000018CC | 00000010           |          |          | 1094+      | DC    | A(RE17+16)                              | address of v2 source                |
| 000018D0 | 00001914           |          |          | 1095+      | DC    | A(16)                                   | result length                       |
| 000018D8 | 00000000 00000000  |          |          | 1096+REA17 | DC    | A(RE17)                                 | result address                      |
| 000018D8 | 00000000 00000000  |          |          | 1097+      | DS    | FD                                      | gap                                 |
| 000018E0 | 00000000 00000000  |          |          | 1098+V1017 | DS    | XL16                                    | V1 output                           |
| 000018E8 | 00000000 00000000  |          |          |            |       |                                         |                                     |
| 000018F0 | 00000000 00000000  |          |          | 1099+      | DS    | FD                                      | gap                                 |
|          |                    |          |          | 1100+*     |       |                                         |                                     |
| 000018F8 |                    |          |          | 1101+X17   | DS    | 0F                                      |                                     |
| 000018F8 | E310 5010 0014     |          | 00000010 | 1102+      | LGF   | R1,V2ADDR                               | load v2 source                      |
| 000018FE | E721 0000 0006     |          | 00000000 | 1103+      | VL    | v2,0(R1)                                |                                     |
| 00001904 |                    |          |          | 1106+      | DS    | 0H                                      | E754 VGEM                           |
| 00001904 | E7                 |          |          | 1107+      | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00001905 | 12                 |          |          | 1108+      | DC    | X'12'                                   | v1, v2                              |
| 00001906 | 0000               |          |          | 1109+      | DC    | x'0000'                                 | reserved                            |
| 00001908 | 00                 |          |          | 1110+      | DC    | HL1'0'                                  | M3, RXB                             |

| LOC      | OBJECT CODE       | ADDR1    | ADDR2    | STMT       |       |                                         |                                     |
|----------|-------------------|----------|----------|------------|-------|-----------------------------------------|-------------------------------------|
| 00001909 | 54                |          |          | 1111+      | DC    | X'54'                                   |                                     |
| 0000190A | E710 5028 000E    |          | 000018E0 | 1113+      | VST   | V1,V1017                                | save v1 output                      |
| 00001910 | 07FB              |          |          | 1114+      | BR    | R11                                     | return                              |
| 00001914 |                   |          |          | 1115+RE17  | DC    | 0F                                      | xl16 expected result                |
| 00001914 |                   |          |          | 1116+      | DROP  | R5                                      |                                     |
| 00001914 | 00000000 00000000 |          |          | 1117       | DC    | XL16'0000000000000000 00000000FFFFFFFF' | result                              |
| 0000191C | 00000000 FFFFFFFF |          |          |            |       |                                         |                                     |
| 00001924 | 000F0000 00000000 |          |          | 1118       | DC    | XL16'000F000000000000 0000000000000000' | v2                                  |
| 0000192C | 00000000 00000000 |          |          |            |       |                                         |                                     |
|          |                   |          |          | 1119       |       |                                         |                                     |
| 00001938 |                   |          |          | 1120       | VRR_A | VGEM,0                                  |                                     |
| 00001938 |                   | 00001938 |          | 1121+      | DS    | 0FD                                     |                                     |
| 00001938 | 00001978          |          |          | 1122+      | USING | *,R5                                    | base for test data and test routine |
| 0000193C | 0012              |          |          | 1123+T18   | DC    | A(X18)                                  | address of test routine             |
| 0000193E | 00                |          |          | 1124+      | DC    | H'18'                                   | test number                         |
| 0000193E | 00                |          |          | 1125+      | DC    | X'00'                                   |                                     |
| 0000193F | 00                |          |          | 1126+      | DC    | HL1'0'                                  | M3                                  |
| 00001940 | E5C7C5D4 40404040 |          |          | 1127+      | DC    | CL8'VGEM'                               | instruction name                    |
| 00001948 | 000019A4          |          |          | 1128+      | DC    | A(RE18+16)                              | address of v2 source                |
| 0000194C | 00000010          |          |          | 1129+      | DC    | A(16)                                   | result length                       |
| 00001950 | 00001994          |          |          | 1130+REA18 | DC    | A(RE18)                                 | result address                      |
| 00001958 | 00000000 00000000 |          |          | 1131+      | DS    | FD                                      | gap                                 |
| 00001960 | 00000000 00000000 |          |          | 1132+V1018 | DS    | XL16                                    | V1 output                           |
| 00001968 | 00000000 00000000 |          |          |            |       |                                         |                                     |
| 00001970 | 00000000 00000000 |          |          | 1133+      | DS    | FD                                      | gap                                 |
|          |                   |          |          | 1134+*     |       |                                         |                                     |
| 00001978 |                   |          |          | 1135+X18   | DS    | 0F                                      |                                     |
| 00001978 | E310 5010 0014    |          | 00000010 | 1136+      | LGF   | R1,V2ADDR                               | load v2 source                      |
| 0000197E | E721 0000 0006    |          | 00000000 | 1137+      | VL    | v2,0(R1)                                |                                     |
|          |                   |          |          |            |       |                                         |                                     |
| 00001984 |                   |          |          | 1140+      | DS    | 0H                                      | E754 VGEM                           |
| 00001984 | E7                |          |          | 1141+      | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00001985 | 12                |          |          | 1142+      | DC    | X'12'                                   | v1, v2                              |
| 00001986 | 0000              |          |          | 1143+      | DC    | x'0000'                                 | reserved                            |
| 00001988 | 00                |          |          | 1144+      | DC    | HL1'0'                                  | M3, RXB                             |
| 00001989 | 54                |          |          | 1145+      | DC    | X'54'                                   |                                     |
|          |                   |          |          |            |       |                                         |                                     |
| 0000198A | E710 5028 000E    |          | 00001960 | 1147+      | VST   | V1,V1018                                | save v1 output                      |
| 00001990 | 07FB              |          |          | 1148+      | BR    | R11                                     | return                              |
| 00001994 |                   |          |          | 1149+RE18  | DC    | 0F                                      | xl16 expected result                |
| 00001994 |                   |          |          | 1150+      | DROP  | R5                                      |                                     |
| 00001994 | 00000000 00000000 |          |          | 1151       | DC    | XL16'0000000000000000 000000FF00000000' | result                              |
| 0000199C | 000000FF 00000000 |          |          |            |       |                                         |                                     |
| 000019A4 | 00100000 00000000 |          |          | 1152       | DC    | XL16'0010000000000000 0000000000000000' | v2                                  |
| 000019AC | 00000000 00000000 |          |          |            |       |                                         |                                     |
|          |                   |          |          | 1153       |       |                                         |                                     |
| 000019B8 |                   |          |          | 1154       | VRR_A | VGEM,0                                  |                                     |
| 000019B8 |                   | 000019B8 |          | 1155+      | DS    | 0FD                                     |                                     |
| 000019B8 | 000019F8          |          |          | 1156+      | USING | *,R5                                    | base for test data and test routine |
| 000019BC | 0013              |          |          | 1157+T19   | DC    | A(X19)                                  | address of test routine             |
| 000019BE | 00                |          |          | 1158+      | DC    | H'19'                                   | test number                         |
| 000019BE | 00                |          |          | 1159+      | DC    | X'00'                                   |                                     |
| 000019BF | 00                |          |          | 1160+      | DC    | HL1'0'                                  | M3                                  |
| 000019C0 | E5C7C5D4 40404040 |          |          | 1161+      | DC    | CL8'VGEM'                               | instruction name                    |
| 000019C8 | 00001A24          |          |          | 1162+      | DC    | A(RE19+16)                              | address of v2 source                |

| LOC      | OBJECT CODE       | ADDR1    | ADDR2    | STMT       |       |                                         |                                     |
|----------|-------------------|----------|----------|------------|-------|-----------------------------------------|-------------------------------------|
| 000019CC | 00000010          |          |          | 1163+      | DC    | A(16)                                   | result length                       |
| 000019D0 | 00001A14          |          |          | 1164+REA19 | DC    | A(RE19)                                 | result address                      |
| 000019D8 | 00000000 00000000 |          |          | 1165+      | DS    | FD                                      | gap                                 |
| 000019E0 | 00000000 00000000 |          |          | 1166+V1019 | DS    | XL16                                    | V1 output                           |
| 000019E8 | 00000000 00000000 |          |          |            |       |                                         |                                     |
| 000019F0 | 00000000 00000000 |          |          | 1167+      | DS    | FD                                      | gap                                 |
|          |                   |          |          | 1168+*     |       |                                         |                                     |
| 000019F8 |                   |          |          | 1169+X19   | DS    | 0F                                      |                                     |
| 000019F8 | E310 5010 0014    |          | 00000010 | 1170+      | LGF   | R1,V2ADDR                               | load v2 source                      |
| 000019FE | E721 0000 0006    |          | 00000000 | 1171+      | VL    | v2,0(R1)                                |                                     |
| 00001A04 |                   |          |          | 1174+      | DS    | 0H                                      | E754 VGEM                           |
| 00001A04 | E7                |          |          | 1175+      | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00001A05 | 12                |          |          | 1176+      | DC    | X'12'                                   | v1, v2                              |
| 00001A06 | 0000              |          |          | 1177+      | DC    | x'0000'                                 | reserved                            |
| 00001A08 | 00                |          |          | 1178+      | DC    | HL1'0'                                  | M3, RXB                             |
| 00001A09 | 54                |          |          | 1179+      | DC    | X'54'                                   |                                     |
| 00001A0A | E710 5028 000E    |          | 000019E0 | 1181+      | VST   | V1,V1019                                | save v1 output                      |
| 00001A10 | 07FB              |          |          | 1182+      | BR    | R11                                     | return                              |
| 00001A14 |                   |          |          | 1183+RE19  | DC    | 0F                                      | xl16 expected result                |
| 00001A14 |                   |          |          | 1184+      | DROP  | R5                                      |                                     |
| 00001A14 | 00000000 00000000 |          |          | 1185       | DC    | XL16'0000000000000000 0000FF0000000000' | result                              |
| 00001A1C | 0000FF00 00000000 |          |          |            |       |                                         |                                     |
| 00001A24 | 00200000 00000000 |          |          | 1186       | DC    | XL16'0020000000000000 0000000000000000' | v2                                  |
| 00001A2C | 00000000 00000000 |          |          |            |       |                                         |                                     |
|          |                   |          |          | 1187       |       |                                         |                                     |
|          |                   |          |          | 1188       | VRR_A | VGEM,0                                  |                                     |
| 00001A38 |                   |          |          | 1189+      | DS    | 0FD                                     |                                     |
| 00001A38 |                   | 00001A38 |          | 1190+      | USING | *,R5                                    | base for test data and test routine |
| 00001A38 | 00001A78          |          |          | 1191+T20   | DC    | A(X20)                                  | address of test routine             |
| 00001A3C | 0014              |          |          | 1192+      | DC    | H'20'                                   | test number                         |
| 00001A3E | 00                |          |          | 1193+      | DC    | X'00'                                   |                                     |
| 00001A3F | 00                |          |          | 1194+      | DC    | HL1'0'                                  | M3                                  |
| 00001A40 | E5C7C5D4 40404040 |          |          | 1195+      | DC    | CL8'VGEM'                               | instruction name                    |
| 00001A48 | 00001AA4          |          |          | 1196+      | DC    | A(RE20+16)                              | address of v2 source                |
| 00001A4C | 00000010          |          |          | 1197+      | DC    | A(16)                                   | result length                       |
| 00001A50 | 00001A94          |          |          | 1198+REA20 | DC    | A(RE20)                                 | result address                      |
| 00001A58 | 00000000 00000000 |          |          | 1199+      | DS    | FD                                      | gap                                 |
| 00001A60 | 00000000 00000000 |          |          | 1200+V1020 | DS    | XL16                                    | V1 output                           |
| 00001A68 | 00000000 00000000 |          |          |            |       |                                         |                                     |
| 00001A70 | 00000000 00000000 |          |          | 1201+      | DS    | FD                                      | gap                                 |
|          |                   |          |          | 1202+*     |       |                                         |                                     |
| 00001A78 |                   |          |          | 1203+X20   | DS    | 0F                                      |                                     |
| 00001A78 | E310 5010 0014    |          | 00000010 | 1204+      | LGF   | R1,V2ADDR                               | load v2 source                      |
| 00001A7E | E721 0000 0006    |          | 00000000 | 1205+      | VL    | v2,0(R1)                                |                                     |
| 00001A84 |                   |          |          | 1208+      | DS    | 0H                                      | E754 VGEM                           |
| 00001A84 | E7                |          |          | 1209+      | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00001A85 | 12                |          |          | 1210+      | DC    | X'12'                                   | v1, v2                              |
| 00001A86 | 0000              |          |          | 1211+      | DC    | x'0000'                                 | reserved                            |
| 00001A88 | 00                |          |          | 1212+      | DC    | HL1'0'                                  | M3, RXB                             |
| 00001A89 | 54                |          |          | 1213+      | DC    | X'54'                                   |                                     |
| 00001A8A | E710 5028 000E    |          | 00001A60 | 1215+      | VST   | V1,V1020                                | save v1 output                      |
| 00001A90 | 07FB              |          |          | 1216+      | BR    | R11                                     | return                              |

| LOC      | OBJECT CODE       | ADDR1    | ADDR2    | STMT       |       |                                         |                                     |
|----------|-------------------|----------|----------|------------|-------|-----------------------------------------|-------------------------------------|
| 00001A94 |                   |          |          | 1217+RE20  | DC    | 0F                                      | xl16 expected result                |
| 00001A94 |                   |          |          | 1218+      | DROP  | R5                                      |                                     |
| 00001A94 | 00000000 00000000 |          |          | 1219       | DC    | XL16'0000000000000000 00FF000000000000' | result                              |
| 00001A9C | 00FF0000 00000000 |          |          |            |       |                                         |                                     |
| 00001AA4 | 00400000 00000000 |          |          | 1220       | DC    | XL16'0040000000000000 0000000000000000' | v2                                  |
| 00001AAC | 00000000 00000000 |          |          |            |       |                                         |                                     |
|          |                   |          |          | 1221       |       |                                         |                                     |
|          |                   |          |          | 1222       | VRR_A | VGEM, 0                                 |                                     |
| 00001AB8 |                   |          |          | 1223+      | DS    | 0FD                                     |                                     |
| 00001AB8 |                   | 00001AB8 |          | 1224+      | USING | *, R5                                   | base for test data and test routine |
| 00001AB8 | 00001AF8          |          |          | 1225+T21   | DC    | A(X21)                                  | address of test routine             |
| 00001ABC | 0015              |          |          | 1226+      | DC    | H'21'                                   | test number                         |
| 00001ABE | 00                |          |          | 1227+      | DC    | X'00'                                   |                                     |
| 00001ABF | 00                |          |          | 1228+      | DC    | HL1'0'                                  | M3                                  |
| 00001AC0 | E5C7C5D4 40404040 |          |          | 1229+      | DC    | CL8'VGEM'                               | instruction name                    |
| 00001AC8 | 00001B24          |          |          | 1230+      | DC    | A(RE21+16)                              | address of v2 source                |
| 00001ACC | 00000010          |          |          | 1231+      | DC    | A(16)                                   | result length                       |
| 00001AD0 | 00001B14          |          |          | 1232+REA21 | DC    | A(RE21)                                 | result address                      |
| 00001AD8 | 00000000 00000000 |          |          | 1233+      | DS    | FD                                      | gap                                 |
| 00001AE0 | 00000000 00000000 |          |          | 1234+V1021 | DS    | XL16                                    | V1 output                           |
| 00001AE8 | 00000000 00000000 |          |          |            |       |                                         |                                     |
| 00001AF0 | 00000000 00000000 |          |          | 1235+      | DS    | FD                                      | gap                                 |
|          |                   |          |          | 1236+*     |       |                                         |                                     |
| 00001AF8 |                   |          |          | 1237+X21   | DS    | 0F                                      |                                     |
| 00001AF8 | E310 5010 0014    |          | 00000010 | 1238+      | LGF   | R1, V2ADDR                              | load v2 source                      |
| 00001AFE | E721 0000 0006    |          | 00000000 | 1239+      | VL    | v2, 0(R1)                               |                                     |
|          |                   |          |          |            |       |                                         |                                     |
| 00001B04 |                   |          |          | 1242+      | DS    | 0H                                      | E754 VGEM                           |
| 00001B04 | E7                |          |          | 1243+      | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00001B05 | 12                |          |          | 1244+      | DC    | X'12'                                   | v1, v2                              |
| 00001B06 | 0000              |          |          | 1245+      | DC    | x'0000'                                 | reserved                            |
| 00001B08 | 00                |          |          | 1246+      | DC    | HL1'0'                                  | M3, RXB                             |
| 00001B09 | 54                |          |          | 1247+      | DC    | X'54'                                   |                                     |
|          |                   |          |          |            |       |                                         |                                     |
| 00001B0A | E710 5028 000E    |          | 00001AE0 | 1249+      | VST   | V1, V1021                               | save v1 output                      |
| 00001B10 | 07FB              |          |          | 1250+      | BR    | R11                                     | return                              |
| 00001B14 |                   |          |          | 1251+RE21  | DC    | 0F                                      | xl16 expected result                |
| 00001B14 |                   |          |          | 1252+      | DROP  | R5                                      |                                     |
| 00001B14 | 00000000 00000000 |          |          | 1253       | DC    | XL16'0000000000000000 FF00000000000000' | result                              |
| 00001B1C | FF000000 00000000 |          |          |            |       |                                         |                                     |
| 00001B24 | 00800000 00000000 |          |          | 1254       | DC    | XL16'0080000000000000 0000000000000000' | v2                                  |
| 00001B2C | 00000000 00000000 |          |          |            |       |                                         |                                     |
|          |                   |          |          | 1255       |       |                                         |                                     |
|          |                   |          |          | 1256       | VRR_A | VGEM, 0                                 |                                     |
| 00001B38 |                   |          |          | 1257+      | DS    | 0FD                                     |                                     |
| 00001B38 |                   | 00001B38 |          | 1258+      | USING | *, R5                                   | base for test data and test routine |
| 00001B38 | 00001B78          |          |          | 1259+T22   | DC    | A(X22)                                  | address of test routine             |
| 00001B3C | 0016              |          |          | 1260+      | DC    | H'22'                                   | test number                         |
| 00001B3E | 00                |          |          | 1261+      | DC    | X'00'                                   |                                     |
| 00001B3F | 00                |          |          | 1262+      | DC    | HL1'0'                                  | M3                                  |
| 00001B40 | E5C7C5D4 40404040 |          |          | 1263+      | DC    | CL8'VGEM'                               | instruction name                    |
| 00001B48 | 00001BA4          |          |          | 1264+      | DC    | A(RE22+16)                              | address of v2 source                |
| 00001B4C | 00000010          |          |          | 1265+      | DC    | A(16)                                   | result length                       |
| 00001B50 | 00001B94          |          |          | 1266+REA22 | DC    | A(RE22)                                 | result address                      |
| 00001B58 | 00000000 00000000 |          |          | 1267+      | DS    | FD                                      | gap                                 |
| 00001B60 | 00000000 00000000 |          |          | 1268+V1022 | DS    | XL16                                    | V1 output                           |

| LOC      | OBJECT CODE       | ADDR1    | ADDR2    | STMT       |       |                                         |                                     |
|----------|-------------------|----------|----------|------------|-------|-----------------------------------------|-------------------------------------|
| 00001B68 | 00000000 00000000 |          |          |            |       |                                         |                                     |
| 00001B70 | 00000000 00000000 |          |          | 1269+      | DS    | FD                                      | gap                                 |
|          |                   |          |          | 1270+*     |       |                                         |                                     |
| 00001B78 |                   |          |          | 1271+X22   | DS    | 0F                                      |                                     |
| 00001B78 | E310 5010 0014    |          | 00000010 | 1272+      | LGF   | R1,V2ADDR                               | load v2 source                      |
| 00001B7E | E721 0000 0006    |          | 00000000 | 1273+      | VL    | v2,0(R1)                                |                                     |
| 00001B84 |                   |          |          | 1276+      | DS    | 0H                                      | E754 VGEM                           |
| 00001B84 | E7                |          |          | 1277+      | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00001B85 | 12                |          |          | 1278+      | DC    | X'12'                                   | v1, v2                              |
| 00001B86 | 0000              |          |          | 1279+      | DC    | x'0000'                                 | reserved                            |
| 00001B88 | 00                |          |          | 1280+      | DC    | HL1'0'                                  | M3, RXB                             |
| 00001B89 | 54                |          |          | 1281+      | DC    | X'54'                                   |                                     |
| 00001B8A | E710 5028 000E    |          | 00001B60 | 1283+      | VST   | V1,V1022                                | save v1 output                      |
| 00001B90 | 07FB              |          |          | 1284+      | BR    | R11                                     | return                              |
| 00001B94 |                   |          |          | 1285+RE22  | DC    | 0F                                      | xl16 expected result                |
| 00001B94 |                   |          |          | 1286+      | DROP  | R5                                      |                                     |
| 00001B94 | 00000000 000000FF |          |          | 1287       | DC    | XL16'00000000000000FF 0000000000000000' | result                              |
| 00001B9C | 00000000 00000000 |          |          |            |       |                                         |                                     |
| 00001BA4 | 01000000 00000000 |          |          | 1288       | DC    | XL16'0100000000000000 0000000000000000' | v2                                  |
| 00001BAC | 00000000 00000000 |          |          |            |       |                                         |                                     |
|          |                   |          |          | 1289       |       |                                         |                                     |
| 00001BB8 |                   |          |          | 1290       | VRR_A | VGEM,0                                  |                                     |
| 00001BB8 |                   | 00001BB8 |          | 1291+      | DS    | 0FD                                     |                                     |
| 00001BB8 | 00001BF8          |          |          | 1292+      | USING | *,R5                                    | base for test data and test routine |
| 00001BBC | 0017              |          |          | 1293+T23   | DC    | A(X23)                                  | address of test routine             |
| 00001BBE | 00                |          |          | 1294+      | DC    | H'23'                                   | test number                         |
| 00001BBE | 00                |          |          | 1295+      | DC    | X'00'                                   |                                     |
| 00001BBF | 00                |          |          | 1296+      | DC    | HL1'0'                                  | M3                                  |
| 00001BC0 | E5C7C5D4 40404040 |          |          | 1297+      | DC    | CL8'VGEM'                               | instruction name                    |
| 00001BC8 | 00001C24          |          |          | 1298+      | DC    | A(RE23+16)                              | address of v2 source                |
| 00001BCC | 00000010          |          |          | 1299+      | DC    | A(16)                                   | result length                       |
| 00001BD0 | 00001C14          |          |          | 1300+REA23 | DC    | A(RE23)                                 | result address                      |
| 00001BD8 | 00000000 00000000 |          |          | 1301+      | DS    | FD                                      | gap                                 |
| 00001BE0 | 00000000 00000000 |          |          | 1302+V1023 | DS    | XL16                                    | V1 output                           |
| 00001BE8 | 00000000 00000000 |          |          |            |       |                                         |                                     |
| 00001BF0 | 00000000 00000000 |          |          | 1303+      | DS    | FD                                      | gap                                 |
|          |                   |          |          | 1304+*     |       |                                         |                                     |
| 00001BF8 |                   |          |          | 1305+X23   | DS    | 0F                                      |                                     |
| 00001BF8 | E310 5010 0014    |          | 00000010 | 1306+      | LGF   | R1,V2ADDR                               | load v2 source                      |
| 00001BFE | E721 0000 0006    |          | 00000000 | 1307+      | VL    | v2,0(R1)                                |                                     |
| 00001C04 |                   |          |          | 1310+      | DS    | 0H                                      | E754 VGEM                           |
| 00001C04 | E7                |          |          | 1311+      | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00001C05 | 12                |          |          | 1312+      | DC    | X'12'                                   | v1, v2                              |
| 00001C06 | 0000              |          |          | 1313+      | DC    | x'0000'                                 | reserved                            |
| 00001C08 | 00                |          |          | 1314+      | DC    | HL1'0'                                  | M3, RXB                             |
| 00001C09 | 54                |          |          | 1315+      | DC    | X'54'                                   |                                     |
| 00001C0A | E710 5028 000E    |          | 00001BE0 | 1317+      | VST   | V1,V1023                                | save v1 output                      |
| 00001C10 | 07FB              |          |          | 1318+      | BR    | R11                                     | return                              |
| 00001C14 |                   |          |          | 1319+RE23  | DC    | 0F                                      | xl16 expected result                |
| 00001C14 |                   |          |          | 1320+      | DROP  | R5                                      |                                     |
| 00001C14 | 00000000 0000FF00 |          |          | 1321       | DC    | XL16'000000000000FF00 0000000000000000' | result                              |
| 00001C1C | 00000000 00000000 |          |          |            |       |                                         |                                     |



| LOC      | OBJECT CODE       | ADDR1    | ADDR2    | STMT       |       |                                         |                                     |
|----------|-------------------|----------|----------|------------|-------|-----------------------------------------|-------------------------------------|
| 00001C24 | 02000000 00000000 |          |          | 1322       | DC    | XL16'0200000000000000 0000000000000000' | v2                                  |
| 00001C2C | 00000000 00000000 |          |          |            |       |                                         |                                     |
|          |                   |          |          | 1323       |       |                                         |                                     |
| 00001C38 |                   |          |          | 1324       | VRR_A | VGEM, 0                                 |                                     |
| 00001C38 |                   | 00001C38 |          | 1325+      | DS    | 0FD                                     |                                     |
| 00001C38 | 00001C78          |          |          | 1326+      | USING | *, R5                                   | base for test data and test routine |
| 00001C3C | 0018              |          |          | 1327+T24   | DC    | A(X24)                                  | address of test routine             |
| 00001C3E | 00                |          |          | 1328+      | DC    | H'24'                                   | test number                         |
| 00001C3F | 00                |          |          | 1329+      | DC    | X'00'                                   |                                     |
| 00001C40 | E5C7C5D4 40404040 |          |          | 1330+      | DC    | HL1'0'                                  | M3                                  |
| 00001C48 | 00001CA4          |          |          | 1331+      | DC    | CL8'VGEM'                               | instruction name                    |
| 00001C4C | 00000010          |          |          | 1332+      | DC    | A(RE24+16)                              | address of v2 source                |
| 00001C50 | 00001C94          |          |          | 1333+      | DC    | A(16)                                   | result length                       |
| 00001C58 | 00000000 00000000 |          |          | 1334+REA24 | DC    | A(RE24)                                 | result address                      |
| 00001C60 | 00000000 00000000 |          |          | 1335+      | DS    | FD                                      | gap                                 |
| 00001C68 | 00000000 00000000 |          |          | 1336+V1024 | DS    | XL16                                    | V1 output                           |
| 00001C70 | 00000000 00000000 |          |          |            |       |                                         |                                     |
|          |                   |          |          | 1337+      | DS    | FD                                      | gap                                 |
|          |                   |          |          | 1338+*     |       |                                         |                                     |
| 00001C78 |                   |          |          | 1339+X24   | DS    | 0F                                      |                                     |
| 00001C78 | E310 5010 0014    |          | 00000010 | 1340+      | LGF   | R1, V2ADDR                              | load v2 source                      |
| 00001C7E | E721 0000 0006    |          | 00000000 | 1341+      | VL    | v2, 0(R1)                               |                                     |
|          |                   |          |          |            |       |                                         |                                     |
| 00001C84 |                   |          |          | 1344+      | DS    | 0H                                      | E754 VGEM                           |
| 00001C84 | E7                |          |          | 1345+      | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00001C85 | 12                |          |          | 1346+      | DC    | X'12'                                   | v1, v2                              |
| 00001C86 | 0000              |          |          | 1347+      | DC    | x'0000'                                 | reserved                            |
| 00001C88 | 00                |          |          | 1348+      | DC    | HL1'0'                                  | M3, RXB                             |
| 00001C89 | 54                |          |          | 1349+      | DC    | X'54'                                   |                                     |
|          |                   |          |          |            |       |                                         |                                     |
| 00001C8A | E710 5028 000E    |          | 00001C60 | 1351+      | VST   | V1, V1024                               | save v1 output                      |
| 00001C90 | 07FB              |          |          | 1352+      | BR    | R11                                     | return                              |
| 00001C94 |                   |          |          | 1353+RE24  | DC    | 0F                                      | xl16 expected result                |
| 00001C94 |                   |          |          | 1354+      | DROP  | R5                                      |                                     |
| 00001C94 | 00000000 00FF0000 |          |          | 1355       | DC    | XL16'0000000000FF0000 0000000000000000' | result                              |
| 00001C9C | 00000000 00000000 |          |          |            |       |                                         |                                     |
| 00001CA4 | 04000000 00000000 |          |          | 1356       | DC    | XL16'0400000000000000 0000000000000000' | v2                                  |
| 00001CAC | 00000000 00000000 |          |          |            |       |                                         |                                     |
|          |                   |          |          | 1357       |       |                                         |                                     |
| 00001CB8 |                   |          |          | 1358       | VRR_A | VGEM, 0                                 |                                     |
| 00001CB8 |                   | 00001CB8 |          | 1359+      | DS    | 0FD                                     |                                     |
| 00001CB8 | 00001CF8          |          |          | 1360+      | USING | *, R5                                   | base for test data and test routine |
| 00001CBC | 0019              |          |          | 1361+T25   | DC    | A(X25)                                  | address of test routine             |
| 00001CBE | 00                |          |          | 1362+      | DC    | H'25'                                   | test number                         |
| 00001CBF | 00                |          |          | 1363+      | DC    | X'00'                                   |                                     |
| 00001CC0 | E5C7C5D4 40404040 |          |          | 1364+      | DC    | HL1'0'                                  | M3                                  |
| 00001CC8 | 00001D24          |          |          | 1365+      | DC    | CL8'VGEM'                               | instruction name                    |
| 00001CCC | 00000010          |          |          | 1366+      | DC    | A(RE25+16)                              | address of v2 source                |
| 00001CD0 | 00001D14          |          |          | 1367+      | DC    | A(16)                                   | result length                       |
| 00001CD8 | 00000000 00000000 |          |          | 1368+REA25 | DC    | A(RE25)                                 | result address                      |
| 00001CE0 | 00000000 00000000 |          |          | 1369+      | DS    | FD                                      | gap                                 |
| 00001CE8 | 00000000 00000000 |          |          | 1370+V1025 | DS    | XL16                                    | V1 output                           |
| 00001CF0 | 00000000 00000000 |          |          |            |       |                                         |                                     |
|          |                   |          |          | 1371+      | DS    | FD                                      | gap                                 |
|          |                   |          |          | 1372+*     |       |                                         |                                     |
| 00001CF8 |                   |          |          | 1373+X25   | DS    | 0F                                      |                                     |

| LOC      | OBJECT CODE       | ADDR1    | ADDR2    | STMT       |       |                                         |                                     |
|----------|-------------------|----------|----------|------------|-------|-----------------------------------------|-------------------------------------|
| 00001CF8 | E310 5010 0014    |          | 00000010 | 1374+      | LGF   | R1,V2ADDR                               | load v2 source                      |
| 00001CFE | E721 0000 0006    |          | 00000000 | 1375+      | VL    | v2,0(R1)                                |                                     |
| 00001D04 |                   |          |          | 1378+      | DS    | 0H                                      | E754 VGEM                           |
| 00001D04 | E7                |          |          | 1379+      | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00001D05 | 12                |          |          | 1380+      | DC    | X'12'                                   | v1, v2                              |
| 00001D06 | 0000              |          |          | 1381+      | DC    | x'0000'                                 | reserved                            |
| 00001D08 | 00                |          |          | 1382+      | DC    | HL1'0'                                  | M3, RXB                             |
| 00001D09 | 54                |          |          | 1383+      | DC    | X'54'                                   |                                     |
| 00001D0A | E710 5028 000E    |          | 00001CE0 | 1385+      | VST   | V1,V1025                                | save v1 output                      |
| 00001D10 | 07FB              |          |          | 1386+      | BR    | R11                                     | return                              |
| 00001D14 |                   |          |          | 1387+RE25  | DC    | 0F                                      | xl16 expected result                |
| 00001D14 |                   |          |          | 1388+      | DROP  | R5                                      |                                     |
| 00001D14 | 00000000 FF000000 |          |          | 1389       | DC    | XL16'00000000FF000000 0000000000000000' | result                              |
| 00001D1C | 00000000 00000000 |          |          |            |       |                                         |                                     |
| 00001D24 | 08000000 00000000 |          |          | 1390       | DC    | XL16'0800000000000000 0000000000000000' | v2                                  |
| 00001D2C | 00000000 00000000 |          |          |            |       |                                         |                                     |
| 00001D38 |                   |          |          | 1391       |       |                                         |                                     |
| 00001D38 |                   |          |          | 1392       | VRR_A | VGEM,0                                  |                                     |
| 00001D38 |                   | 00001D38 |          | 1393+      | DS    | 0FD                                     |                                     |
| 00001D38 | 00001D78          |          |          | 1394+      | USING | *,R5                                    | base for test data and test routine |
| 00001D3C | 001A              |          |          | 1395+T26   | DC    | A(X26)                                  | address of test routine             |
| 00001D3E | 00                |          |          | 1396+      | DC    | H'26'                                   | test number                         |
| 00001D3F | 00                |          |          | 1397+      | DC    | X'00'                                   |                                     |
| 00001D40 | E5C7C5D4 40404040 |          |          | 1398+      | DC    | HL1'0'                                  | M3                                  |
| 00001D48 | 00001DA4          |          |          | 1399+      | DC    | CL8'VGEM'                               | instruction name                    |
| 00001D4C | 00000010          |          |          | 1400+      | DC    | A(RE26+16)                              | address of v2 source                |
| 00001D50 | 00001D94          |          |          | 1401+      | DC    | A(16)                                   | result length                       |
| 00001D58 | 00000000 00000000 |          |          | 1402+REA26 | DC    | A(RE26)                                 | result address                      |
| 00001D58 | 00000000 00000000 |          |          | 1403+      | DS    | FD                                      | gap                                 |
| 00001D60 | 00000000 00000000 |          |          | 1404+V1026 | DS    | XL16                                    | V1 output                           |
| 00001D68 | 00000000 00000000 |          |          |            |       |                                         |                                     |
| 00001D70 | 00000000 00000000 |          |          | 1405+      | DS    | FD                                      | gap                                 |
| 00001D78 |                   |          |          | 1406+*     |       |                                         |                                     |
| 00001D78 | E310 5010 0014    |          | 00000010 | 1407+X26   | DS    | 0F                                      |                                     |
| 00001D7E | E721 0000 0006    |          | 00000000 | 1408+      | LGF   | R1,V2ADDR                               | load v2 source                      |
|          |                   |          |          | 1409+      | VL    | v2,0(R1)                                |                                     |
| 00001D84 |                   |          |          | 1412+      | DS    | 0H                                      | E754 VGEM                           |
| 00001D84 | E7                |          |          | 1413+      | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00001D85 | 12                |          |          | 1414+      | DC    | X'12'                                   | v1, v2                              |
| 00001D86 | 0000              |          |          | 1415+      | DC    | x'0000'                                 | reserved                            |
| 00001D88 | 00                |          |          | 1416+      | DC    | HL1'0'                                  | M3, RXB                             |
| 00001D89 | 54                |          |          | 1417+      | DC    | X'54'                                   |                                     |
| 00001D8A | E710 5028 000E    |          | 00001D60 | 1419+      | VST   | V1,V1026                                | save v1 output                      |
| 00001D90 | 07FB              |          |          | 1420+      | BR    | R11                                     | return                              |
| 00001D94 |                   |          |          | 1421+RE26  | DC    | 0F                                      | xl16 expected result                |
| 00001D94 |                   |          |          | 1422+      | DROP  | R5                                      |                                     |
| 00001D94 | 000000FF 00000000 |          |          | 1423       | DC    | XL16'000000FF00000000 0000000000000000' | result                              |
| 00001D9C | 00000000 00000000 |          |          |            |       |                                         |                                     |
| 00001DA4 | 10000000 00000000 |          |          | 1424       | DC    | XL16'1000000000000000 0000000000000000' | v2                                  |
| 00001DAC | 00000000 00000000 |          |          |            |       |                                         |                                     |
|          |                   |          |          | 1425       |       |                                         |                                     |
|          |                   |          |          | 1426       | VRR_A | VGEM,0                                  |                                     |



| LOC      | OBJECT CODE       | ADDR1    | ADDR2    | STMT       |       |                                         |                                     |
|----------|-------------------|----------|----------|------------|-------|-----------------------------------------|-------------------------------------|
| 00001DB8 |                   |          |          | 1427+      | DS    | 0FD                                     |                                     |
| 00001DB8 |                   | 00001DB8 |          | 1428+      | USING | *,R5                                    | base for test data and test routine |
| 00001DB8 | 00001DF8          |          |          | 1429+T27   | DC    | A(X27)                                  | address of test routine             |
| 00001DBC | 001B              |          |          | 1430+      | DC    | H'27'                                   | test number                         |
| 00001DBE | 00                |          |          | 1431+      | DC    | X'00'                                   |                                     |
| 00001DBF | 00                |          |          | 1432+      | DC    | HL1'0'                                  | M3                                  |
| 00001DC0 | E5C7C5D4 40404040 |          |          | 1433+      | DC    | CL8'VGEM'                               | instruction name                    |
| 00001DC8 | 00001E24          |          |          | 1434+      | DC    | A(RE27+16)                              | address of v2 source                |
| 00001DCC | 00000010          |          |          | 1435+      | DC    | A(16)                                   | result length                       |
| 00001DD0 | 00001E14          |          |          | 1436+REA27 | DC    | A(RE27)                                 | result address                      |
| 00001DD8 | 00000000 00000000 |          |          | 1437+      | DS    | FD                                      | gap                                 |
| 00001DE0 | 00000000 00000000 |          |          | 1438+V1027 | DS    | XL16                                    | V1 output                           |
| 00001DE8 | 00000000 00000000 |          |          |            |       |                                         |                                     |
| 00001DF0 | 00000000 00000000 |          |          | 1439+      | DS    | FD                                      | gap                                 |
|          |                   |          |          | 1440+*     |       |                                         |                                     |
| 00001DF8 |                   |          |          | 1441+X27   | DS    | 0F                                      |                                     |
| 00001DF8 | E310 5010 0014    |          | 00000010 | 1442+      | LGF   | R1,V2ADDR                               | load v2 source                      |
| 00001DFE | E721 0000 0006    |          | 00000000 | 1443+      | VL    | v2,0(R1)                                |                                     |
| 00001E04 |                   |          |          | 1446+      | DS    | 0H                                      | E754 VGEM                           |
| 00001E04 | E7                |          |          | 1447+      | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00001E05 | 12                |          |          | 1448+      | DC    | X'12'                                   | v1, v2                              |
| 00001E06 | 0000              |          |          | 1449+      | DC    | x'0000'                                 | reserved                            |
| 00001E08 | 00                |          |          | 1450+      | DC    | HL1'0'                                  | M3, RXB                             |
| 00001E09 | 54                |          |          | 1451+      | DC    | X'54'                                   |                                     |
| 00001E0A | E710 5028 000E    |          | 00001DE0 | 1453+      | VST   | V1,V1027                                | save v1 output                      |
| 00001E10 | 07FB              |          |          | 1454+      | BR    | R11                                     | return                              |
| 00001E14 |                   |          |          | 1455+RE27  | DC    | 0F                                      | xl16 expected result                |
| 00001E14 |                   |          |          | 1456+      | DROP  | R5                                      |                                     |
| 00001E14 | 0000FF00 00000000 |          |          | 1457       | DC    | XL16'0000FF0000000000 0000000000000000' | result                              |
| 00001E1C | 00000000 00000000 |          |          |            |       |                                         |                                     |
| 00001E24 | 20000000 00000000 |          |          | 1458       | DC    | XL16'2000000000000000 0000000000000000' | v2                                  |
| 00001E2C | 00000000 00000000 |          |          |            |       |                                         |                                     |
|          |                   |          |          | 1459       |       |                                         |                                     |
| 00001E38 |                   |          |          | 1460       | VRR_A | VGEM,0                                  |                                     |
| 00001E38 |                   | 00001E38 |          | 1461+      | DS    | 0FD                                     |                                     |
| 00001E38 | 00001E78          |          |          | 1462+      | USING | *,R5                                    | base for test data and test routine |
| 00001E3C | 001C              |          |          | 1463+T28   | DC    | A(X28)                                  | address of test routine             |
| 00001E3E | 00                |          |          | 1464+      | DC    | H'28'                                   | test number                         |
| 00001E3F | 00                |          |          | 1465+      | DC    | X'00'                                   |                                     |
| 00001E3F | 00                |          |          | 1466+      | DC    | HL1'0'                                  | M3                                  |
| 00001E40 | E5C7C5D4 40404040 |          |          | 1467+      | DC    | CL8'VGEM'                               | instruction name                    |
| 00001E48 | 00001EA4          |          |          | 1468+      | DC    | A(RE28+16)                              | address of v2 source                |
| 00001E4C | 00000010          |          |          | 1469+      | DC    | A(16)                                   | result length                       |
| 00001E50 | 00001E94          |          |          | 1470+REA28 | DC    | A(RE28)                                 | result address                      |
| 00001E58 | 00000000 00000000 |          |          | 1471+      | DS    | FD                                      | gap                                 |
| 00001E60 | 00000000 00000000 |          |          | 1472+V1028 | DS    | XL16                                    | V1 output                           |
| 00001E68 | 00000000 00000000 |          |          |            |       |                                         |                                     |
| 00001E70 | 00000000 00000000 |          |          | 1473+      | DS    | FD                                      | gap                                 |
|          |                   |          |          | 1474+*     |       |                                         |                                     |
| 00001E78 |                   |          |          | 1475+X28   | DS    | 0F                                      |                                     |
| 00001E78 | E310 5010 0014    |          | 00000010 | 1476+      | LGF   | R1,V2ADDR                               | load v2 source                      |
| 00001E7E | E721 0000 0006    |          | 00000000 | 1477+      | VL    | v2,0(R1)                                |                                     |
| 00001E84 |                   |          |          | 1480+      | DS    | 0H                                      | E754 VGEM                           |

| LOC      | OBJECT CODE       | ADDR1    | ADDR2    | STMT       |       |                                         |                                     |
|----------|-------------------|----------|----------|------------|-------|-----------------------------------------|-------------------------------------|
| 00001E84 | E7                |          |          | 1481+      | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00001E85 | 12                |          |          | 1482+      | DC    | X'12'                                   | v1, v2                              |
| 00001E86 | 0000              |          |          | 1483+      | DC    | x'0000'                                 | reserved                            |
| 00001E88 | 00                |          |          | 1484+      | DC    | HL1'0'                                  | M3, RXB                             |
| 00001E89 | 54                |          |          | 1485+      | DC    | X'54'                                   |                                     |
| 00001E8A | E710 5028 000E    |          | 00001E60 | 1487+      | VST   | V1,V1028                                | save v1 output                      |
| 00001E90 | 07FB              |          |          | 1488+      | BR    | R11                                     | return                              |
| 00001E94 |                   |          |          | 1489+RE28  | DC    | 0F                                      | xl16 expected result                |
| 00001E94 |                   |          |          | 1490+      | DROP  | R5                                      |                                     |
| 00001E94 | 00FF0000 00000000 |          |          | 1491       | DC    | XL16'00FF000000000000 0000000000000000' | result                              |
| 00001E9C | 00000000 00000000 |          |          |            |       |                                         |                                     |
| 00001EA4 | 40000000 00000000 |          |          | 1492       | DC    | XL16'4000000000000000 0000000000000000' | v2                                  |
| 00001EAC | 00000000 00000000 |          |          |            |       |                                         |                                     |
|          |                   |          |          | 1493       |       |                                         |                                     |
| 00001EB8 |                   |          |          | 1494       | VRR_A | VGEM, 0                                 |                                     |
| 00001EB8 |                   | 00001EB8 |          | 1495+      | DS    | 0FD                                     |                                     |
| 00001EB8 | 00001EF8          |          |          | 1496+      | USING | *, R5                                   | base for test data and test routine |
| 00001EBC | 001D              |          |          | 1497+T29   | DC    | A(X29)                                  | address of test routine             |
| 00001EBE | 00                |          |          | 1498+      | DC    | H'29'                                   | test number                         |
| 00001EBF | 00                |          |          | 1499+      | DC    | X'00'                                   |                                     |
| 00001EC0 | E5C7C5D4 40404040 |          |          | 1500+      | DC    | HL1'0'                                  | M3                                  |
| 00001EC8 | 00001F24          |          |          | 1501+      | DC    | CL8'VGEM'                               | instruction name                    |
| 00001ECC | 00000010          |          |          | 1502+      | DC    | A(RE29+16)                              | address of v2 source                |
| 00001ED0 | 00001F14          |          |          | 1503+      | DC    | A(16)                                   | result length                       |
| 00001ED8 | 00000000 00000000 |          |          | 1504+REA29 | DC    | A(RE29)                                 | result address                      |
| 00001EE0 | 00000000 00000000 |          |          | 1505+      | DS    | FD                                      | gap                                 |
| 00001EE8 | 00000000 00000000 |          |          | 1506+V1029 | DS    | XL16                                    | V1 output                           |
| 00001EF0 | 00000000 00000000 |          |          | 1507+      | DS    | FD                                      | gap                                 |
|          |                   |          |          | 1508+*     |       |                                         |                                     |
| 00001EF8 |                   |          |          | 1509+X29   | DS    | 0F                                      |                                     |
| 00001EF8 | E310 5010 0014    |          | 00000010 | 1510+      | LGF   | R1,V2ADDR                               | load v2 source                      |
| 00001EFE | E721 0000 0006    |          | 00000000 | 1511+      | VL    | v2, 0(R1)                               |                                     |
| 00001F04 |                   |          |          | 1514+      | DS    | 0H                                      | E754 VGEM                           |
| 00001F04 | E7                |          |          | 1515+      | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00001F05 | 12                |          |          | 1516+      | DC    | X'12'                                   | v1, v2                              |
| 00001F06 | 0000              |          |          | 1517+      | DC    | x'0000'                                 | reserved                            |
| 00001F08 | 00                |          |          | 1518+      | DC    | HL1'0'                                  | M3, RXB                             |
| 00001F09 | 54                |          |          | 1519+      | DC    | X'54'                                   |                                     |
| 00001F0A | E710 5028 000E    |          | 00001EE0 | 1521+      | VST   | V1,V1029                                | save v1 output                      |
| 00001F10 | 07FB              |          |          | 1522+      | BR    | R11                                     | return                              |
| 00001F14 |                   |          |          | 1523+RE29  | DC    | 0F                                      | xl16 expected result                |
| 00001F14 |                   |          |          | 1524+      | DROP  | R5                                      |                                     |
| 00001F14 | FF000000 00000000 |          |          | 1525       | DC    | XL16'FF00000000000000 0000000000000000' | result                              |
| 00001F1C | 00000000 00000000 |          |          |            |       |                                         |                                     |
| 00001F24 | 80000000 00000000 |          |          | 1526       | DC    | XL16'8000000000000000 0000000000000000' | v2                                  |
| 00001F2C | 00000000 00000000 |          |          |            |       |                                         |                                     |
|          |                   |          |          | 1527       |       |                                         |                                     |
| 00001F38 |                   |          |          | 1528       | VRR_A | VGEM, 0                                 |                                     |
| 00001F38 |                   | 00001F38 |          | 1529+      | DS    | 0FD                                     |                                     |
| 00001F38 | 00001F78          |          |          | 1530+      | USING | *, R5                                   | base for test data and test routine |
| 00001F3C | 001E              |          |          | 1531+T30   | DC    | A(X30)                                  | address of test routine             |
|          |                   |          |          | 1532+      | DC    | H'30'                                   | test number                         |

| LOC      | OBJECT CODE       | ADDR1    | ADDR2    | STMT            |       |                                           |                                     |
|----------|-------------------|----------|----------|-----------------|-------|-------------------------------------------|-------------------------------------|
| 00001F3E | 00                |          |          | 1533+           | DC    | X'00'                                     |                                     |
| 00001F3F | 00                |          |          | 1534+           | DC    | HL1'0'                                    | M3                                  |
| 00001F40 | E5C7C5D4 40404040 |          |          | 1535+           | DC    | CL8'VGEM'                                 | instruction name                    |
| 00001F48 | 00001FA4          |          |          | 1536+           | DC    | A(RE30+16)                                | address of v2 source                |
| 00001F4C | 00000010          |          |          | 1537+           | DC    | A(16)                                     | result length                       |
| 00001F50 | 00001F94          |          |          | 1538+REA30      | DC    | A(RE30)                                   | result address                      |
| 00001F58 | 00000000 00000000 |          |          | 1539+           | DS    | FD                                        | gap                                 |
| 00001F60 | 00000000 00000000 |          |          | 1540+V1030      | DS    | XL16                                      | V1 output                           |
| 00001F68 | 00000000 00000000 |          |          |                 |       |                                           |                                     |
| 00001F70 | 00000000 00000000 |          |          | 1541+           | DS    | FD                                        | gap                                 |
|          |                   |          |          | 1542+*          |       |                                           |                                     |
| 00001F78 |                   |          |          | 1543+X30        | DS    | 0F                                        |                                     |
| 00001F78 | E310 5010 0014    |          | 00000010 | 1544+           | LGF   | R1,V2ADDR                                 | load v2 source                      |
| 00001F7E | E721 0000 0006    |          | 00000000 | 1545+           | VL    | v2,0(R1)                                  |                                     |
| 00001F84 |                   |          |          | 1548+           | DS    | 0H                                        | E754 VGEM                           |
| 00001F84 | E7                |          |          | 1549+           | DC    | X'E7'                                     | - Vector Generate Element Masks     |
| 00001F85 | 12                |          |          | 1550+           | DC    | X'12'                                     | v1, v2                              |
| 00001F86 | 0000              |          |          | 1551+           | DC    | x'0000'                                   | reserved                            |
| 00001F88 | 00                |          |          | 1552+           | DC    | HL1'0'                                    | M3, RXB                             |
| 00001F89 | 54                |          |          | 1553+           | DC    | X'54'                                     |                                     |
| 00001F8A | E710 5028 000E    |          | 00001F60 | 1555+           | VST   | V1,V1030                                  | save v1 output                      |
| 00001F90 | 07FB              |          |          | 1556+           | BR    | R11                                       | return                              |
| 00001F94 |                   |          |          | 1557+RE30       | DC    | 0F                                        | xl16 expected result                |
| 00001F94 |                   |          |          | 1558+           | DROP  | R5                                        |                                     |
| 00001F94 | FFFFFFFF FFFFFFFF |          |          | 1559            | DC    | XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFF' | result                              |
| 00001F9C | FFFFFFFF FFFFFFFF |          |          |                 |       |                                           |                                     |
| 00001FA4 | FFFF0000 00000000 |          |          | 1560            | DC    | XL16'FFFF000000000000 0000000000000000'   | v2                                  |
| 00001FAC | 00000000 00000000 |          |          |                 |       |                                           |                                     |
|          |                   |          |          | 1561            |       |                                           |                                     |
|          |                   |          |          | 1562 * Halfword |       | M3 = 1                                    |                                     |
|          |                   |          |          | 1563            | VRR_A | VGEM,1                                    |                                     |
| 00001FB8 |                   |          |          | 1564+           | DS    | 0FD                                       |                                     |
| 00001FB8 |                   | 00001FB8 |          | 1565+           | USING | *,R5                                      | base for test data and test routine |
| 00001FB8 | 00001FF8          |          |          | 1566+T31        | DC    | A(X31)                                    | address of test routine             |
| 00001FBC | 001F              |          |          | 1567+           | DC    | H'31'                                     | test number                         |
| 00001FBE | 00                |          |          | 1568+           | DC    | X'00'                                     |                                     |
| 00001FBF | 01                |          |          | 1569+           | DC    | HL1'1'                                    | M3                                  |
| 00001FC0 | E5C7C5D4 40404040 |          |          | 1570+           | DC    | CL8'VGEM'                                 | instruction name                    |
| 00001FC8 | 00002024          |          |          | 1571+           | DC    | A(RE31+16)                                | address of v2 source                |
| 00001FCC | 00000010          |          |          | 1572+           | DC    | A(16)                                     | result length                       |
| 00001FD0 | 00002014          |          |          | 1573+REA31      | DC    | A(RE31)                                   | result address                      |
| 00001FD8 | 00000000 00000000 |          |          | 1574+           | DS    | FD                                        | gap                                 |
| 00001FE0 | 00000000 00000000 |          |          | 1575+V1031      | DS    | XL16                                      | V1 output                           |
| 00001FE8 | 00000000 00000000 |          |          |                 |       |                                           |                                     |
| 00001FF0 | 00000000 00000000 |          |          | 1576+           | DS    | FD                                        | gap                                 |
|          |                   |          |          | 1577+*          |       |                                           |                                     |
| 00001FF8 |                   |          |          | 1578+X31        | DS    | 0F                                        |                                     |
| 00001FF8 | E310 5010 0014    |          | 00000010 | 1579+           | LGF   | R1,V2ADDR                                 | load v2 source                      |
| 00001FFE | E721 0000 0006    |          | 00000000 | 1580+           | VL    | v2,0(R1)                                  |                                     |
| 00002004 |                   |          |          | 1583+           | DS    | 0H                                        | E754 VGEM                           |
| 00002004 | E7                |          |          | 1584+           | DC    | X'E7'                                     | - Vector Generate Element Masks     |
| 00002005 | 12                |          |          | 1585+           | DC    | X'12'                                     | v1, v2                              |
| 00002006 | 0000              |          |          | 1586+           | DC    | x'0000'                                   | reserved                            |

| LOC      | OBJECT CODE       | ADDR1    | ADDR2    | STMT       |       |                                         |                                     |
|----------|-------------------|----------|----------|------------|-------|-----------------------------------------|-------------------------------------|
| 00002008 | 10                |          |          | 1587+      | DC    | HL1'16'                                 | M3, RXB                             |
| 00002009 | 54                |          |          | 1588+      | DC    | X'54'                                   |                                     |
| 0000200A | E710 5028 000E    |          | 00001FE0 | 1590+      | VST   | V1,V1031                                | save v1 output                      |
| 00002010 | 07FB              |          |          | 1591+      | BR    | R11                                     | return                              |
| 00002014 |                   |          |          | 1592+RE31  | DC    | 0F                                      | xl16 expected result                |
| 00002014 |                   |          |          | 1593+      | DROP  | R5                                      |                                     |
| 00002014 | 00000000 00000000 |          |          | 1594       | DC    | XL16'0000000000000000 0000000000000000' | result                              |
| 0000201C | 00000000 00000000 |          |          |            |       |                                         |                                     |
| 00002024 | 00000000 00000000 |          |          | 1595       | DC    | XL16'0000000000000000 0000000000000000' | v2                                  |
| 0000202C | 00000000 00000000 |          |          |            |       |                                         |                                     |
| 00002038 |                   |          |          | 1596       |       |                                         |                                     |
| 00002038 |                   |          |          | 1597       | VRR_A | VGEM,1                                  |                                     |
| 00002038 | 00002078          | 00002038 |          | 1598+      | DS    | 0FD                                     |                                     |
| 00002038 |                   |          |          | 1599+      | USING | *,R5                                    | base for test data and test routine |
| 0000203C | 0020              |          |          | 1600+T32   | DC    | A(X32)                                  | address of test routine             |
| 0000203E | 00                |          |          | 1601+      | DC    | H'32'                                   | test number                         |
| 0000203F | 01                |          |          | 1602+      | DC    | X'00'                                   |                                     |
| 00002040 | E5C7C5D4 40404040 |          |          | 1603+      | DC    | HL1'1'                                  | M3                                  |
| 00002048 | 000020A4          |          |          | 1604+      | DC    | CL8'VGEM'                               | instruction name                    |
| 0000204C | 00000010          |          |          | 1605+      | DC    | A(RE32+16)                              | address of v2 source                |
| 00002050 | 00002094          |          |          | 1606+      | DC    | A(16)                                   | result length                       |
| 00002058 | 00000000 00000000 |          |          | 1607+REA32 | DC    | A(RE32)                                 | result address                      |
| 00002060 | 00000000 00000000 |          |          | 1608+      | DS    | FD                                      | gap                                 |
| 00002068 | 00000000 00000000 |          |          | 1609+V1032 | DS    | XL16                                    | V1 output                           |
| 00002070 | 00000000 00000000 |          |          |            |       |                                         |                                     |
| 00002078 |                   |          |          | 1610+      | DS    | FD                                      | gap                                 |
| 00002078 | E310 5010 0014    |          |          | 1611+*     |       |                                         |                                     |
| 00002078 | E721 0000 0006    |          |          | 1612+X32   | DS    | 0F                                      |                                     |
| 0000207E |                   |          | 00000010 | 1613+      | LGF   | R1,V2ADDR                               | load v2 source                      |
| 00002084 |                   |          | 00000000 | 1614+      | VL    | v2,0(R1)                                |                                     |
| 00002084 | E7                |          |          | 1617+      | DS    | 0H                                      | E754 VGEM                           |
| 00002084 | 12                |          |          | 1618+      | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00002085 |                   |          |          | 1619+      | DC    | X'12'                                   | v1, v2                              |
| 00002086 | 0000              |          |          | 1620+      | DC    | x'0000'                                 | reserved                            |
| 00002088 | 10                |          |          | 1621+      | DC    | HL1'16'                                 | M3, RXB                             |
| 00002089 | 54                |          |          | 1622+      | DC    | X'54'                                   |                                     |
| 0000208A | E710 5028 000E    |          | 00002060 | 1624+      | VST   | V1,V1032                                | save v1 output                      |
| 00002090 | 07FB              |          |          | 1625+      | BR    | R11                                     | return                              |
| 00002094 |                   |          |          | 1626+RE32  | DC    | 0F                                      | xl16 expected result                |
| 00002094 |                   |          |          | 1627+      | DROP  | R5                                      |                                     |
| 00002094 | 00000000 00000000 |          |          | 1628       | DC    | XL16'0000000000000000 0000000000000000' | result                              |
| 0000209C | 00000000 00000000 |          |          |            |       |                                         |                                     |
| 000020A4 | 00FFFFFF FFFFFFFF |          |          | 1629       | DC    | XL16'00FFFFFFF FFFFFFFF'                | v2                                  |
| 000020AC | FFFFFFF FFFFFFFF  |          |          |            |       |                                         |                                     |
| 000020B8 |                   |          |          | 1630       |       |                                         |                                     |
| 000020B8 |                   |          |          | 1631       | VRR_A | VGEM,1                                  |                                     |
| 000020B8 | 000020F8          | 000020B8 |          | 1632+      | DS    | 0FD                                     |                                     |
| 000020B8 |                   |          |          | 1633+      | USING | *,R5                                    | base for test data and test routine |
| 000020BC | 0021              |          |          | 1634+T33   | DC    | A(X33)                                  | address of test routine             |
| 000020BE | 00                |          |          | 1635+      | DC    | H'33'                                   | test number                         |
| 000020BF | 01                |          |          | 1636+      | DC    | X'00'                                   |                                     |
| 000020C0 | E5C7C5D4 40404040 |          |          | 1637+      | DC    | HL1'1'                                  | M3                                  |
|          |                   |          |          | 1638+      | DC    | CL8'VGEM'                               | instruction name                    |

| LOC      | OBJECT CODE       | ADDR1    | ADDR2    | STMT       |       |                                         |                                     |
|----------|-------------------|----------|----------|------------|-------|-----------------------------------------|-------------------------------------|
| 000020C8 | 00002124          |          |          | 1639+      | DC    | A(RE33+16)                              | address of v2 source                |
| 000020CC | 00000010          |          |          | 1640+      | DC    | A(16)                                   | result length                       |
| 000020D0 | 00002114          |          |          | 1641+REA33 | DC    | A(RE33)                                 | result address                      |
| 000020D8 | 00000000 00000000 |          |          | 1642+      | DS    | FD                                      | gap                                 |
| 000020E0 | 00000000 00000000 |          |          | 1643+V1033 | DS    | XL16                                    | V1 output                           |
| 000020E8 | 00000000 00000000 |          |          |            |       |                                         |                                     |
| 000020F0 | 00000000 00000000 |          |          | 1644+      | DS    | FD                                      | gap                                 |
|          |                   |          |          | 1645+*     |       |                                         |                                     |
| 000020F8 |                   |          |          | 1646+X33   | DS    | 0F                                      |                                     |
| 000020F8 | E310 5010 0014    |          | 00000010 | 1647+      | LGF   | R1,V2ADDR                               | load v2 source                      |
| 000020FE | E721 0000 0006    |          | 00000000 | 1648+      | VL    | v2,0(R1)                                |                                     |
| 00002104 |                   |          |          | 1651+      | DS    | 0H                                      | E754 VGEM                           |
| 00002104 | E7                |          |          | 1652+      | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00002105 | 12                |          |          | 1653+      | DC    | X'12'                                   | v1, v2                              |
| 00002106 | 0000              |          |          | 1654+      | DC    | x'0000'                                 | reserved                            |
| 00002108 | 10                |          |          | 1655+      | DC    | HL1'16'                                 | M3, RXB                             |
| 00002109 | 54                |          |          | 1656+      | DC    | X'54'                                   |                                     |
| 0000210A | E710 5028 000E    |          | 000020E0 | 1658+      | VST   | V1,V1033                                | save v1 output                      |
| 00002110 | 07FB              |          |          | 1659+      | BR    | R11                                     | return                              |
| 00002114 |                   |          |          | 1660+RE33  | DC    | 0F                                      | xl16 expected result                |
| 00002114 |                   |          |          | 1661+      | DROP  | R5                                      |                                     |
| 00002114 | 00000000 00000000 |          |          | 1662       | DC    | XL16'0000000000000000 000000000000FFFF' | result                              |
| 0000211C | 00000000 0000FFFF |          |          |            |       |                                         |                                     |
| 00002124 | 01000000 00000000 |          |          | 1663       | DC    | XL16'0100000000000000 0000000000000000' | v2                                  |
| 0000212C | 00000000 00000000 |          |          |            |       |                                         |                                     |
|          |                   |          |          | 1664       |       |                                         |                                     |
|          |                   |          |          | 1665       | VRR_A | VGEM,1                                  |                                     |
| 00002138 |                   |          |          | 1666+      | DS    | 0FD                                     |                                     |
| 00002138 |                   | 00002138 |          | 1667+      | USING | *,R5                                    | base for test data and test routine |
| 00002138 | 00002178          |          |          | 1668+T34   | DC    | A(X34)                                  | address of test routine             |
| 0000213C | 0022              |          |          | 1669+      | DC    | H'34'                                   | test number                         |
| 0000213E | 00                |          |          | 1670+      | DC    | X'00'                                   |                                     |
| 0000213F | 01                |          |          | 1671+      | DC    | HL1'1'                                  | M3                                  |
| 00002140 | E5C7C5D4 40404040 |          |          | 1672+      | DC    | CL8'VGEM'                               | instruction name                    |
| 00002148 | 000021A4          |          |          | 1673+      | DC    | A(RE34+16)                              | address of v2 source                |
| 0000214C | 00000010          |          |          | 1674+      | DC    | A(16)                                   | result length                       |
| 00002150 | 00002194          |          |          | 1675+REA34 | DC    | A(RE34)                                 | result address                      |
| 00002158 | 00000000 00000000 |          |          | 1676+      | DS    | FD                                      | gap                                 |
| 00002160 | 00000000 00000000 |          |          | 1677+V1034 | DS    | XL16                                    | V1 output                           |
| 00002168 | 00000000 00000000 |          |          |            |       |                                         |                                     |
| 00002170 | 00000000 00000000 |          |          | 1678+      | DS    | FD                                      | gap                                 |
|          |                   |          |          | 1679+*     |       |                                         |                                     |
| 00002178 |                   |          |          | 1680+X34   | DS    | 0F                                      |                                     |
| 00002178 | E310 5010 0014    |          | 00000010 | 1681+      | LGF   | R1,V2ADDR                               | load v2 source                      |
| 0000217E | E721 0000 0006    |          | 00000000 | 1682+      | VL    | v2,0(R1)                                |                                     |
| 00002184 |                   |          |          | 1685+      | DS    | 0H                                      | E754 VGEM                           |
| 00002184 | E7                |          |          | 1686+      | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00002185 | 12                |          |          | 1687+      | DC    | X'12'                                   | v1, v2                              |
| 00002186 | 0000              |          |          | 1688+      | DC    | x'0000'                                 | reserved                            |
| 00002188 | 10                |          |          | 1689+      | DC    | HL1'16'                                 | M3, RXB                             |
| 00002189 | 54                |          |          | 1690+      | DC    | X'54'                                   |                                     |
| 0000218A | E710 5028 000E    |          | 00002160 | 1692+      | VST   | V1,V1034                                | save v1 output                      |



| LOC      | OBJECT CODE       | ADDR1    | ADDR2    | STMT       |       |                                         |                                     |
|----------|-------------------|----------|----------|------------|-------|-----------------------------------------|-------------------------------------|
| 00002190 | 07FB              |          |          | 1693+      | BR    | R11                                     | return                              |
| 00002194 |                   |          |          | 1694+RE34  | DC    | 0F                                      | xl16 expected result                |
| 00002194 |                   |          |          | 1695+      | DROP  | R5                                      |                                     |
| 00002194 | 00000000 00000000 |          |          | 1696       | DC    | XL16'0000000000000000 00000000FFFF0000' | result                              |
| 0000219C | 00000000 FFFF0000 |          |          |            |       |                                         |                                     |
| 000021A4 | 02000000 00000000 |          |          | 1697       | DC    | XL16'0200000000000000 0000000000000000' | v2                                  |
| 000021AC | 00000000 00000000 |          |          |            |       |                                         |                                     |
|          |                   |          |          | 1698       |       |                                         |                                     |
| 000021B8 |                   |          |          | 1699       | VRR_A | VGEM,1                                  |                                     |
| 000021B8 |                   | 000021B8 |          | 1700+      | DS    | 0FD                                     |                                     |
| 000021B8 | 000021F8          |          |          | 1701+      | USING | *,R5                                    | base for test data and test routine |
| 000021BC | 0023              |          |          | 1702+T35   | DC    | A(X35)                                  | address of test routine             |
| 000021BE | 00                |          |          | 1703+      | DC    | H'35'                                   | test number                         |
| 000021BF | 01                |          |          | 1704+      | DC    | X'00'                                   |                                     |
| 000021C0 | E5C7C5D4 40404040 |          |          | 1705+      | DC    | HL1'1'                                  | M3                                  |
| 000021C8 | 00002224          |          |          | 1706+      | DC    | CL8'VGEM'                               | instruction name                    |
| 000021CC | 00000010          |          |          | 1707+      | DC    | A(RE35+16)                              | address of v2 source                |
| 000021D0 | 00002214          |          |          | 1708+      | DC    | A(16)                                   | result length                       |
| 000021D8 | 00000000 00000000 |          |          | 1709+REA35 | DC    | A(RE35)                                 | result address                      |
| 000021E0 | 00000000 00000000 |          |          | 1710+      | DS    | FD                                      | gap                                 |
| 000021E8 | 00000000 00000000 |          |          | 1711+V1035 | DS    | XL16                                    | V1 output                           |
| 000021F0 | 00000000 00000000 |          |          |            |       |                                         |                                     |
|          |                   |          |          | 1712+      | DS    | FD                                      | gap                                 |
|          |                   |          |          | 1713+*     |       |                                         |                                     |
| 000021F8 |                   |          |          | 1714+X35   | DS    | 0F                                      |                                     |
| 000021F8 | E310 5010 0014    |          | 00000010 | 1715+      | LGF   | R1,V2ADDR                               | load v2 source                      |
| 000021FE | E721 0000 0006    |          | 00000000 | 1716+      | VL    | v2,0(R1)                                |                                     |
|          |                   |          |          |            |       |                                         |                                     |
| 00002204 |                   |          |          | 1719+      | DS    | 0H                                      | E754 VGEM                           |
| 00002204 | E7                |          |          | 1720+      | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00002205 | 12                |          |          | 1721+      | DC    | X'12'                                   | v1, v2                              |
| 00002206 | 0000              |          |          | 1722+      | DC    | x'0000'                                 | reserved                            |
| 00002208 | 10                |          |          | 1723+      | DC    | HL1'16'                                 | M3, RXB                             |
| 00002209 | 54                |          |          | 1724+      | DC    | X'54'                                   |                                     |
|          |                   |          |          |            |       |                                         |                                     |
| 0000220A | E710 5028 000E    |          | 000021E0 | 1726+      | VST   | V1,V1035                                | save v1 output                      |
| 00002210 | 07FB              |          |          | 1727+      | BR    | R11                                     | return                              |
| 00002214 |                   |          |          | 1728+RE35  | DC    | 0F                                      | xl16 expected result                |
| 00002214 |                   |          |          | 1729+      | DROP  | R5                                      |                                     |
| 00002214 | 00000000 00000000 |          |          | 1730       | DC    | XL16'0000000000000000 00000000FFFFFFFF' | result                              |
| 0000221C | 00000000 FFFFFFFF |          |          |            |       |                                         |                                     |
| 00002224 | 03000000 00000000 |          |          | 1731       | DC    | XL16'0300000000000000 0000000000000000' | v2                                  |
| 0000222C | 00000000 00000000 |          |          |            |       |                                         |                                     |
|          |                   |          |          | 1732       |       |                                         |                                     |
| 00002238 |                   |          |          | 1733       | VRR_A | VGEM,1                                  |                                     |
| 00002238 |                   | 00002238 |          | 1734+      | DS    | 0FD                                     |                                     |
| 00002238 | 00002278          |          |          | 1735+      | USING | *,R5                                    | base for test data and test routine |
| 0000223C | 0024              |          |          | 1736+T36   | DC    | A(X36)                                  | address of test routine             |
| 0000223E | 00                |          |          | 1737+      | DC    | H'36'                                   | test number                         |
| 0000223F | 01                |          |          | 1738+      | DC    | X'00'                                   |                                     |
| 00002240 | E5C7C5D4 40404040 |          |          | 1739+      | DC    | HL1'1'                                  | M3                                  |
| 00002248 | 000022A4          |          |          | 1740+      | DC    | CL8'VGEM'                               | instruction name                    |
| 0000224C | 00000010          |          |          | 1741+      | DC    | A(RE36+16)                              | address of v2 source                |
| 00002250 | 00002294          |          |          | 1742+      | DC    | A(16)                                   | result length                       |
| 00002258 | 00000000 00000000 |          |          | 1743+REA36 | DC    | A(RE36)                                 | result address                      |
|          |                   |          |          | 1744+      | DS    | FD                                      | gap                                 |

| LOC      | OBJECT CODE       | ADDR1    | ADDR2    | STMT            |       |                                         |                                     |
|----------|-------------------|----------|----------|-----------------|-------|-----------------------------------------|-------------------------------------|
| 00002260 | 00000000 00000000 |          |          | 1745+V1036      | DS    | XL16                                    | V1 output                           |
| 00002268 | 00000000 00000000 |          |          |                 |       |                                         |                                     |
| 00002270 | 00000000 00000000 |          |          | 1746+<br>1747+* | DS    | FD                                      | gap                                 |
| 00002278 |                   |          |          | 1748+X36        | DS    | 0F                                      |                                     |
| 00002278 | E310 5010 0014    |          | 00000010 | 1749+           | LGF   | R1,V2ADDR                               | load v2 source                      |
| 0000227E | E721 0000 0006    |          | 00000000 | 1750+           | VL    | v2,0(R1)                                |                                     |
| 00002284 |                   |          |          | 1753+           | DS    | 0H                                      | E754 VGEM                           |
| 00002284 | E7                |          |          | 1754+           | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00002285 | 12                |          |          | 1755+           | DC    | X'12'                                   | v1, v2                              |
| 00002286 | 0000              |          |          | 1756+           | DC    | x'0000'                                 | reserved                            |
| 00002288 | 10                |          |          | 1757+           | DC    | HL1'16'                                 | M3, RXB                             |
| 00002289 | 54                |          |          | 1758+           | DC    | X'54'                                   |                                     |
| 0000228A | E710 5028 000E    |          | 00002260 | 1760+           | VST   | V1,V1036                                | save v1 output                      |
| 00002290 | 07FB              |          |          | 1761+           | BR    | R11                                     | return                              |
| 00002294 |                   |          |          | 1762+RE36       | DC    | 0F                                      | xl16 expected result                |
| 00002294 |                   |          |          | 1763+           | DROP  | R5                                      |                                     |
| 00002294 | 00000000 00000000 |          |          | 1764            | DC    | XL16'0000000000000000 0000FFFF00000000' | result                              |
| 0000229C | 0000FFFF 00000000 |          |          |                 |       |                                         |                                     |
| 000022A4 | 04000000 00000000 |          |          | 1765            | DC    | XL16'0400000000000000 0000000000000000' | v2                                  |
| 000022AC | 00000000 00000000 |          |          |                 |       |                                         |                                     |
| 000022B8 |                   |          |          | 1766            |       |                                         |                                     |
| 000022B8 |                   | 000022B8 |          | 1767            | VRR_A | VGEM,1                                  |                                     |
| 000022B8 | 000022F8          |          |          | 1768+           | DS    | 0FD                                     |                                     |
| 000022BC | 0025              |          |          | 1769+           | USING | *,R5                                    | base for test data and test routine |
| 000022BE | 00                |          |          | 1770+T37        | DC    | A(X37)                                  | address of test routine             |
| 000022BF | 01                |          |          | 1771+           | DC    | H'37'                                   | test number                         |
| 000022C0 | E5C7C5D4 40404040 |          |          | 1772+           | DC    | X'00'                                   |                                     |
| 000022C8 | 00002324          |          |          | 1773+           | DC    | HL1'1'                                  | M3                                  |
| 000022CC | 00000010          |          |          | 1774+           | DC    | CL8'VGEM'                               | instruction name                    |
| 000022D0 | 00002314          |          |          | 1775+           | DC    | A(RE37+16)                              | address of v2 source                |
| 000022D8 | 00000000 00000000 |          |          | 1776+           | DC    | A(16)                                   | result length                       |
| 000022E0 | 00000000 00000000 |          |          | 1777+REA37      | DC    | A(RE37)                                 | result address                      |
| 000022E8 | 00000000 00000000 |          |          | 1778+           | DS    | FD                                      | gap                                 |
| 000022F0 | 00000000 00000000 |          |          | 1779+V1037      | DS    | XL16                                    | V1 output                           |
| 000022F8 |                   |          |          | 1780+           | DS    | FD                                      | gap                                 |
| 000022F8 | E310 5010 0014    |          | 00000010 | 1781+*          |       |                                         |                                     |
| 000022FE | E721 0000 0006    |          | 00000000 | 1782+X37        | DS    | 0F                                      |                                     |
|          |                   |          |          | 1783+           | LGF   | R1,V2ADDR                               | load v2 source                      |
|          |                   |          |          | 1784+           | VL    | v2,0(R1)                                |                                     |
| 00002304 |                   |          |          | 1787+           | DS    | 0H                                      | E754 VGEM                           |
| 00002304 | E7                |          |          | 1788+           | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00002305 | 12                |          |          | 1789+           | DC    | X'12'                                   | v1, v2                              |
| 00002306 | 0000              |          |          | 1790+           | DC    | x'0000'                                 | reserved                            |
| 00002308 | 10                |          |          | 1791+           | DC    | HL1'16'                                 | M3, RXB                             |
| 00002309 | 54                |          |          | 1792+           | DC    | X'54'                                   |                                     |
| 0000230A | E710 5028 000E    |          | 000022E0 | 1794+           | VST   | V1,V1037                                | save v1 output                      |
| 00002310 | 07FB              |          |          | 1795+           | BR    | R11                                     | return                              |
| 00002314 |                   |          |          | 1796+RE37       | DC    | 0F                                      | xl16 expected result                |
| 00002314 |                   |          |          | 1797+           | DROP  | R5                                      |                                     |
| 00002314 | 00000000 00000000 |          |          | 1798            | DC    | XL16'0000000000000000 0000FFFF0000FFFF' | result                              |



| LOC      | OBJECT CODE       | ADDR1    | ADDR2    | STMT       |       |                                         |                                     |
|----------|-------------------|----------|----------|------------|-------|-----------------------------------------|-------------------------------------|
| 0000231C | 0000FFFF 0000FFFF |          |          |            |       |                                         |                                     |
| 00002324 | 05000000 00000000 |          |          | 1799       | DC    | XL16'0500000000000000 0000000000000000' | v2                                  |
| 0000232C | 00000000 00000000 |          |          |            |       |                                         |                                     |
|          |                   |          |          | 1800       |       |                                         |                                     |
|          |                   |          |          | 1801       | VRR_A | VGEM,1                                  |                                     |
| 00002338 |                   |          |          | 1802+      | DS    | 0FD                                     |                                     |
| 00002338 |                   | 00002338 |          | 1803+      | USING | *,R5                                    | base for test data and test routine |
| 00002338 | 00002378          |          |          | 1804+T38   | DC    | A(X38)                                  | address of test routine             |
| 0000233C | 0026              |          |          | 1805+      | DC    | H'38'                                   | test number                         |
| 0000233E | 00                |          |          | 1806+      | DC    | X'00'                                   |                                     |
| 0000233F | 01                |          |          | 1807+      | DC    | HL1'1'                                  | M3                                  |
| 00002340 | E5C7C5D4 40404040 |          |          | 1808+      | DC    | CL8'VGEM'                               | instruction name                    |
| 00002348 | 000023A4          |          |          | 1809+      | DC    | A(RE38+16)                              | address of v2 source                |
| 0000234C | 00000010          |          |          | 1810+      | DC    | A(16)                                   | result length                       |
| 00002350 | 00002394          |          |          | 1811+REA38 | DC    | A(RE38)                                 | result address                      |
| 00002358 | 00000000 00000000 |          |          | 1812+      | DS    | FD                                      | gap                                 |
| 00002360 | 00000000 00000000 |          |          | 1813+V1038 | DS    | XL16                                    | V1 output                           |
| 00002368 | 00000000 00000000 |          |          |            |       |                                         |                                     |
| 00002370 | 00000000 00000000 |          |          | 1814+      | DS    | FD                                      | gap                                 |
|          |                   |          |          | 1815+*     |       |                                         |                                     |
| 00002378 |                   |          |          | 1816+X38   | DS    | 0F                                      |                                     |
| 00002378 | E310 5010 0014    |          | 00000010 | 1817+      | LGF   | R1,V2ADDR                               | load v2 source                      |
| 0000237E | E721 0000 0006    |          | 00000000 | 1818+      | VL    | v2,0(R1)                                |                                     |
|          |                   |          |          |            |       |                                         |                                     |
| 00002384 |                   |          |          | 1821+      | DS    | 0H                                      | E754 VGEM                           |
| 00002384 | E7                |          |          | 1822+      | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00002385 | 12                |          |          | 1823+      | DC    | X'12'                                   | v1, v2                              |
| 00002386 | 0000              |          |          | 1824+      | DC    | x'0000'                                 | reserved                            |
| 00002388 | 10                |          |          | 1825+      | DC    | HL1'16'                                 | M3, RXB                             |
| 00002389 | 54                |          |          | 1826+      | DC    | X'54'                                   |                                     |
|          |                   |          |          |            |       |                                         |                                     |
| 0000238A | E710 5028 000E    |          | 00002360 | 1828+      | VST   | V1,V1038                                | save v1 output                      |
| 00002390 | 07FB              |          |          | 1829+      | BR    | R11                                     | return                              |
| 00002394 |                   |          |          | 1830+RE38  | DC    | 0F                                      | xl16 expected result                |
| 00002394 |                   |          |          | 1831+      | DROP  | R5                                      |                                     |
| 00002394 | 00000000 00000000 |          |          | 1832       | DC    | XL16'0000000000000000 0000FFFFFFFF0000' | result                              |
| 0000239C | 0000FFFF FFFF0000 |          |          |            |       |                                         |                                     |
| 000023A4 | 06000000 00000000 |          |          | 1833       | DC    | XL16'0600000000000000 0000000000000000' | v2                                  |
| 000023AC | 00000000 00000000 |          |          |            |       |                                         |                                     |
|          |                   |          |          | 1834       |       |                                         |                                     |
|          |                   |          |          | 1835       | VRR_A | VGEM,1                                  |                                     |
| 000023B8 |                   |          |          | 1836+      | DS    | 0FD                                     |                                     |
| 000023B8 |                   | 000023B8 |          | 1837+      | USING | *,R5                                    | base for test data and test routine |
| 000023B8 | 000023F8          |          |          | 1838+T39   | DC    | A(X39)                                  | address of test routine             |
| 000023BC | 0027              |          |          | 1839+      | DC    | H'39'                                   | test number                         |
| 000023BE | 00                |          |          | 1840+      | DC    | X'00'                                   |                                     |
| 000023BF | 01                |          |          | 1841+      | DC    | HL1'1'                                  | M3                                  |
| 000023C0 | E5C7C5D4 40404040 |          |          | 1842+      | DC    | CL8'VGEM'                               | instruction name                    |
| 000023C8 | 00002424          |          |          | 1843+      | DC    | A(RE39+16)                              | address of v2 source                |
| 000023CC | 00000010          |          |          | 1844+      | DC    | A(16)                                   | result length                       |
| 000023D0 | 00002414          |          |          | 1845+REA39 | DC    | A(RE39)                                 | result address                      |
| 000023D8 | 00000000 00000000 |          |          | 1846+      | DS    | FD                                      | gap                                 |
| 000023E0 | 00000000 00000000 |          |          | 1847+V1039 | DS    | XL16                                    | V1 output                           |
| 000023E8 | 00000000 00000000 |          |          |            |       |                                         |                                     |
| 000023F0 | 00000000 00000000 |          |          | 1848+      | DS    | FD                                      | gap                                 |
|          |                   |          |          | 1849+*     |       |                                         |                                     |

| LOC      | OBJECT CODE       | ADDR1    | ADDR2    | STMT       |       |                                         |                                     |
|----------|-------------------|----------|----------|------------|-------|-----------------------------------------|-------------------------------------|
| 000023F8 |                   |          |          | 1850+X39   | DS    | 0F                                      |                                     |
| 000023F8 | E310 5010 0014    |          | 00000010 | 1851+      | LGF   | R1,V2ADDR                               | load v2 source                      |
| 000023FE | E721 0000 0006    |          | 00000000 | 1852+      | VL    | v2,0(R1)                                |                                     |
| 00002404 |                   |          |          | 1855+      | DS    | 0H                                      | E754 VGEM                           |
| 00002404 | E7                |          |          | 1856+      | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00002405 | 12                |          |          | 1857+      | DC    | X'12'                                   | v1, v2                              |
| 00002406 | 0000              |          |          | 1858+      | DC    | x'0000'                                 | reserved                            |
| 00002408 | 10                |          |          | 1859+      | DC    | HL1'16'                                 | M3, RXB                             |
| 00002409 | 54                |          |          | 1860+      | DC    | X'54'                                   |                                     |
| 0000240A | E710 5028 000E    |          | 000023E0 | 1862+      | VST   | V1,V1039                                | save v1 output                      |
| 00002410 | 07FB              |          |          | 1863+      | BR    | R11                                     | return                              |
| 00002414 |                   |          |          | 1864+RE39  | DC    | 0F                                      | xl16 expected result                |
| 00002414 |                   |          |          | 1865+      | DROP  | R5                                      |                                     |
| 00002414 | 00000000 00000000 |          |          | 1866       | DC    | XL16'0000000000000000 0000FFFFFFFFFFFF' | result                              |
| 0000241C | 0000FFFF FFFFFFFF |          |          |            |       |                                         |                                     |
| 00002424 | 07000000 00000000 |          |          | 1867       | DC    | XL16'0700000000000000 0000000000000000' | v2                                  |
| 0000242C | 00000000 00000000 |          |          |            |       |                                         |                                     |
|          |                   |          |          | 1868       |       |                                         |                                     |
|          |                   |          |          | 1869       | VRR_A | VGEM,1                                  |                                     |
| 00002438 |                   |          |          | 1870+      | DS    | 0FD                                     |                                     |
| 00002438 |                   | 00002438 |          | 1871+      | USING | *,R5                                    | base for test data and test routine |
| 00002438 | 00002478          |          |          | 1872+T40   | DC    | A(X40)                                  | address of test routine             |
| 0000243C | 0028              |          |          | 1873+      | DC    | H'40'                                   | test number                         |
| 0000243E | 00                |          |          | 1874+      | DC    | X'00'                                   |                                     |
| 0000243F | 01                |          |          | 1875+      | DC    | HL1'1'                                  | M3                                  |
| 00002440 | E5C7C5D4 40404040 |          |          | 1876+      | DC    | CL8'VGEM'                               | instruction name                    |
| 00002448 | 000024A4          |          |          | 1877+      | DC    | A(RE40+16)                              | address of v2 source                |
| 0000244C | 00000010          |          |          | 1878+      | DC    | A(16)                                   | result length                       |
| 00002450 | 00002494          |          |          | 1879+REA40 | DC    | A(RE40)                                 | result address                      |
| 00002458 | 00000000 00000000 |          |          | 1880+      | DS    | FD                                      | gap                                 |
| 00002460 | 00000000 00000000 |          |          | 1881+V1040 | DS    | XL16                                    | V1 output                           |
| 00002468 | 00000000 00000000 |          |          |            |       |                                         |                                     |
| 00002470 | 00000000 00000000 |          |          |            |       |                                         |                                     |
|          |                   |          |          | 1882+      | DS    | FD                                      | gap                                 |
|          |                   |          |          | 1883+*     |       |                                         |                                     |
| 00002478 |                   |          |          | 1884+X40   | DS    | 0F                                      |                                     |
| 00002478 | E310 5010 0014    |          | 00000010 | 1885+      | LGF   | R1,V2ADDR                               | load v2 source                      |
| 0000247E | E721 0000 0006    |          | 00000000 | 1886+      | VL    | v2,0(R1)                                |                                     |
| 00002484 |                   |          |          | 1889+      | DS    | 0H                                      | E754 VGEM                           |
| 00002484 | E7                |          |          | 1890+      | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00002485 | 12                |          |          | 1891+      | DC    | X'12'                                   | v1, v2                              |
| 00002486 | 0000              |          |          | 1892+      | DC    | x'0000'                                 | reserved                            |
| 00002488 | 10                |          |          | 1893+      | DC    | HL1'16'                                 | M3, RXB                             |
| 00002489 | 54                |          |          | 1894+      | DC    | X'54'                                   |                                     |
| 0000248A | E710 5028 000E    |          | 00002460 | 1896+      | VST   | V1,V1040                                | save v1 output                      |
| 00002490 | 07FB              |          |          | 1897+      | BR    | R11                                     | return                              |
| 00002494 |                   |          |          | 1898+RE40  | DC    | 0F                                      | xl16 expected result                |
| 00002494 |                   |          |          | 1899+      | DROP  | R5                                      |                                     |
| 00002494 | 00000000 00000000 |          |          | 1900       | DC    | XL16'0000000000000000 FFFF000000000000' | result                              |
| 0000249C | FFFF0000 00000000 |          |          |            |       |                                         |                                     |
| 000024A4 | 08000000 00000000 |          |          | 1901       | DC    | XL16'0800000000000000 0000000000000000' | v2                                  |
| 000024AC | 00000000 00000000 |          |          |            |       |                                         |                                     |
|          |                   |          |          | 1902       |       |                                         |                                     |

| LOC      | OBJECT CODE       | ADDR1    | ADDR2    | STMT       |       |                                         |                                     |
|----------|-------------------|----------|----------|------------|-------|-----------------------------------------|-------------------------------------|
| 000024B8 |                   |          |          | 1903       | VRR_A | VGEM,1                                  |                                     |
| 000024B8 |                   |          |          | 1904+      | DS    | 0FD                                     |                                     |
| 000024B8 |                   | 000024B8 |          | 1905+      | USING | *,R5                                    | base for test data and test routine |
| 000024B8 | 000024F8          |          |          | 1906+T41   | DC    | A(X41)                                  | address of test routine             |
| 000024BC | 0029              |          |          | 1907+      | DC    | H'41'                                   | test number                         |
| 000024BE | 00                |          |          | 1908+      | DC    | X'00'                                   |                                     |
| 000024BF | 01                |          |          | 1909+      | DC    | HL1'1'                                  | M3                                  |
| 000024C0 | E5C7C5D4 40404040 |          |          | 1910+      | DC    | CL8'VGEM'                               | instruction name                    |
| 000024C8 | 00002524          |          |          | 1911+      | DC    | A(RE41+16)                              | address of v2 source                |
| 000024CC | 00000010          |          |          | 1912+      | DC    | A(16)                                   | result length                       |
| 000024D0 | 00002514          |          |          | 1913+REA41 | DC    | A(RE41)                                 | result address                      |
| 000024D8 | 00000000 00000000 |          |          | 1914+      | DS    | FD                                      | gap                                 |
| 000024E0 | 00000000 00000000 |          |          | 1915+V1041 | DS    | XL16                                    | V1 output                           |
| 000024E8 | 00000000 00000000 |          |          |            |       |                                         |                                     |
| 000024F0 | 00000000 00000000 |          |          | 1916+      | DS    | FD                                      | gap                                 |
|          |                   |          |          | 1917+*     |       |                                         |                                     |
| 000024F8 |                   |          |          | 1918+X41   | DS    | 0F                                      |                                     |
| 000024F8 | E310 5010 0014    |          | 00000010 | 1919+      | LGF   | R1,V2ADDR                               | load v2 source                      |
| 000024FE | E721 0000 0006    |          | 00000000 | 1920+      | VL    | v2,0(R1)                                |                                     |
| 00002504 |                   |          |          | 1923+      | DS    | 0H                                      | E754 VGEM                           |
| 00002504 | E7                |          |          | 1924+      | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00002505 | 12                |          |          | 1925+      | DC    | X'12'                                   | v1, v2                              |
| 00002506 | 0000              |          |          | 1926+      | DC    | x'0000'                                 | reserved                            |
| 00002508 | 10                |          |          | 1927+      | DC    | HL1'16'                                 | M3, RXB                             |
| 00002509 | 54                |          |          | 1928+      | DC    | X'54'                                   |                                     |
| 0000250A | E710 5028 000E    |          | 000024E0 | 1930+      | VST   | V1,V1041                                | save v1 output                      |
| 00002510 | 07FB              |          |          | 1931+      | BR    | R11                                     | return                              |
| 00002514 |                   |          |          | 1932+RE41  | DC    | 0F                                      | xl16 expected result                |
| 00002514 |                   |          |          | 1933+      | DROP  | R5                                      |                                     |
| 00002514 | 00000000 0000FFFF |          |          | 1934       | DC    | XL16'000000000000FFFF 0000000000000000' | result                              |
| 0000251C | 00000000 00000000 |          |          |            |       |                                         |                                     |
| 00002524 | 10000000 00000000 |          |          | 1935       | DC    | XL16'1000000000000000 0000000000000000' | v2                                  |
| 0000252C | 00000000 00000000 |          |          |            |       |                                         |                                     |
| 00002538 |                   |          |          | 1936       | VRR_A | VGEM,1                                  |                                     |
| 00002538 |                   |          |          | 1937       | DS    | 0FD                                     |                                     |
| 00002538 |                   | 00002538 |          | 1938+      | USING | *,R5                                    | base for test data and test routine |
| 00002538 | 00002578          |          |          | 1939+      | DC    | A(X42)                                  | address of test routine             |
| 0000253C | 002A              |          |          | 1940+T42   | DC    | H'42'                                   | test number                         |
| 0000253E | 00                |          |          | 1941+      | DC    | X'00'                                   |                                     |
| 0000253F | 01                |          |          | 1942+      | DC    | HL1'1'                                  | M3                                  |
| 00002540 | E5C7C5D4 40404040 |          |          | 1943+      | DC    | CL8'VGEM'                               | instruction name                    |
| 00002548 | 000025A4          |          |          | 1944+      | DC    | A(RE42+16)                              | address of v2 source                |
| 0000254C | 00000010          |          |          | 1945+      | DC    | A(16)                                   | result length                       |
| 00002550 | 00002594          |          |          | 1946+      | DC    | A(RE42)                                 | result address                      |
| 00002558 | 00000000 00000000 |          |          | 1947+REA42 | DC    | A(RE42)                                 |                                     |
| 00002558 | 00000000 00000000 |          |          | 1948+      | DS    | FD                                      | gap                                 |
| 00002560 | 00000000 00000000 |          |          | 1949+V1042 | DS    | XL16                                    | V1 output                           |
| 00002568 | 00000000 00000000 |          |          |            |       |                                         |                                     |
| 00002570 | 00000000 00000000 |          |          | 1950+      | DS    | FD                                      | gap                                 |
|          |                   |          |          | 1951+*     |       |                                         |                                     |
| 00002578 |                   |          |          | 1952+X42   | DS    | 0F                                      |                                     |
| 00002578 | E310 5010 0014    |          | 00000010 | 1953+      | LGF   | R1,V2ADDR                               | load v2 source                      |
| 0000257E | E721 0000 0006    |          | 00000000 | 1954+      | VL    | v2,0(R1)                                |                                     |

| LOC      | OBJECT CODE       | ADDR1    | ADDR2    | STMT       |       |                                         |                                     |
|----------|-------------------|----------|----------|------------|-------|-----------------------------------------|-------------------------------------|
| 00002584 |                   |          |          | 1957+      | DS    | 0H                                      | E754 VGEM                           |
| 00002584 | E7                |          |          | 1958+      | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00002585 | 12                |          |          | 1959+      | DC    | X'12'                                   | v1, v2                              |
| 00002586 | 0000              |          |          | 1960+      | DC    | x'0000'                                 | reserved                            |
| 00002588 | 10                |          |          | 1961+      | DC    | HL1'16'                                 | M3, RXB                             |
| 00002589 | 54                |          |          | 1962+      | DC    | X'54'                                   |                                     |
| 0000258A | E710 5028 000E    |          | 00002560 | 1964+      | VST   | V1,V1042                                | save v1 output                      |
| 00002590 | 07FB              |          |          | 1965+      | BR    | R11                                     | return                              |
| 00002594 |                   |          |          | 1966+RE42  | DC    | 0F                                      | xl16 expected result                |
| 00002594 |                   |          |          | 1967+      | DROP  | R5                                      |                                     |
| 00002594 | 00000000 FFFF0000 |          |          | 1968       | DC    | XL16'00000000FFFF0000 0000000000000000' | result                              |
| 0000259C | 00000000 00000000 |          |          |            |       |                                         |                                     |
| 000025A4 | 20000000 00000000 |          |          | 1969       | DC    | XL16'2000000000000000 0000000000000000' | v2                                  |
| 000025AC | 00000000 00000000 |          |          |            |       |                                         |                                     |
|          |                   |          |          | 1970       |       |                                         |                                     |
|          |                   |          |          | 1971       | VRR_A | VGEM,1                                  |                                     |
| 000025B8 |                   |          |          | 1972+      | DS    | 0FD                                     |                                     |
| 000025B8 |                   | 000025B8 |          | 1973+      | USING | *,R5                                    | base for test data and test routine |
| 000025B8 | 000025F8          |          |          | 1974+T43   | DC    | A(X43)                                  | address of test routine             |
| 000025BC | 002B              |          |          | 1975+      | DC    | H'43'                                   | test number                         |
| 000025BE | 00                |          |          | 1976+      | DC    | X'00'                                   |                                     |
| 000025BF | 01                |          |          | 1977+      | DC    | HL1'1'                                  | M3                                  |
| 000025C0 | E5C7C5D4 40404040 |          |          | 1978+      | DC    | CL8'VGEM'                               | instruction name                    |
| 000025C8 | 00002624          |          |          | 1979+      | DC    | A(RE43+16)                              | address of v2 source                |
| 000025CC | 00000010          |          |          | 1980+      | DC    | A(16)                                   | result length                       |
| 000025D0 | 00002614          |          |          | 1981+REA43 | DC    | A(RE43)                                 | result address                      |
| 000025D8 | 00000000 00000000 |          |          | 1982+      | DS    | FD                                      | gap                                 |
| 000025E0 | 00000000 00000000 |          |          | 1983+V1043 | DS    | XL16                                    | V1 output                           |
| 000025E8 | 00000000 00000000 |          |          |            |       |                                         |                                     |
| 000025F0 | 00000000 00000000 |          |          | 1984+      | DS    | FD                                      | gap                                 |
|          |                   |          |          | 1985+*     |       |                                         |                                     |
| 000025F8 |                   |          |          | 1986+X43   | DS    | 0F                                      |                                     |
| 000025F8 | E310 5010 0014    |          | 00000010 | 1987+      | LGF   | R1,V2ADDR                               | load v2 source                      |
| 000025FE | E721 0000 0006    |          | 00000000 | 1988+      | VL    | v2,0(R1)                                |                                     |
| 00002604 |                   |          |          | 1991+      | DS    | 0H                                      | E754 VGEM                           |
| 00002604 | E7                |          |          | 1992+      | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00002605 | 12                |          |          | 1993+      | DC    | X'12'                                   | v1, v2                              |
| 00002606 | 0000              |          |          | 1994+      | DC    | x'0000'                                 | reserved                            |
| 00002608 | 10                |          |          | 1995+      | DC    | HL1'16'                                 | M3, RXB                             |
| 00002609 | 54                |          |          | 1996+      | DC    | X'54'                                   |                                     |
| 0000260A | E710 5028 000E    |          | 000025E0 | 1998+      | VST   | V1,V1043                                | save v1 output                      |
| 00002610 | 07FB              |          |          | 1999+      | BR    | R11                                     | return                              |
| 00002614 |                   |          |          | 2000+RE43  | DC    | 0F                                      | xl16 expected result                |
| 00002614 |                   |          |          | 2001+      | DROP  | R5                                      |                                     |
| 00002614 | 0000FFFF 00000000 |          |          | 2002       | DC    | XL16'0000FFFF00000000 0000000000000000' | result                              |
| 0000261C | 00000000 00000000 |          |          |            |       |                                         |                                     |
| 00002624 | 40000000 00000000 |          |          | 2003       | DC    | XL16'4000000000000000 0000000000000000' | v2                                  |
| 0000262C | 00000000 00000000 |          |          |            |       |                                         |                                     |
|          |                   |          |          | 2004       |       |                                         |                                     |
|          |                   |          |          | 2005       | VRR_A | VGEM,1                                  |                                     |
| 00002638 |                   |          |          | 2006+      | DS    | 0FD                                     |                                     |
| 00002638 |                   | 00002638 |          | 2007+      | USING | *,R5                                    | base for test data and test routine |
| 00002638 | 00002678          |          |          | 2008+T44   | DC    | A(X44)                                  | address of test routine             |

| LOC      | OBJECT CODE       | ADDR1    | ADDR2    | STMT       |       |                                         |                                     |
|----------|-------------------|----------|----------|------------|-------|-----------------------------------------|-------------------------------------|
| 0000263C | 002C              |          |          | 2009+      | DC    | H'44'                                   | test number                         |
| 0000263E | 00                |          |          | 2010+      | DC    | X'00'                                   |                                     |
| 0000263F | 01                |          |          | 2011+      | DC    | HL1'1'                                  | M3                                  |
| 00002640 | E5C7C5D4 40404040 |          |          | 2012+      | DC    | CL8'VGEM'                               | instruction name                    |
| 00002648 | 000026A4          |          |          | 2013+      | DC    | A(RE44+16)                              | address of v2 source                |
| 0000264C | 00000010          |          |          | 2014+      | DC    | A(16)                                   | result length                       |
| 00002650 | 00002694          |          |          | 2015+REA44 | DC    | A(RE44)                                 | result address                      |
| 00002658 | 00000000 00000000 |          |          | 2016+      | DS    | FD                                      | gap                                 |
| 00002660 | 00000000 00000000 |          |          | 2017+V1044 | DS    | XL16                                    | V1 output                           |
| 00002668 | 00000000 00000000 |          |          |            |       |                                         |                                     |
| 00002670 | 00000000 00000000 |          |          | 2018+      | DS    | FD                                      | gap                                 |
|          |                   |          |          | 2019+*     |       |                                         |                                     |
| 00002678 |                   |          |          | 2020+X44   | DS    | 0F                                      |                                     |
| 00002678 | E310 5010 0014    |          | 00000010 | 2021+      | LGF   | R1,V2ADDR                               | load v2 source                      |
| 0000267E | E721 0000 0006    |          | 00000000 | 2022+      | VL    | v2,0(R1)                                |                                     |
|          |                   |          |          |            |       |                                         |                                     |
| 00002684 |                   |          |          | 2025+      | DS    | 0H                                      | E754 VGEM                           |
| 00002684 | E7                |          |          | 2026+      | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00002685 | 12                |          |          | 2027+      | DC    | X'12'                                   | v1, v2                              |
| 00002686 | 0000              |          |          | 2028+      | DC    | x'0000'                                 | reserved                            |
| 00002688 | 10                |          |          | 2029+      | DC    | HL1'16'                                 | M3, RXB                             |
| 00002689 | 54                |          |          | 2030+      | DC    | X'54'                                   |                                     |
|          |                   |          |          |            |       |                                         |                                     |
| 0000268A | E710 5028 000E    |          | 00002660 | 2032+      | VST   | V1,V1044                                | save v1 output                      |
| 00002690 | 07FB              |          |          | 2033+      | BR    | R11                                     | return                              |
| 00002694 |                   |          |          | 2034+RE44  | DC    | 0F                                      | xl16 expected result                |
| 00002694 |                   |          |          | 2035+      | DROP  | R5                                      |                                     |
| 00002694 | FFFF0000 00000000 |          |          | 2036       | DC    | XL16'FFFF000000000000 0000000000000000' | result                              |
| 0000269C | 00000000 00000000 |          |          |            |       |                                         |                                     |
| 000026A4 | 80000000 00000000 |          |          | 2037       | DC    | XL16'8000000000000000 0000000000000000' | v2                                  |
| 000026AC | 00000000 00000000 |          |          |            |       |                                         |                                     |
|          |                   |          |          | 2038       |       |                                         |                                     |
|          |                   |          |          | 2039       | VRR_A | VGEM,1                                  |                                     |
| 000026B8 |                   |          |          | 2040+      | DS    | 0FD                                     |                                     |
| 000026B8 |                   | 000026B8 |          | 2041+      | USING | *,R5                                    | base for test data and test routine |
| 000026B8 | 000026F8          |          |          | 2042+T45   | DC    | A(X45)                                  | address of test routine             |
| 000026BC | 002D              |          |          | 2043+      | DC    | H'45'                                   | test number                         |
| 000026BE | 00                |          |          | 2044+      | DC    | X'00'                                   |                                     |
| 000026BF | 01                |          |          | 2045+      | DC    | HL1'1'                                  | M3                                  |
| 000026C0 | E5C7C5D4 40404040 |          |          | 2046+      | DC    | CL8'VGEM'                               | instruction name                    |
| 000026C8 | 00002724          |          |          | 2047+      | DC    | A(RE45+16)                              | address of v2 source                |
| 000026CC | 00000010          |          |          | 2048+      | DC    | A(16)                                   | result length                       |
| 000026D0 | 00002714          |          |          | 2049+REA45 | DC    | A(RE45)                                 | result address                      |
| 000026D8 | 00000000 00000000 |          |          | 2050+      | DS    | FD                                      | gap                                 |
| 000026E0 | 00000000 00000000 |          |          | 2051+V1045 | DS    | XL16                                    | V1 output                           |
| 000026E8 | 00000000 00000000 |          |          |            |       |                                         |                                     |
| 000026F0 | 00000000 00000000 |          |          | 2052+      | DS    | FD                                      | gap                                 |
|          |                   |          |          | 2053+*     |       |                                         |                                     |
| 000026F8 |                   |          |          | 2054+X45   | DS    | 0F                                      |                                     |
| 000026F8 | E310 5010 0014    |          | 00000010 | 2055+      | LGF   | R1,V2ADDR                               | load v2 source                      |
| 000026FE | E721 0000 0006    |          | 00000000 | 2056+      | VL    | v2,0(R1)                                |                                     |
|          |                   |          |          |            |       |                                         |                                     |
| 00002704 |                   |          |          | 2059+      | DS    | 0H                                      | E754 VGEM                           |
| 00002704 | E7                |          |          | 2060+      | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00002705 | 12                |          |          | 2061+      | DC    | X'12'                                   | v1, v2                              |
| 00002706 | 0000              |          |          | 2062+      | DC    | x'0000'                                 | reserved                            |



| LOC      | OBJECT CODE       | ADDR1    | ADDR2    | STMT        |       |                                           |                                     |
|----------|-------------------|----------|----------|-------------|-------|-------------------------------------------|-------------------------------------|
| 00002708 | 10                |          |          | 2063+       | DC    | HL1'16'                                   | M3, RXB                             |
| 00002709 | 54                |          |          | 2064+       | DC    | X'54'                                     |                                     |
| 0000270A | E710 5028 000E    |          | 000026E0 | 2066+       | VST   | V1,V1045                                  | save v1 output                      |
| 00002710 | 07FB              |          |          | 2067+       | BR    | R11                                       | return                              |
| 00002714 |                   |          |          | 2068+RE45   | DC    | 0F                                        | xl16 expected result                |
| 00002714 |                   |          |          | 2069+       | DROP  | R5                                        |                                     |
| 00002714 | FFFFFFFF FFFFFFFF |          |          | 2070        | DC    | XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFF' | result                              |
| 0000271C | FFFFFFFF FFFFFFFF |          |          |             |       |                                           |                                     |
| 00002724 | FF000000 00000000 |          |          | 2071        | DC    | XL16'FF00000000000000 0000000000000000'   | v2                                  |
| 0000272C | 00000000 00000000 |          |          |             |       |                                           |                                     |
|          |                   |          |          | 2072        |       |                                           |                                     |
|          |                   |          |          | 2073 * Word |       | M3 = 2                                    |                                     |
|          |                   |          |          | 2074        | VRR_A | VGEM, 2                                   |                                     |
| 00002738 |                   |          |          | 2075+       | DS    | 0FD                                       |                                     |
| 00002738 |                   | 00002738 |          | 2076+       | USING | *, R5                                     | base for test data and test routine |
| 00002738 | 00002778          |          |          | 2077+T46    | DC    | A(X46)                                    | address of test routine             |
| 0000273C | 002E              |          |          | 2078+       | DC    | H'46'                                     | test number                         |
| 0000273E | 00                |          |          | 2079+       | DC    | X'00'                                     |                                     |
| 0000273F | 02                |          |          | 2080+       | DC    | HL1'2'                                    | M3                                  |
| 00002740 | E5C7C5D4 40404040 |          |          | 2081+       | DC    | CL8'VGEM'                                 | instruction name                    |
| 00002748 | 000027A4          |          |          | 2082+       | DC    | A(RE46+16)                                | address of v2 source                |
| 0000274C | 00000010          |          |          | 2083+       | DC    | A(16)                                     | result length                       |
| 00002750 | 00002794          |          |          | 2084+REA46  | DC    | A(RE46)                                   | result address                      |
| 00002758 | 00000000 00000000 |          |          | 2085+       | DS    | FD                                        | gap                                 |
| 00002760 | 00000000 00000000 |          |          | 2086+V1046  | DS    | XL16                                      | V1 output                           |
| 00002768 | 00000000 00000000 |          |          |             |       |                                           |                                     |
| 00002770 | 00000000 00000000 |          |          | 2087+       | DS    | FD                                        | gap                                 |
|          |                   |          |          | 2088+*      |       |                                           |                                     |
| 00002778 |                   |          |          | 2089+X46    | DS    | 0F                                        |                                     |
| 00002778 | E310 5010 0014    |          | 00000010 | 2090+       | LGF   | R1,V2ADDR                                 | load v2 source                      |
| 0000277E | E721 0000 0006    |          | 00000000 | 2091+       | VL    | v2, 0(R1)                                 |                                     |
|          |                   |          |          |             |       |                                           |                                     |
| 00002784 |                   |          |          | 2094+       | DS    | 0H                                        | E754 VGEM                           |
| 00002784 | E7                |          |          | 2095+       | DC    | X'E7'                                     | - Vector Generate Element Masks     |
| 00002785 | 12                |          |          | 2096+       | DC    | X'12'                                     | v1, v2                              |
| 00002786 | 0000              |          |          | 2097+       | DC    | x'0000'                                   | reserved                            |
| 00002788 | 20                |          |          | 2098+       | DC    | HL1'32'                                   | M3, RXB                             |
| 00002789 | 54                |          |          | 2099+       | DC    | X'54'                                     |                                     |
|          |                   |          |          |             |       |                                           |                                     |
| 0000278A | E710 5028 000E    |          | 00002760 | 2101+       | VST   | V1,V1046                                  | save v1 output                      |
| 00002790 | 07FB              |          |          | 2102+       | BR    | R11                                       | return                              |
| 00002794 |                   |          |          | 2103+RE46   | DC    | 0F                                        | xl16 expected result                |
| 00002794 |                   |          |          | 2104+       | DROP  | R5                                        |                                     |
| 00002794 | 00000000 00000000 |          |          | 2105        | DC    | XL16'0000000000000000 0000000000000000'   | result                              |
| 0000279C | 00000000 00000000 |          |          |             |       |                                           |                                     |
| 000027A4 | 00000000 00000000 |          |          | 2106        | DC    | XL16'0000000000000000 0000000000000000'   | v2                                  |
| 000027AC | 00000000 00000000 |          |          |             |       |                                           |                                     |
|          |                   |          |          | 2107        |       |                                           |                                     |
|          |                   |          |          | 2108        | VRR_A | VGEM, 2                                   |                                     |
| 000027B8 |                   |          |          | 2109+       | DS    | 0FD                                       |                                     |
| 000027B8 |                   | 000027B8 |          | 2110+       | USING | *, R5                                     | base for test data and test routine |
| 000027B8 | 000027F8          |          |          | 2111+T47    | DC    | A(X47)                                    | address of test routine             |
| 000027BC | 002F              |          |          | 2112+       | DC    | H'47'                                     | test number                         |
| 000027BE | 00                |          |          | 2113+       | DC    | X'00'                                     |                                     |
| 000027BF | 02                |          |          | 2114+       | DC    | HL1'2'                                    | M3                                  |



| LOC      | OBJECT CODE       | ADDR1    | ADDR2    | STMT       |       |                                         |                                     |
|----------|-------------------|----------|----------|------------|-------|-----------------------------------------|-------------------------------------|
| 000027C0 | E5C7C5D4 40404040 |          |          | 2115+      | DC    | CL8'VGEM'                               | instruction name                    |
| 000027C8 | 00002824          |          |          | 2116+      | DC    | A(RE47+16)                              | address of v2 source                |
| 000027CC | 00000010          |          |          | 2117+      | DC    | A(16)                                   | result length                       |
| 000027D0 | 00002814          |          |          | 2118+REA47 | DC    | A(RE47)                                 | result address                      |
| 000027D8 | 00000000 00000000 |          |          | 2119+      | DS    | FD                                      | gap                                 |
| 000027E0 | 00000000 00000000 |          |          | 2120+V1047 | DS    | XL16                                    | V1 output                           |
| 000027E8 | 00000000 00000000 |          |          |            |       |                                         |                                     |
| 000027F0 | 00000000 00000000 |          |          | 2121+      | DS    | FD                                      | gap                                 |
|          |                   |          |          | 2122+*     |       |                                         |                                     |
| 000027F8 |                   |          |          | 2123+X47   | DS    | 0F                                      |                                     |
| 000027F8 | E310 5010 0014    |          | 00000010 | 2124+      | LGF   | R1,V2ADDR                               | load v2 source                      |
| 000027FE | E721 0000 0006    |          | 00000000 | 2125+      | VL    | v2,0(R1)                                |                                     |
| 00002804 |                   |          |          | 2128+      | DS    | 0H                                      | E754 VGEM                           |
| 00002804 | E7                |          |          | 2129+      | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00002805 | 12                |          |          | 2130+      | DC    | X'12'                                   | v1, v2                              |
| 00002806 | 0000              |          |          | 2131+      | DC    | x'0000'                                 | reserved                            |
| 00002808 | 20                |          |          | 2132+      | DC    | HL1'32'                                 | M3, RXB                             |
| 00002809 | 54                |          |          | 2133+      | DC    | X'54'                                   |                                     |
| 0000280A | E710 5028 000E    |          | 000027E0 | 2135+      | VST   | V1,V1047                                | save v1 output                      |
| 00002810 | 07FB              |          |          | 2136+      | BR    | R11                                     | return                              |
| 00002814 |                   |          |          | 2137+RE47  | DC    | 0F                                      | xl16 expected result                |
| 00002814 |                   |          |          | 2138+      | DROP  | R5                                      |                                     |
| 00002814 | 00000000 00000000 |          |          | 2139       | DC    | XL16'0000000000000000 0000000000000000' | result                              |
| 0000281C | 00000000 00000000 |          |          |            |       |                                         |                                     |
| 00002824 | 0FFFFFFF FFFFFFFF |          |          | 2140       | DC    | XL16'0FFFFFFF0FFFFFFF FFFFFFFF0FFFFFFF' | v2                                  |
| 0000282C | FFFFFFF FFFFFFFF  |          |          |            |       |                                         |                                     |
|          |                   |          |          | 2141       |       |                                         |                                     |
| 00002838 |                   |          |          | 2142       | VRR_A | VGEM, 2                                 |                                     |
| 00002838 |                   | 00002838 |          | 2143+      | DS    | 0FD                                     |                                     |
| 00002838 | 00002878          |          |          | 2144+      | USING | *,R5                                    | base for test data and test routine |
| 0000283C | 0030              |          |          | 2145+T48   | DC    | A(X48)                                  | address of test routine             |
| 0000283E | 00                |          |          | 2146+      | DC    | H'48'                                   | test number                         |
| 0000283F | 02                |          |          | 2147+      | DC    | X'00'                                   |                                     |
| 00002840 | E5C7C5D4 40404040 |          |          | 2148+      | DC    | HL1'2'                                  | M3                                  |
| 00002848 | 000028A4          |          |          | 2149+      | DC    | CL8'VGEM'                               | instruction name                    |
| 0000284C | 00000010          |          |          | 2150+      | DC    | A(RE48+16)                              | address of v2 source                |
| 00002850 | 00002894          |          |          | 2151+      | DC    | A(16)                                   | result length                       |
| 00002858 | 00000000 00000000 |          |          | 2152+REA48 | DC    | A(RE48)                                 | result address                      |
| 00002860 | 00000000 00000000 |          |          | 2153+      | DS    | FD                                      | gap                                 |
| 00002868 | 00000000 00000000 |          |          | 2154+V1048 | DS    | XL16                                    | V1 output                           |
| 00002870 | 00000000 00000000 |          |          |            |       |                                         |                                     |
|          |                   |          |          | 2155+      | DS    | FD                                      | gap                                 |
|          |                   |          |          | 2156+*     |       |                                         |                                     |
| 00002878 |                   |          |          | 2157+X48   | DS    | 0F                                      |                                     |
| 00002878 | E310 5010 0014    |          | 00000010 | 2158+      | LGF   | R1,V2ADDR                               | load v2 source                      |
| 0000287E | E721 0000 0006    |          | 00000000 | 2159+      | VL    | v2,0(R1)                                |                                     |
| 00002884 |                   |          |          | 2162+      | DS    | 0H                                      | E754 VGEM                           |
| 00002884 | E7                |          |          | 2163+      | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00002885 | 12                |          |          | 2164+      | DC    | X'12'                                   | v1, v2                              |
| 00002886 | 0000              |          |          | 2165+      | DC    | x'0000'                                 | reserved                            |
| 00002888 | 20                |          |          | 2166+      | DC    | HL1'32'                                 | M3, RXB                             |
| 00002889 | 54                |          |          | 2167+      | DC    | X'54'                                   |                                     |

| LOC      | OBJECT CODE       | ADDR1    | ADDR2    | STMT       |       |                                         |                                     |
|----------|-------------------|----------|----------|------------|-------|-----------------------------------------|-------------------------------------|
| 0000288A | E710 5028 000E    |          | 00002860 | 2169+      | VST   | V1,V1048                                | save v1 output                      |
| 00002890 | 07FB              |          |          | 2170+      | BR    | R11                                     | return                              |
| 00002894 |                   |          |          | 2171+RE48  | DC    | 0F                                      | xl16 expected result                |
| 00002894 |                   |          |          | 2172+      | DROP  | R5                                      |                                     |
| 00002894 | 00000000 00000000 |          |          | 2173       | DC    | XL16'0000000000000000 00000000FFFFFFFF' | result                              |
| 0000289C | 00000000 FFFFFFFF |          |          |            |       |                                         |                                     |
| 000028A4 | 10000000 00000000 |          |          | 2174       | DC    | XL16'1000000000000000 0000000000000000' | v2                                  |
| 000028AC | 00000000 00000000 |          |          |            |       |                                         |                                     |
|          |                   |          |          | 2175       |       |                                         |                                     |
| 000028B8 |                   |          |          | 2176       | VRR_A | VGEM, 2                                 |                                     |
| 000028B8 |                   | 000028B8 |          | 2177+      | DS    | 0FD                                     |                                     |
| 000028B8 | 000028F8          |          |          | 2178+      | USING | *,R5                                    | base for test data and test routine |
| 000028BC | 0031              |          |          | 2179+T49   | DC    | A(X49)                                  | address of test routine             |
| 000028BE | 00                |          |          | 2180+      | DC    | H'49'                                   | test number                         |
| 000028BF | 02                |          |          | 2181+      | DC    | X'00'                                   |                                     |
| 000028C0 | E5C7C5D4 40404040 |          |          | 2182+      | DC    | HL1'2'                                  | M3                                  |
| 000028C8 | 00002924          |          |          | 2183+      | DC    | CL8'VGEM'                               | instruction name                    |
| 000028CC | 00000010          |          |          | 2184+      | DC    | A(RE49+16)                              | address of v2 source                |
| 000028D0 | 00002914          |          |          | 2185+      | DC    | A(16)                                   | result length                       |
| 000028D8 | 00000000 00000000 |          |          | 2186+REA49 | DC    | A(RE49)                                 | result address                      |
| 000028E0 | 00000000 00000000 |          |          | 2187+      | DS    | FD                                      | gap                                 |
| 000028E8 | 00000000 00000000 |          |          | 2188+V1049 | DS    | XL16                                    | V1 output                           |
| 000028F0 | 00000000 00000000 |          |          |            |       |                                         |                                     |
|          |                   |          |          | 2189+      | DS    | FD                                      | gap                                 |
|          |                   |          |          | 2190+*     |       |                                         |                                     |
| 000028F8 |                   |          |          | 2191+X49   | DS    | 0F                                      |                                     |
| 000028F8 | E310 5010 0014    |          | 00000010 | 2192+      | LGF   | R1,V2ADDR                               | load v2 source                      |
| 000028FE | E721 0000 0006    |          | 00000000 | 2193+      | VL    | v2,0(R1)                                |                                     |
|          |                   |          |          |            |       |                                         |                                     |
| 00002904 |                   |          |          | 2196+      | DS    | 0H                                      | E754 VGEM                           |
| 00002904 | E7                |          |          | 2197+      | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00002905 | 12                |          |          | 2198+      | DC    | X'12'                                   | v1, v2                              |
| 00002906 | 0000              |          |          | 2199+      | DC    | x'0000'                                 | reserved                            |
| 00002908 | 20                |          |          | 2200+      | DC    | HL1'32'                                 | M3, RXB                             |
| 00002909 | 54                |          |          | 2201+      | DC    | X'54'                                   |                                     |
|          |                   |          |          |            |       |                                         |                                     |
| 0000290A | E710 5028 000E    |          | 000028E0 | 2203+      | VST   | V1,V1049                                | save v1 output                      |
| 00002910 | 07FB              |          |          | 2204+      | BR    | R11                                     | return                              |
| 00002914 |                   |          |          | 2205+RE49  | DC    | 0F                                      | xl16 expected result                |
| 00002914 |                   |          |          | 2206+      | DROP  | R5                                      |                                     |
| 00002914 | 00000000 00000000 |          |          | 2207       | DC    | XL16'0000000000000000 FFFFFFFF00000000' | result                              |
| 0000291C | FFFFFFFF 00000000 |          |          |            |       |                                         |                                     |
| 00002924 | 20000000 00000000 |          |          | 2208       | DC    | XL16'2000000000000000 0000000000000000' | v2                                  |
| 0000292C | 00000000 00000000 |          |          |            |       |                                         |                                     |
|          |                   |          |          | 2209       |       |                                         |                                     |
| 00002938 |                   |          |          | 2210       | VRR_A | VGEM, 2                                 |                                     |
| 00002938 |                   | 00002938 |          | 2211+      | DS    | 0FD                                     |                                     |
| 00002938 | 00002978          |          |          | 2212+      | USING | *,R5                                    | base for test data and test routine |
| 0000293C | 0032              |          |          | 2213+T50   | DC    | A(X50)                                  | address of test routine             |
| 0000293E | 00                |          |          | 2214+      | DC    | H'50'                                   | test number                         |
| 0000293F | 02                |          |          | 2215+      | DC    | X'00'                                   |                                     |
| 00002940 | E5C7C5D4 40404040 |          |          | 2216+      | DC    | HL1'2'                                  | M3                                  |
| 00002948 | 000029A4          |          |          | 2217+      | DC    | CL8'VGEM'                               | instruction name                    |
| 0000294C | 00000010          |          |          | 2218+      | DC    | A(RE50+16)                              | address of v2 source                |
| 00002950 | 00002994          |          |          | 2219+      | DC    | A(16)                                   | result length                       |
|          |                   |          |          | 2220+REA50 | DC    | A(RE50)                                 | result address                      |

| LOC      | OBJECT CODE       | ADDR1    | ADDR2    | STMT       |       |                                           |                                     |
|----------|-------------------|----------|----------|------------|-------|-------------------------------------------|-------------------------------------|
| 00002958 | 00000000 00000000 |          |          | 2221+      | DS    | FD                                        | gap                                 |
| 00002960 | 00000000 00000000 |          |          | 2222+V1050 | DS    | XL16                                      | V1 output                           |
| 00002968 | 00000000 00000000 |          |          |            |       |                                           |                                     |
| 00002970 | 00000000 00000000 |          |          | 2223+      | DS    | FD                                        | gap                                 |
|          |                   |          |          | 2224+*     |       |                                           |                                     |
| 00002978 |                   |          |          | 2225+X50   | DS    | 0F                                        |                                     |
| 00002978 | E310 5010 0014    |          | 00000010 | 2226+      | LGF   | R1,V2ADDR                                 | load v2 source                      |
| 0000297E | E721 0000 0006    |          | 00000000 | 2227+      | VL    | v2,0(R1)                                  |                                     |
| 00002984 |                   |          |          | 2230+      | DS    | 0H                                        | E754 VGEM                           |
| 00002984 | E7                |          |          | 2231+      | DC    | X'E7'                                     | - Vector Generate Element Masks     |
| 00002985 | 12                |          |          | 2232+      | DC    | X'12'                                     | v1, v2                              |
| 00002986 | 0000              |          |          | 2233+      | DC    | x'0000'                                   | reserved                            |
| 00002988 | 20                |          |          | 2234+      | DC    | HL1'32'                                   | M3, RXB                             |
| 00002989 | 54                |          |          | 2235+      | DC    | X'54'                                     |                                     |
| 0000298A | E710 5028 000E    |          | 00002960 | 2237+      | VST   | V1,V1050                                  | save v1 output                      |
| 00002990 | 07FB              |          |          | 2238+      | BR    | R11                                       | return                              |
| 00002994 |                   |          |          | 2239+RE50  | DC    | 0F                                        | xl16 expected result                |
| 00002994 |                   |          |          | 2240+      | DROP  | R5                                        |                                     |
| 00002994 | 00000000 00000000 |          |          | 2241       | DC    | XL16'0000000000000000 FFFFFFFFFFFFFFFFFF' | result                              |
| 0000299C | FFFFFFFF FFFFFFFF |          |          |            |       |                                           |                                     |
| 000029A4 | 30000000 00000000 |          |          | 2242       | DC    | XL16'3000000000000000 0000000000000000'   | v2                                  |
| 000029AC | 00000000 00000000 |          |          |            |       |                                           |                                     |
|          |                   |          |          | 2243       |       |                                           |                                     |
|          |                   |          |          | 2244       | VRR_A | VGEM, 2                                   |                                     |
| 000029B8 |                   |          |          | 2245+      | DS    | 0FD                                       |                                     |
| 000029B8 |                   | 000029B8 |          | 2246+      | USING | *,R5                                      | base for test data and test routine |
| 000029B8 | 000029F8          |          |          | 2247+T51   | DC    | A(X51)                                    | address of test routine             |
| 000029BC | 0033              |          |          | 2248+      | DC    | H'51'                                     | test number                         |
| 000029BE | 00                |          |          | 2249+      | DC    | X'00'                                     |                                     |
| 000029BF | 02                |          |          | 2250+      | DC    | HL1'2'                                    | M3                                  |
| 000029C0 | E5C7C5D4 40404040 |          |          | 2251+      | DC    | CL8'VGEM'                                 | instruction name                    |
| 000029C8 | 00002A24          |          |          | 2252+      | DC    | A(RE51+16)                                | address of v2 source                |
| 000029CC | 00000010          |          |          | 2253+      | DC    | A(16)                                     | result length                       |
| 000029D0 | 00002A14          |          |          | 2254+REA51 | DC    | A(RE51)                                   | result address                      |
| 000029D8 | 00000000 00000000 |          |          | 2255+      | DS    | FD                                        | gap                                 |
| 000029E0 | 00000000 00000000 |          |          | 2256+V1051 | DS    | XL16                                      | V1 output                           |
| 000029E8 | 00000000 00000000 |          |          |            |       |                                           |                                     |
| 000029F0 | 00000000 00000000 |          |          | 2257+      | DS    | FD                                        | gap                                 |
|          |                   |          |          | 2258+*     |       |                                           |                                     |
| 000029F8 |                   |          |          | 2259+X51   | DS    | 0F                                        |                                     |
| 000029F8 | E310 5010 0014    |          | 00000010 | 2260+      | LGF   | R1,V2ADDR                                 | load v2 source                      |
| 000029FE | E721 0000 0006    |          | 00000000 | 2261+      | VL    | v2,0(R1)                                  |                                     |
| 00002A04 |                   |          |          | 2264+      | DS    | 0H                                        | E754 VGEM                           |
| 00002A04 | E7                |          |          | 2265+      | DC    | X'E7'                                     | - Vector Generate Element Masks     |
| 00002A05 | 12                |          |          | 2266+      | DC    | X'12'                                     | v1, v2                              |
| 00002A06 | 0000              |          |          | 2267+      | DC    | x'0000'                                   | reserved                            |
| 00002A08 | 20                |          |          | 2268+      | DC    | HL1'32'                                   | M3, RXB                             |
| 00002A09 | 54                |          |          | 2269+      | DC    | X'54'                                     |                                     |
| 00002A0A | E710 5028 000E    |          | 000029E0 | 2271+      | VST   | V1,V1051                                  | save v1 output                      |
| 00002A10 | 07FB              |          |          | 2272+      | BR    | R11                                       | return                              |
| 00002A14 |                   |          |          | 2273+RE51  | DC    | 0F                                        | xl16 expected result                |
| 00002A14 |                   |          |          | 2274+      | DROP  | R5                                        |                                     |

| LOC      | OBJECT CODE       | ADDR1    | ADDR2    | STMT       |       |                                         |                                     |
|----------|-------------------|----------|----------|------------|-------|-----------------------------------------|-------------------------------------|
| 00002A14 | 00000000 FFFFFFFF |          |          | 2275       | DC    | XL16'00000000FFFFFFFF 0000000000000000' | result                              |
| 00002A1C | 00000000 00000000 |          |          |            |       |                                         |                                     |
| 00002A24 | 40000000 00000000 |          |          | 2276       | DC    | XL16'4000000000000000 0000000000000000' | v2                                  |
| 00002A2C | 00000000 00000000 |          |          |            |       |                                         |                                     |
|          |                   |          |          | 2277       |       |                                         |                                     |
| 00002A38 |                   |          |          | 2278       | VRR_A | VGEM, 2                                 |                                     |
| 00002A38 |                   | 00002A38 |          | 2279+      | DS    | 0FD                                     |                                     |
| 00002A38 | 00002A78          |          |          | 2280+      | USING | *, R5                                   | base for test data and test routine |
| 00002A3C | 0034              |          |          | 2281+T52   | DC    | A(X52)                                  | address of test routine             |
| 00002A3E | 00                |          |          | 2282+      | DC    | H'52'                                   | test number                         |
| 00002A3F | 02                |          |          | 2283+      | DC    | X'00'                                   |                                     |
| 00002A40 | E5C7C5D4 40404040 |          |          | 2284+      | DC    | HL1'2'                                  | M3                                  |
| 00002A48 | 00002AA4          |          |          | 2285+      | DC    | CL8'VGEM'                               | instruction name                    |
| 00002A4C | 00000010          |          |          | 2286+      | DC    | A(RE52+16)                              | address of v2 source                |
| 00002A50 | 00002A94          |          |          | 2287+      | DC    | A(16)                                   | result length                       |
| 00002A58 | 00000000 00000000 |          |          | 2288+REA52 | DC    | A(RE52)                                 | result address                      |
| 00002A60 | 00000000 00000000 |          |          | 2289+      | DS    | FD                                      | gap                                 |
| 00002A68 | 00000000 00000000 |          |          | 2290+V1052 | DS    | XL16                                    | V1 output                           |
| 00002A70 | 00000000 00000000 |          |          |            |       |                                         |                                     |
|          |                   |          |          | 2291+      | DS    | FD                                      | gap                                 |
|          |                   |          |          | 2292+*     |       |                                         |                                     |
| 00002A78 |                   |          |          | 2293+X52   | DS    | 0F                                      |                                     |
| 00002A78 | E310 5010 0014    |          | 00000010 | 2294+      | LGF   | R1, V2ADDR                              | load v2 source                      |
| 00002A7E | E721 0000 0006    |          | 00000000 | 2295+      | VL    | v2, 0(R1)                               |                                     |
|          |                   |          |          |            |       |                                         |                                     |
| 00002A84 |                   |          |          | 2298+      | DS    | 0H                                      | E754 VGEM                           |
| 00002A84 | E7                |          |          | 2299+      | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00002A85 | 12                |          |          | 2300+      | DC    | X'12'                                   | v1, v2                              |
| 00002A86 | 0000              |          |          | 2301+      | DC    | x'0000'                                 | reserved                            |
| 00002A88 | 20                |          |          | 2302+      | DC    | HL1'32'                                 | M3, RXB                             |
| 00002A89 | 54                |          |          | 2303+      | DC    | X'54'                                   |                                     |
|          |                   |          |          |            |       |                                         |                                     |
| 00002A8A | E710 5028 000E    |          | 00002A60 | 2305+      | VST   | V1, V1052                               | save v1 output                      |
| 00002A90 | 07FB              |          |          | 2306+      | BR    | R11                                     | return                              |
| 00002A94 |                   |          |          | 2307+RE52  | DC    | 0F                                      | xl16 expected result                |
| 00002A94 |                   |          |          | 2308+      | DROP  | R5                                      |                                     |
| 00002A94 | 00000000 FFFFFFFF |          |          | 2309       | DC    | XL16'00000000FFFFFFFF 00000000FFFFFFFF' | result                              |
| 00002A9C | 00000000 FFFFFFFF |          |          |            |       |                                         |                                     |
| 00002AA4 | 50000000 00000000 |          |          | 2310       | DC    | XL16'5000000000000000 0000000000000000' | v2                                  |
| 00002AAC | 00000000 00000000 |          |          |            |       |                                         |                                     |
|          |                   |          |          | 2311       |       |                                         |                                     |
| 00002AB8 |                   |          |          | 2312       | VRR_A | VGEM, 2                                 |                                     |
| 00002AB8 |                   | 00002AB8 |          | 2313+      | DS    | 0FD                                     |                                     |
| 00002AB8 | 00002AF8          |          |          | 2314+      | USING | *, R5                                   | base for test data and test routine |
| 00002ABC | 0035              |          |          | 2315+T53   | DC    | A(X53)                                  | address of test routine             |
| 00002ABE | 00                |          |          | 2316+      | DC    | H'53'                                   | test number                         |
| 00002ABF | 02                |          |          | 2317+      | DC    | X'00'                                   |                                     |
| 00002AC0 | E5C7C5D4 40404040 |          |          | 2318+      | DC    | HL1'2'                                  | M3                                  |
| 00002AC8 | 00002B24          |          |          | 2319+      | DC    | CL8'VGEM'                               | instruction name                    |
| 00002ACC | 00000010          |          |          | 2320+      | DC    | A(RE53+16)                              | address of v2 source                |
| 00002AD0 | 00002B14          |          |          | 2321+      | DC    | A(16)                                   | result length                       |
| 00002AD8 | 00000000 00000000 |          |          | 2322+REA53 | DC    | A(RE53)                                 | result address                      |
| 00002AE0 | 00000000 00000000 |          |          | 2323+      | DS    | FD                                      | gap                                 |
| 00002AE8 | 00000000 00000000 |          |          | 2324+V1053 | DS    | XL16                                    | V1 output                           |
| 00002AF0 | 00000000 00000000 |          |          |            |       |                                         |                                     |
|          |                   |          |          | 2325+      | DS    | FD                                      | gap                                 |

| LOC      | OBJECT CODE       | ADDR1    | ADDR2    | STMT       |       |                                           |                                     |  |
|----------|-------------------|----------|----------|------------|-------|-------------------------------------------|-------------------------------------|--|
| 00002AF8 |                   |          |          | 2326+*     |       |                                           |                                     |  |
| 00002AF8 | E310 5010 0014    |          | 00000010 | 2327+X53   | DS    | 0F                                        |                                     |  |
| 00002AFE | E721 0000 0006    |          | 00000000 | 2328+      | LGF   | R1,V2ADDR                                 | load v2 source                      |  |
|          |                   |          |          | 2329+      | VL    | v2,0(R1)                                  |                                     |  |
| 00002B04 |                   |          |          | 2332+      | DS    | 0H                                        | E754 VGEM                           |  |
| 00002B04 | E7                |          |          | 2333+      | DC    | X'E7'                                     | - Vector Generate Element Masks     |  |
| 00002B05 | 12                |          |          | 2334+      | DC    | X'12'                                     | v1, v2                              |  |
| 00002B06 | 0000              |          |          | 2335+      | DC    | x'0000'                                   | reserved                            |  |
| 00002B08 | 20                |          |          | 2336+      | DC    | HL1'32'                                   | M3, RXB                             |  |
| 00002B09 | 54                |          |          | 2337+      | DC    | X'54'                                     |                                     |  |
| 00002B0A | E710 5028 000E    |          | 00002AE0 | 2339+      | VST   | V1,V1053                                  | save v1 output                      |  |
| 00002B10 | 07FB              |          |          | 2340+      | BR    | R11                                       | return                              |  |
| 00002B14 |                   |          |          | 2341+RE53  | DC    | 0F                                        | xl16 expected result                |  |
| 00002B14 | 00000000 FFFFFFFF |          |          | 2342+      | DROP  | R5                                        |                                     |  |
| 00002B14 |                   |          |          | 2343       | DC    | XL16'00000000FFFFFFFF FFFFFFFF00000000'   | result                              |  |
| 00002B1C | FFFFFFFF 00000000 |          |          |            |       |                                           |                                     |  |
| 00002B24 | 60000000 00000000 |          |          | 2344       | DC    | XL16'6000000000000000 0000000000000000'   | v2                                  |  |
| 00002B2C | 00000000 00000000 |          |          |            |       |                                           |                                     |  |
|          |                   |          |          | 2345       |       |                                           |                                     |  |
| 00002B38 |                   |          |          | 2346       | VRR_A | VGEM, 2                                   |                                     |  |
| 00002B38 |                   | 00002B38 |          | 2347+      | DS    | 0FD                                       |                                     |  |
| 00002B38 | 00002B78          |          |          | 2348+      | USING | *,R5                                      | base for test data and test routine |  |
| 00002B3C | 0036              |          |          | 2349+T54   | DC    | A(X54)                                    | address of test routine             |  |
| 00002B3E | 00                |          |          | 2350+      | DC    | H'54'                                     | test number                         |  |
| 00002B3F | 02                |          |          | 2351+      | DC    | X'00'                                     |                                     |  |
| 00002B40 | E5C7C5D4 40404040 |          |          | 2352+      | DC    | HL1'2'                                    | M3                                  |  |
| 00002B48 | 00002BA4          |          |          | 2353+      | DC    | CL8'VGEM'                                 | instruction name                    |  |
| 00002B4C | 00000010          |          |          | 2354+      | DC    | A(RE54+16)                                | address of v2 source                |  |
| 00002B50 | 00002B94          |          |          | 2355+      | DC    | A(16)                                     | result length                       |  |
| 00002B58 | 00000000 00000000 |          |          | 2356+REA54 | DC    | A(RE54)                                   | result address                      |  |
| 00002B60 | 00000000 00000000 |          |          | 2357+      | DS    | FD                                        | gap                                 |  |
| 00002B68 | 00000000 00000000 |          |          | 2358+V1054 | DS    | XL16                                      | V1 output                           |  |
| 00002B70 | 00000000 00000000 |          |          | 2359+      | DS    | FD                                        | gap                                 |  |
|          |                   |          |          | 2360+*     |       |                                           |                                     |  |
| 00002B78 |                   |          |          | 2361+X54   | DS    | 0F                                        |                                     |  |
| 00002B78 | E310 5010 0014    |          | 00000010 | 2362+      | LGF   | R1,V2ADDR                                 | load v2 source                      |  |
| 00002B7E | E721 0000 0006    |          | 00000000 | 2363+      | VL    | v2,0(R1)                                  |                                     |  |
| 00002B84 |                   |          |          | 2366+      | DS    | 0H                                        | E754 VGEM                           |  |
| 00002B84 | E7                |          |          | 2367+      | DC    | X'E7'                                     | - Vector Generate Element Masks     |  |
| 00002B85 | 12                |          |          | 2368+      | DC    | X'12'                                     | v1, v2                              |  |
| 00002B86 | 0000              |          |          | 2369+      | DC    | x'0000'                                   | reserved                            |  |
| 00002B88 | 20                |          |          | 2370+      | DC    | HL1'32'                                   | M3, RXB                             |  |
| 00002B89 | 54                |          |          | 2371+      | DC    | X'54'                                     |                                     |  |
| 00002B8A | E710 5028 000E    |          | 00002B60 | 2373+      | VST   | V1,V1054                                  | save v1 output                      |  |
| 00002B90 | 07FB              |          |          | 2374+      | BR    | R11                                       | return                              |  |
| 00002B94 |                   |          |          | 2375+RE54  | DC    | 0F                                        | xl16 expected result                |  |
| 00002B94 | 00000000 FFFFFFFF |          |          | 2376+      | DROP  | R5                                        |                                     |  |
| 00002B94 |                   |          |          | 2377       | DC    | XL16'00000000FFFFFFFF FFFFFFFFFFFFFFFFFF' | result                              |  |
| 00002B9C | FFFFFFFF FFFFFFFF |          |          |            |       |                                           |                                     |  |
| 00002BA4 | 70000000 00000000 |          |          | 2378       | DC    | XL16'7000000000000000 0000000000000000'   | v2                                  |  |
| 00002BAC | 00000000 00000000 |          |          |            |       |                                           |                                     |  |



| LOC      | OBJECT CODE       | ADDR1    | ADDR2    | STMT       |                                            |                                     |
|----------|-------------------|----------|----------|------------|--------------------------------------------|-------------------------------------|
|          |                   |          |          | 2379       |                                            |                                     |
|          |                   |          |          | 2380       | VRR_A VGEM, 2                              |                                     |
| 00002BB8 |                   |          |          | 2381+      | DS 0FD                                     |                                     |
| 00002BB8 |                   | 00002BB8 |          | 2382+      | USING *, R5                                | base for test data and test routine |
| 00002BB8 | 00002BF8          |          |          | 2383+T55   | DC A(X55)                                  | address of test routine             |
| 00002BBC | 0037              |          |          | 2384+      | DC H'55'                                   | test number                         |
| 00002BBE | 00                |          |          | 2385+      | DC X'00'                                   |                                     |
| 00002BBF | 02                |          |          | 2386+      | DC HL1'2'                                  | M3                                  |
| 00002BC0 | E5C7C5D4 40404040 |          |          | 2387+      | DC CL8'VGEM'                               | instruction name                    |
| 00002BC8 | 00002C24          |          |          | 2388+      | DC A(RE55+16)                              | address of v2 source                |
| 00002BCC | 00000010          |          |          | 2389+      | DC A(16)                                   | result length                       |
| 00002BD0 | 00002C14          |          |          | 2390+REA55 | DC A(RE55)                                 | result address                      |
| 00002BD8 | 00000000 00000000 |          |          | 2391+      | DS FD                                      | gap                                 |
| 00002BE0 | 00000000 00000000 |          |          | 2392+V1055 | DS XL16                                    | V1 output                           |
| 00002BE8 | 00000000 00000000 |          |          |            |                                            |                                     |
| 00002BF0 | 00000000 00000000 |          |          | 2393+      | DS FD                                      | gap                                 |
|          |                   |          |          | 2394+*     |                                            |                                     |
| 00002BF8 |                   |          |          | 2395+X55   | DS 0F                                      |                                     |
| 00002BF8 | E310 5010 0014    |          | 00000010 | 2396+      | LGF R1, V2ADDR                             | load v2 source                      |
| 00002BFE | E721 0000 0006    |          | 00000000 | 2397+      | VL v2, 0(R1)                               |                                     |
|          |                   |          |          |            |                                            |                                     |
| 00002C04 |                   |          |          | 2400+      | DS 0H                                      | E754 VGEM                           |
| 00002C04 | E7                |          |          | 2401+      | DC X'E7'                                   | - Vector Generate Element Masks     |
| 00002C05 | 12                |          |          | 2402+      | DC X'12'                                   | v1, v2                              |
| 00002C06 | 0000              |          |          | 2403+      | DC x'0000'                                 | reserved                            |
| 00002C08 | 20                |          |          | 2404+      | DC HL1'32'                                 | M3, RXB                             |
| 00002C09 | 54                |          |          | 2405+      | DC X'54'                                   |                                     |
|          |                   |          |          |            |                                            |                                     |
| 00002C0A | E710 5028 000E    |          | 00002BE0 | 2407+      | VST V1, V1055                              | save v1 output                      |
| 00002C10 | 07FB              |          |          | 2408+      | BR R11                                     | return                              |
| 00002C14 |                   |          |          | 2409+RE55  | DC 0F                                      | xl16 expected result                |
| 00002C14 |                   |          |          | 2410+      | DROP R5                                    |                                     |
| 00002C14 | FFFFFFFF 00000000 |          |          | 2411       | DC XL16'FFFFFFFF00000000 0000000000000000' | result                              |
| 00002C1C | 00000000 00000000 |          |          |            |                                            |                                     |
| 00002C24 | 80000000 00000000 |          |          | 2412       | DC XL16'8000000000000000 0000000000000000' | v2                                  |
| 00002C2C | 00000000 00000000 |          |          |            |                                            |                                     |
|          |                   |          |          | 2413       |                                            |                                     |
|          |                   |          |          | 2414       | VRR_A VGEM, 2                              |                                     |
| 00002C38 |                   |          |          | 2415+      | DS 0FD                                     |                                     |
| 00002C38 |                   | 00002C38 |          | 2416+      | USING *, R5                                | base for test data and test routine |
| 00002C38 | 00002C78          |          |          | 2417+T56   | DC A(X56)                                  | address of test routine             |
| 00002C3C | 0038              |          |          | 2418+      | DC H'56'                                   | test number                         |
| 00002C3E | 00                |          |          | 2419+      | DC X'00'                                   |                                     |
| 00002C3F | 02                |          |          | 2420+      | DC HL1'2'                                  | M3                                  |
| 00002C40 | E5C7C5D4 40404040 |          |          | 2421+      | DC CL8'VGEM'                               | instruction name                    |
| 00002C48 | 00002CA4          |          |          | 2422+      | DC A(RE56+16)                              | address of v2 source                |
| 00002C4C | 00000010          |          |          | 2423+      | DC A(16)                                   | result length                       |
| 00002C50 | 00002C94          |          |          | 2424+REA56 | DC A(RE56)                                 | result address                      |
| 00002C58 | 00000000 00000000 |          |          | 2425+      | DS FD                                      | gap                                 |
| 00002C60 | 00000000 00000000 |          |          | 2426+V1056 | DS XL16                                    | V1 output                           |
| 00002C68 | 00000000 00000000 |          |          |            |                                            |                                     |
| 00002C70 | 00000000 00000000 |          |          | 2427+      | DS FD                                      | gap                                 |
|          |                   |          |          | 2428+*     |                                            |                                     |
| 00002C78 |                   |          |          | 2429+X56   | DS 0F                                      |                                     |
| 00002C78 | E310 5010 0014    |          | 00000010 | 2430+      | LGF R1, V2ADDR                             | load v2 source                      |
| 00002C7E | E721 0000 0006    |          | 00000000 | 2431+      | VL v2, 0(R1)                               |                                     |



| LOC      | OBJECT CODE       | ADDR1    | ADDR2    | STMT              |       |                                           |                                     |
|----------|-------------------|----------|----------|-------------------|-------|-------------------------------------------|-------------------------------------|
| 00002C84 |                   |          |          | 2434+             | DS    | 0H                                        | E754 VGEM                           |
| 00002C84 | E7                |          |          | 2435+             | DC    | X'E7'                                     | - Vector Generate Element Masks     |
| 00002C85 | 12                |          |          | 2436+             | DC    | X'12'                                     | v1, v2                              |
| 00002C86 | 0000              |          |          | 2437+             | DC    | x'0000'                                   | reserved                            |
| 00002C88 | 20                |          |          | 2438+             | DC    | HL1'32'                                   | M3, RXB                             |
| 00002C89 | 54                |          |          | 2439+             | DC    | X'54'                                     |                                     |
| 00002C8A | E710 5028 000E    |          | 00002C60 | 2441+             | VST   | V1,V1056                                  | save v1 output                      |
| 00002C90 | 07FB              |          |          | 2442+             | BR    | R11                                       | return                              |
| 00002C94 |                   |          |          | 2443+RE56         | DC    | 0F                                        | xl16 expected result                |
| 00002C94 |                   |          |          | 2444+             | DROP  | R5                                        |                                     |
| 00002C94 | FFFFFFFF FFFFFFFF |          |          | 2445              | DC    | XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFF' | result                              |
| 00002C9C | FFFFFFFF FFFFFFFF |          |          |                   |       |                                           |                                     |
| 00002CA4 | F0000000 00000000 |          |          | 2446              | DC    | XL16'F000000000000000 0000000000000000'   | v2                                  |
| 00002CAC | 00000000 00000000 |          |          |                   |       |                                           |                                     |
|          |                   |          |          | 2447              |       |                                           |                                     |
|          |                   |          |          | 2448 * Doubleword |       | M3 = 3                                    |                                     |
|          |                   |          |          | 2449              | VRR_A | VGEM, 3                                   |                                     |
| 00002CB8 |                   |          |          | 2450+             | DS    | 0FD                                       |                                     |
| 00002CB8 |                   | 00002CB8 |          | 2451+             | USING | *,R5                                      | base for test data and test routine |
| 00002CB8 | 00002CF8          |          |          | 2452+T57          | DC    | A(X57)                                    | address of test routine             |
| 00002CBC | 0039              |          |          | 2453+             | DC    | H'57'                                     | test number                         |
| 00002CBE | 00                |          |          | 2454+             | DC    | X'00'                                     |                                     |
| 00002CBF | 03                |          |          | 2455+             | DC    | HL1'3'                                    | M3                                  |
| 00002CC0 | E5C7C5D4 40404040 |          |          | 2456+             | DC    | CL8'VGEM'                                 | instruction name                    |
| 00002CC8 | 00002D24          |          |          | 2457+             | DC    | A(RE57+16)                                | address of v2 source                |
| 00002CCC | 00000010          |          |          | 2458+             | DC    | A(16)                                     | result length                       |
| 00002CD0 | 00002D14          |          |          | 2459+REA57        | DC    | A(RE57)                                   | result address                      |
| 00002CD8 | 00000000 00000000 |          |          | 2460+             | DS    | FD                                        | gap                                 |
| 00002CE0 | 00000000 00000000 |          |          | 2461+V1057        | DS    | XL16                                      | V1 output                           |
| 00002CE8 | 00000000 00000000 |          |          |                   |       |                                           |                                     |
| 00002CF0 | 00000000 00000000 |          |          | 2462+             | DS    | FD                                        | gap                                 |
|          |                   |          |          | 2463+*            |       |                                           |                                     |
| 00002CF8 |                   |          |          | 2464+X57          | DS    | 0F                                        |                                     |
| 00002CF8 | E310 5010 0014    |          | 00000010 | 2465+             | LGF   | R1,V2ADDR                                 | load v2 source                      |
| 00002CFE | E721 0000 0006    |          | 00000000 | 2466+             | VL    | v2,0(R1)                                  |                                     |
| 00002D04 |                   |          |          | 2469+             | DS    | 0H                                        | E754 VGEM                           |
| 00002D04 | E7                |          |          | 2470+             | DC    | X'E7'                                     | - Vector Generate Element Masks     |
| 00002D05 | 12                |          |          | 2471+             | DC    | X'12'                                     | v1, v2                              |
| 00002D06 | 0000              |          |          | 2472+             | DC    | x'0000'                                   | reserved                            |
| 00002D08 | 30                |          |          | 2473+             | DC    | HL1'48'                                   | M3, RXB                             |
| 00002D09 | 54                |          |          | 2474+             | DC    | X'54'                                     |                                     |
| 00002D0A | E710 5028 000E    |          | 00002CE0 | 2476+             | VST   | V1,V1057                                  | save v1 output                      |
| 00002D10 | 07FB              |          |          | 2477+             | BR    | R11                                       | return                              |
| 00002D14 |                   |          |          | 2478+RE57         | DC    | 0F                                        | xl16 expected result                |
| 00002D14 |                   |          |          | 2479+             | DROP  | R5                                        |                                     |
| 00002D14 | 00000000 00000000 |          |          | 2480              | DC    | XL16'0000000000000000 0000000000000000'   | result                              |
| 00002D1C | 00000000 00000000 |          |          |                   |       |                                           |                                     |
| 00002D24 | 00000000 00000000 |          |          | 2481              | DC    | XL16'0000000000000000 0000000000000000'   | v2                                  |
| 00002D2C | 00000000 00000000 |          |          |                   |       |                                           |                                     |
|          |                   |          |          | 2482              |       |                                           |                                     |
|          |                   |          |          | 2483              | VRR_A | VGEM, 3                                   |                                     |
| 00002D38 |                   |          |          | 2484+             | DS    | 0FD                                       |                                     |

| LOC      | OBJECT CODE       | ADDR1    | ADDR2    | STMT       |                                            |                                     |
|----------|-------------------|----------|----------|------------|--------------------------------------------|-------------------------------------|
| 00002D38 |                   | 00002D38 |          | 2485+      | USING *,R5                                 | base for test data and test routine |
| 00002D38 | 00002D78          |          |          | 2486+T58   | DC A(X58)                                  | address of test routine             |
| 00002D3C | 003A              |          |          | 2487+      | DC H'58'                                   | test number                         |
| 00002D3E | 00                |          |          | 2488+      | DC X'00'                                   |                                     |
| 00002D3F | 03                |          |          | 2489+      | DC HL1'3'                                  | M3                                  |
| 00002D40 | E5C7C5D4 40404040 |          |          | 2490+      | DC CL8'VGEM'                               | instruction name                    |
| 00002D48 | 00002DA4          |          |          | 2491+      | DC A(RE58+16)                              | address of v2 source                |
| 00002D4C | 00000010          |          |          | 2492+      | DC A(16)                                   | result length                       |
| 00002D50 | 00002D94          |          |          | 2493+REA58 | DC A(RE58)                                 | result address                      |
| 00002D58 | 00000000 00000000 |          |          | 2494+      | DS FD                                      | gap                                 |
| 00002D60 | 00000000 00000000 |          |          | 2495+V1058 | DS XL16                                    | V1 output                           |
| 00002D68 | 00000000 00000000 |          |          |            |                                            |                                     |
| 00002D70 | 00000000 00000000 |          |          | 2496+      | DS FD                                      | gap                                 |
|          |                   |          |          | 2497+*     |                                            |                                     |
| 00002D78 |                   |          |          | 2498+X58   | DS 0F                                      |                                     |
| 00002D78 | E310 5010 0014    |          | 00000010 | 2499+      | LGF R1,V2ADDR                              | load v2 source                      |
| 00002D7E | E721 0000 0006    |          | 00000000 | 2500+      | VL v2,0(R1)                                |                                     |
| 00002D84 |                   |          |          | 2503+      | DS 0H                                      | E754 VGEM                           |
| 00002D84 | E7                |          |          | 2504+      | DC X'E7'                                   | - Vector Generate Element Masks     |
| 00002D85 | 12                |          |          | 2505+      | DC X'12'                                   | v1, v2                              |
| 00002D86 | 0000              |          |          | 2506+      | DC x'0000'                                 | reserved                            |
| 00002D88 | 30                |          |          | 2507+      | DC HL1'48'                                 | M3, RXB                             |
| 00002D89 | 54                |          |          | 2508+      | DC X'54'                                   |                                     |
| 00002D8A | E710 5028 000E    |          | 00002D60 | 2510+      | VST V1,V1058                               | save v1 output                      |
| 00002D90 | 07FB              |          |          | 2511+      | BR R11                                     | return                              |
| 00002D94 |                   |          |          | 2512+RE58  | DC 0F                                      | xl16 expected result                |
| 00002D94 |                   |          |          | 2513+      | DROP R5                                    |                                     |
| 00002D94 | 00000000 00000000 |          |          | 2514       | DC XL16'0000000000000000 0000000000000000' | result                              |
| 00002D9C | 00000000 00000000 |          |          |            |                                            |                                     |
| 00002DA4 | 3FFFFFFF FFFFFFFF |          |          | 2515       | DC XL16'3FFFFFFF FFFFFFFF'                 | v2                                  |
| 00002DAC | FFFFFFF FFFFFFFF  |          |          |            |                                            |                                     |
|          |                   |          |          | 2516       |                                            |                                     |
|          |                   |          |          | 2517       | VRR_A VGEM,3                               |                                     |
| 00002DB8 |                   |          |          | 2518+      | DS 0FD                                     |                                     |
| 00002DB8 |                   | 00002DB8 |          | 2519+      | USING *,R5                                 | base for test data and test routine |
| 00002DB8 | 00002DF8          |          |          | 2520+T59   | DC A(X59)                                  | address of test routine             |
| 00002DBC | 003B              |          |          | 2521+      | DC H'59'                                   | test number                         |
| 00002DBE | 00                |          |          | 2522+      | DC X'00'                                   |                                     |
| 00002DBF | 03                |          |          | 2523+      | DC HL1'3'                                  | M3                                  |
| 00002DC0 | E5C7C5D4 40404040 |          |          | 2524+      | DC CL8'VGEM'                               | instruction name                    |
| 00002DC8 | 00002E24          |          |          | 2525+      | DC A(RE59+16)                              | address of v2 source                |
| 00002DCC | 00000010          |          |          | 2526+      | DC A(16)                                   | result length                       |
| 00002DD0 | 00002E14          |          |          | 2527+REA59 | DC A(RE59)                                 | result address                      |
| 00002DD8 | 00000000 00000000 |          |          | 2528+      | DS FD                                      | gap                                 |
| 00002DE0 | 00000000 00000000 |          |          | 2529+V1059 | DS XL16                                    | V1 output                           |
| 00002DE8 | 00000000 00000000 |          |          |            |                                            |                                     |
| 00002DF0 | 00000000 00000000 |          |          | 2530+      | DS FD                                      | gap                                 |
|          |                   |          |          | 2531+*     |                                            |                                     |
| 00002DF8 |                   |          |          | 2532+X59   | DS 0F                                      |                                     |
| 00002DF8 | E310 5010 0014    |          | 00000010 | 2533+      | LGF R1,V2ADDR                              | load v2 source                      |
| 00002DFE | E721 0000 0006    |          | 00000000 | 2534+      | VL v2,0(R1)                                |                                     |
| 00002E04 |                   |          |          | 2537+      | DS 0H                                      | E754 VGEM                           |
| 00002E04 | E7                |          |          | 2538+      | DC X'E7'                                   | - Vector Generate Element Masks     |

| LOC      | OBJECT CODE       | ADDR1    | ADDR2    | STMT       |       |                                         |                                     |
|----------|-------------------|----------|----------|------------|-------|-----------------------------------------|-------------------------------------|
| 00002E05 | 12                |          |          | 2539+      | DC    | X'12'                                   | v1, v2                              |
| 00002E06 | 0000              |          |          | 2540+      | DC    | x'0000'                                 | reserved                            |
| 00002E08 | 30                |          |          | 2541+      | DC    | HL1'48'                                 | M3, RXB                             |
| 00002E09 | 54                |          |          | 2542+      | DC    | X'54'                                   |                                     |
| 00002E0A | E710 5028 000E    |          | 00002DE0 | 2544+      | VST   | V1,V1059                                | save v1 output                      |
| 00002E10 | 07FB              |          |          | 2545+      | BR    | R11                                     | return                              |
| 00002E14 |                   |          |          | 2546+RE59  | DC    | 0F                                      | xl16 expected result                |
| 00002E14 |                   |          |          | 2547+      | DROP  | R5                                      |                                     |
| 00002E14 | 00000000 00000000 |          |          | 2548       | DC    | XL16'0000000000000000 0000000000000000' | result                              |
| 00002E1C | FFFFFFFF FFFFFFFF |          |          |            |       |                                         |                                     |
| 00002E24 | 40000000 00000000 |          |          | 2549       | DC    | XL16'4000000000000000 0000000000000000' | v2                                  |
| 00002E2C | 00000000 00000000 |          |          |            |       |                                         |                                     |
|          |                   |          |          | 2550       |       |                                         |                                     |
| 00002E38 |                   |          |          | 2551       | VRR_A | VGEM, 3                                 |                                     |
| 00002E38 |                   | 00002E38 |          | 2552+      | DS    | 0FD                                     |                                     |
| 00002E38 | 00002E78          |          |          | 2553+      | USING | *,R5                                    | base for test data and test routine |
| 00002E3C | 003C              |          |          | 2554+T60   | DC    | A(X60)                                  | address of test routine             |
| 00002E3E | 00                |          |          | 2555+      | DC    | H'60'                                   | test number                         |
| 00002E3E | 00                |          |          | 2556+      | DC    | X'00'                                   |                                     |
| 00002E3F | 03                |          |          | 2557+      | DC    | HL1'3'                                  | M3                                  |
| 00002E40 | E5C7C5D4 40404040 |          |          | 2558+      | DC    | CL8'VGEM'                               | instruction name                    |
| 00002E48 | 00002EA4          |          |          | 2559+      | DC    | A(RE60+16)                              | address of v2 source                |
| 00002E4C | 00000010          |          |          | 2560+      | DC    | A(16)                                   | result length                       |
| 00002E50 | 00002E94          |          |          | 2561+REA60 | DC    | A(RE60)                                 | result address                      |
| 00002E58 | 00000000 00000000 |          |          | 2562+      | DS    | FD                                      | gap                                 |
| 00002E60 | 00000000 00000000 |          |          | 2563+V1060 | DS    | XL16                                    | V1 output                           |
| 00002E68 | 00000000 00000000 |          |          |            |       |                                         |                                     |
| 00002E70 | 00000000 00000000 |          |          | 2564+      | DS    | FD                                      | gap                                 |
|          |                   |          |          | 2565+*     |       |                                         |                                     |
| 00002E78 |                   |          |          | 2566+X60   | DS    | 0F                                      |                                     |
| 00002E78 | E310 5010 0014    |          | 00000010 | 2567+      | LGF   | R1,V2ADDR                               | load v2 source                      |
| 00002E7E | E721 0000 0006    |          | 00000000 | 2568+      | VL    | v2,0(R1)                                |                                     |
| 00002E84 |                   |          |          | 2571+      | DS    | 0H                                      | E754 VGEM                           |
| 00002E84 | E7                |          |          | 2572+      | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00002E85 | 12                |          |          | 2573+      | DC    | X'12'                                   | v1, v2                              |
| 00002E86 | 0000              |          |          | 2574+      | DC    | x'0000'                                 | reserved                            |
| 00002E88 | 30                |          |          | 2575+      | DC    | HL1'48'                                 | M3, RXB                             |
| 00002E89 | 54                |          |          | 2576+      | DC    | X'54'                                   |                                     |
| 00002E8A | E710 5028 000E    |          | 00002E60 | 2578+      | VST   | V1,V1060                                | save v1 output                      |
| 00002E90 | 07FB              |          |          | 2579+      | BR    | R11                                     | return                              |
| 00002E94 |                   |          |          | 2580+RE60  | DC    | 0F                                      | xl16 expected result                |
| 00002E94 |                   |          |          | 2581+      | DROP  | R5                                      |                                     |
| 00002E94 | FFFFFFFF FFFFFFFF |          |          | 2582       | DC    | XL16'FFFFFFFFFFFFFFFF 0000000000000000' | result                              |
| 00002E9C | 00000000 00000000 |          |          |            |       |                                         |                                     |
| 00002EA4 | 80000000 00000000 |          |          | 2583       | DC    | XL16'8000000000000000 0000000000000000' | v2                                  |
| 00002EAC | 00000000 00000000 |          |          |            |       |                                         |                                     |
|          |                   |          |          | 2584       |       |                                         |                                     |
| 00002EB8 |                   |          |          | 2585       | VRR_A | VGEM, 3                                 |                                     |
| 00002EB8 |                   | 00002EB8 |          | 2586+      | DS    | 0FD                                     |                                     |
| 00002EB8 | 00002EF8          |          |          | 2587+      | USING | *,R5                                    | base for test data and test routine |
| 00002EB8 | 003D              |          |          | 2588+T61   | DC    | A(X61)                                  | address of test routine             |
| 00002EBC | 00                |          |          | 2589+      | DC    | H'61'                                   | test number                         |
| 00002EBE |                   |          |          | 2590+      | DC    | X'00'                                   |                                     |

| LOC      | OBJECT CODE       | ADDR1    | ADDR2    | STMT       |       |                                         |                                     |
|----------|-------------------|----------|----------|------------|-------|-----------------------------------------|-------------------------------------|
| 00002EBF | 03                |          |          | 2591+      | DC    | HL1'3'                                  | M3                                  |
| 00002EC0 | E5C7C5D4 40404040 |          |          | 2592+      | DC    | CL8'VGEM'                               | instruction name                    |
| 00002EC8 | 00002F24          |          |          | 2593+      | DC    | A(RE61+16)                              | address of v2 source                |
| 00002ECC | 00000010          |          |          | 2594+      | DC    | A(16)                                   | result length                       |
| 00002ED0 | 00002F14          |          |          | 2595+REA61 | DC    | A(RE61)                                 | result address                      |
| 00002ED8 | 00000000 00000000 |          |          | 2596+      | DS    | FD                                      | gap                                 |
| 00002EE0 | 00000000 00000000 |          |          | 2597+V1061 | DS    | XL16                                    | V1 output                           |
| 00002EE8 | 00000000 00000000 |          |          |            |       |                                         |                                     |
| 00002EF0 | 00000000 00000000 |          |          | 2598+      | DS    | FD                                      | gap                                 |
|          |                   |          |          | 2599+*     |       |                                         |                                     |
| 00002EF8 |                   |          |          | 2600+X61   | DS    | 0F                                      |                                     |
| 00002EF8 | E310 5010 0014    |          | 00000010 | 2601+      | LGF   | R1,V2ADDR                               | load v2 source                      |
| 00002EFE | E721 0000 0006    |          | 00000000 | 2602+      | VL    | v2,0(R1)                                |                                     |
| 00002F04 |                   |          |          | 2605+      | DS    | 0H                                      | E754 VGEM                           |
| 00002F04 | E7                |          |          | 2606+      | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00002F05 | 12                |          |          | 2607+      | DC    | X'12'                                   | v1, v2                              |
| 00002F06 | 0000              |          |          | 2608+      | DC    | x'0000'                                 | reserved                            |
| 00002F08 | 30                |          |          | 2609+      | DC    | HL1'48'                                 | M3, RXB                             |
| 00002F09 | 54                |          |          | 2610+      | DC    | X'54'                                   |                                     |
| 00002F0A | E710 5028 000E    |          | 00002EE0 | 2612+      | VST   | V1,V1061                                | save v1 output                      |
| 00002F10 | 07FB              |          |          | 2613+      | BR    | R11                                     | return                              |
| 00002F14 |                   |          |          | 2614+RE61  | DC    | 0F                                      | xl16 expected result                |
| 00002F14 |                   |          |          | 2615+      | DROP  | R5                                      |                                     |
| 00002F14 | FFFFFFFF FFFFFFFF |          |          | 2616       | DC    | XL16'FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF'  | result                              |
| 00002F1C | FFFFFFFF FFFFFFFF |          |          |            |       |                                         |                                     |
| 00002F24 | C0000000 00000000 |          |          | 2617       | DC    | XL16'C000000000000000 0000000000000000' | v2                                  |
| 00002F2C | 00000000 00000000 |          |          |            |       |                                         |                                     |
| 00002F38 |                   |          |          | 2618       |       |                                         |                                     |
| 00002F38 |                   |          |          | 2619       | VRR_A | VGEM, 3                                 |                                     |
| 00002F38 |                   | 00002F38 |          | 2620+      | DS    | 0FD                                     |                                     |
| 00002F38 | 00002F78          |          |          | 2621+      | USING | *,R5                                    | base for test data and test routine |
| 00002F3C | 003E              |          |          | 2622+T62   | DC    | A(X62)                                  | address of test routine             |
| 00002F3E | 00                |          |          | 2623+      | DC    | H'62'                                   | test number                         |
| 00002F3F | 03                |          |          | 2624+      | DC    | X'00'                                   |                                     |
| 00002F40 | E5C7C5D4 40404040 |          |          | 2625+      | DC    | HL1'3'                                  | M3                                  |
| 00002F48 | 00002FA4          |          |          | 2626+      | DC    | CL8'VGEM'                               | instruction name                    |
| 00002F4C | 00000010          |          |          | 2627+      | DC    | A(RE62+16)                              | address of v2 source                |
| 00002F50 | 00002F94          |          |          | 2628+      | DC    | A(16)                                   | result length                       |
| 00002F58 | 00000000 00000000 |          |          | 2629+REA62 | DC    | A(RE62)                                 | result address                      |
| 00002F60 | 00000000 00000000 |          |          | 2630+      | DS    | FD                                      | gap                                 |
| 00002F68 | 00000000 00000000 |          |          | 2631+V1062 | DS    | XL16                                    | V1 output                           |
| 00002F70 | 00000000 00000000 |          |          |            |       |                                         |                                     |
|          |                   |          |          | 2632+      | DS    | FD                                      | gap                                 |
|          |                   |          |          | 2633+*     |       |                                         |                                     |
| 00002F78 |                   |          |          | 2634+X62   | DS    | 0F                                      |                                     |
| 00002F78 | E310 5010 0014    |          | 00000010 | 2635+      | LGF   | R1,V2ADDR                               | load v2 source                      |
| 00002F7E | E721 0000 0006    |          | 00000000 | 2636+      | VL    | v2,0(R1)                                |                                     |
| 00002F84 |                   |          |          | 2639+      | DS    | 0H                                      | E754 VGEM                           |
| 00002F84 | E7                |          |          | 2640+      | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00002F85 | 12                |          |          | 2641+      | DC    | X'12'                                   | v1, v2                              |
| 00002F86 | 0000              |          |          | 2642+      | DC    | x'0000'                                 | reserved                            |
| 00002F88 | 30                |          |          | 2643+      | DC    | HL1'48'                                 | M3, RXB                             |
| 00002F89 | 54                |          |          | 2644+      | DC    | X'54'                                   |                                     |

| LOC      | OBJECT CODE       | ADDR1    | ADDR2    | STMT            |       |                                           |                                     |
|----------|-------------------|----------|----------|-----------------|-------|-------------------------------------------|-------------------------------------|
| 00002F8A | E710 5028 000E    |          | 00002F60 | 2646+           | VST   | V1,V1062                                  | save v1 output                      |
| 00002F90 | 07FB              |          |          | 2647+           | BR    | R11                                       | return                              |
| 00002F94 |                   |          |          | 2648+RE62       | DC    | 0F                                        | xl16 expected result                |
| 00002F94 |                   |          |          | 2649+           | DROP  | R5                                        |                                     |
| 00002F94 | FFFFFFFF FFFFFFFF |          |          | 2650            | DC    | XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFF' | result                              |
| 00002F9C | FFFFFFFF FFFFFFFF |          |          |                 |       |                                           |                                     |
| 00002FA4 | F0000000 00000000 |          |          | 2651            | DC    | XL16'F000000000000000 0000000000000000'   | v2                                  |
| 00002FAC | 00000000 00000000 |          |          |                 |       |                                           |                                     |
|          |                   |          |          | 2652            |       |                                           |                                     |
|          |                   |          |          | 2653 * Quadword |       | M3 = 4                                    |                                     |
| 00002FB8 |                   |          |          | 2654            | VRR_A | VGEM, 4                                   |                                     |
| 00002FB8 |                   | 00002FB8 |          | 2655+           | DS    | 0FD                                       |                                     |
| 00002FB8 | 00002FF8          |          |          | 2656+           | USING | *, R5                                     | base for test data and test routine |
| 00002FBC | 003F              |          |          | 2657+T63        | DC    | A(X63)                                    | address of test routine             |
| 00002FBE | 00                |          |          | 2658+           | DC    | H'63'                                     | test number                         |
| 00002FBE | 00                |          |          | 2659+           | DC    | X'00'                                     |                                     |
| 00002FBF | 04                |          |          | 2660+           | DC    | HL1'4'                                    | M3                                  |
| 00002FC0 | E5C7C5D4 40404040 |          |          | 2661+           | DC    | CL8'VGEM'                                 | instruction name                    |
| 00002FC8 | 00003024          |          |          | 2662+           | DC    | A(RE63+16)                                | address of v2 source                |
| 00002FCC | 00000010          |          |          | 2663+           | DC    | A(16)                                     | result length                       |
| 00002FD0 | 00003014          |          |          | 2664+REA63      | DC    | A(RE63)                                   | result address                      |
| 00002FD8 | 00000000 00000000 |          |          | 2665+           | DS    | FD                                        | gap                                 |
| 00002FE0 | 00000000 00000000 |          |          | 2666+V1063      | DS    | XL16                                      | V1 output                           |
| 00002FE8 | 00000000 00000000 |          |          |                 |       |                                           |                                     |
| 00002FF0 | 00000000 00000000 |          |          | 2667+           | DS    | FD                                        | gap                                 |
|          |                   |          |          | 2668+*          |       |                                           |                                     |
| 00002FF8 |                   |          |          | 2669+X63        | DS    | 0F                                        |                                     |
| 00002FF8 | E310 5010 0014    |          | 00000010 | 2670+           | LGF   | R1,V2ADDR                                 | load v2 source                      |
| 00002FFE | E721 0000 0006    |          | 00000000 | 2671+           | VL    | v2, 0(R1)                                 |                                     |
| 00003004 |                   |          |          | 2674+           | DS    | 0H                                        | E754 VGEM                           |
| 00003004 | E7                |          |          | 2675+           | DC    | X'E7'                                     | - Vector Generate Element Masks     |
| 00003005 | 12                |          |          | 2676+           | DC    | X'12'                                     | v1, v2                              |
| 00003006 | 0000              |          |          | 2677+           | DC    | x'0000'                                   | reserved                            |
| 00003008 | 40                |          |          | 2678+           | DC    | HL1'64'                                   | M3, RXB                             |
| 00003009 | 54                |          |          | 2679+           | DC    | X'54'                                     |                                     |
| 0000300A | E710 5028 000E    |          | 00002FE0 | 2681+           | VST   | V1,V1063                                  | save v1 output                      |
| 00003010 | 07FB              |          |          | 2682+           | BR    | R11                                       | return                              |
| 00003014 |                   |          |          | 2683+RE63       | DC    | 0F                                        | xl16 expected result                |
| 00003014 |                   |          |          | 2684+           | DROP  | R5                                        |                                     |
| 00003014 | 00000000 00000000 |          |          | 2685            | DC    | XL16'0000000000000000 0000000000000000'   | result                              |
| 0000301C | 00000000 00000000 |          |          |                 |       |                                           |                                     |
| 00003024 | 00000000 00000000 |          |          | 2686            | DC    | XL16'0000000000000000 0000000000000000'   | v2                                  |
| 0000302C | 00000000 00000000 |          |          |                 |       |                                           |                                     |
|          |                   |          |          | 2687            |       |                                           |                                     |
| 00003038 |                   |          |          | 2688            | VRR_A | VGEM, 4                                   |                                     |
| 00003038 |                   | 00003038 |          | 2689+           | DS    | 0FD                                       |                                     |
| 00003038 | 00003078          |          |          | 2690+           | USING | *, R5                                     | base for test data and test routine |
| 00003038 | 0040              |          |          | 2691+T64        | DC    | A(X64)                                    | address of test routine             |
| 0000303C | 00                |          |          | 2692+           | DC    | H'64'                                     | test number                         |
| 0000303E | 00                |          |          | 2693+           | DC    | X'00'                                     |                                     |
| 0000303F | 04                |          |          | 2694+           | DC    | HL1'4'                                    | M3                                  |
| 00003040 | E5C7C5D4 40404040 |          |          | 2695+           | DC    | CL8'VGEM'                                 | instruction name                    |
| 00003048 | 000030A4          |          |          | 2696+           | DC    | A(RE64+16)                                | address of v2 source                |



| LOC      | OBJECT CODE       | ADDR1    | ADDR2    | STMT       |       |                                         |                                     |
|----------|-------------------|----------|----------|------------|-------|-----------------------------------------|-------------------------------------|
| 0000304C | 00000010          |          |          | 2697+      | DC    | A(16)                                   | result length                       |
| 00003050 | 00003094          |          |          | 2698+REA64 | DC    | A(RE64)                                 | result address                      |
| 00003058 | 00000000 00000000 |          |          | 2699+      | DS    | FD                                      | gap                                 |
| 00003060 | 00000000 00000000 |          |          | 2700+V1064 | DS    | XL16                                    | V1 output                           |
| 00003068 | 00000000 00000000 |          |          |            |       |                                         |                                     |
| 00003070 | 00000000 00000000 |          |          | 2701+      | DS    | FD                                      | gap                                 |
|          |                   |          |          | 2702+*     |       |                                         |                                     |
| 00003078 |                   |          |          | 2703+X64   | DS    | 0F                                      |                                     |
| 00003078 | E310 5010 0014    |          | 00000010 | 2704+      | LGF   | R1,V2ADDR                               | load v2 source                      |
| 0000307E | E721 0000 0006    |          | 00000000 | 2705+      | VL    | v2,0(R1)                                |                                     |
| 00003084 |                   |          |          | 2708+      | DS    | 0H                                      | E754 VGEM                           |
| 00003084 | E7                |          |          | 2709+      | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00003085 | 12                |          |          | 2710+      | DC    | X'12'                                   | v1, v2                              |
| 00003086 | 0000              |          |          | 2711+      | DC    | x'0000'                                 | reserved                            |
| 00003088 | 40                |          |          | 2712+      | DC    | HL1'64'                                 | M3, RXB                             |
| 00003089 | 54                |          |          | 2713+      | DC    | X'54'                                   |                                     |
| 0000308A | E710 5028 000E    |          | 00003060 | 2715+      | VST   | V1,V1064                                | save v1 output                      |
| 00003090 | 07FB              |          |          | 2716+      | BR    | R11                                     | return                              |
| 00003094 |                   |          |          | 2717+RE64  | DC    | 0F                                      | xl16 expected result                |
| 00003094 |                   |          |          | 2718+      | DROP  | R5                                      |                                     |
| 00003094 | 00000000 00000000 |          |          | 2719       | DC    | XL16'0000000000000000 0000000000000000' | result                              |
| 0000309C | 00000000 00000000 |          |          |            |       |                                         |                                     |
| 000030A4 | 7FFFFFFF FFFFFFFF |          |          | 2720       | DC    | XL16'7FFFFFFF FFFFFFFF'                 | v2                                  |
| 000030AC | FFFFFFF FFFFFFFF  |          |          |            |       |                                         |                                     |
|          |                   |          |          | 2721       |       |                                         |                                     |
| 000030B8 |                   |          |          | 2722       | VRR_A | VGEM, 4                                 |                                     |
| 000030B8 |                   | 000030B8 |          | 2723+      | DS    | 0FD                                     |                                     |
| 000030B8 | 000030F8          |          |          | 2724+      | USING | *,R5                                    | base for test data and test routine |
| 000030BC | 0041              |          |          | 2725+T65   | DC    | A(X65)                                  | address of test routine             |
| 000030BE | 00                |          |          | 2726+      | DC    | H'65'                                   | test number                         |
| 000030BF | 04                |          |          | 2727+      | DC    | X'00'                                   |                                     |
| 000030C0 | E5C7C5D4 40404040 |          |          | 2728+      | DC    | HL1'4'                                  | M3                                  |
| 000030C8 | 00003124          |          |          | 2729+      | DC    | CL8'VGEM'                               | instruction name                    |
| 000030CC | 00000010          |          |          | 2730+      | DC    | A(RE65+16)                              | address of v2 source                |
| 000030D0 | 00003114          |          |          | 2731+      | DC    | A(16)                                   | result length                       |
| 000030D8 | 00000000 00000000 |          |          | 2732+REA65 | DC    | A(RE65)                                 | result address                      |
| 000030E0 | 00000000 00000000 |          |          | 2733+      | DS    | FD                                      | gap                                 |
| 000030E8 | 00000000 00000000 |          |          | 2734+V1065 | DS    | XL16                                    | V1 output                           |
| 000030F0 | 00000000 00000000 |          |          |            |       |                                         |                                     |
|          |                   |          |          | 2735+      | DS    | FD                                      | gap                                 |
|          |                   |          |          | 2736+*     |       |                                         |                                     |
| 000030F8 |                   |          |          | 2737+X65   | DS    | 0F                                      |                                     |
| 000030F8 | E310 5010 0014    |          | 00000010 | 2738+      | LGF   | R1,V2ADDR                               | load v2 source                      |
| 000030FE | E721 0000 0006    |          | 00000000 | 2739+      | VL    | v2,0(R1)                                |                                     |
| 00003104 |                   |          |          | 2742+      | DS    | 0H                                      | E754 VGEM                           |
| 00003104 | E7                |          |          | 2743+      | DC    | X'E7'                                   | - Vector Generate Element Masks     |
| 00003105 | 12                |          |          | 2744+      | DC    | X'12'                                   | v1, v2                              |
| 00003106 | 0000              |          |          | 2745+      | DC    | x'0000'                                 | reserved                            |
| 00003108 | 40                |          |          | 2746+      | DC    | HL1'64'                                 | M3, RXB                             |
| 00003109 | 54                |          |          | 2747+      | DC    | X'54'                                   |                                     |
| 0000310A | E710 5028 000E    |          | 000030E0 | 2749+      | VST   | V1,V1065                                | save v1 output                      |
| 00003110 | 07FB              |          |          | 2750+      | BR    | R11                                     | return                              |



| LOC      | OBJECT CODE       | ADDR1    | ADDR2    | STMT         |                                           |                                           |                                     |
|----------|-------------------|----------|----------|--------------|-------------------------------------------|-------------------------------------------|-------------------------------------|
| 00003114 |                   |          |          | 2751+RE65    | DC                                        | 0F                                        | xl16 expected result                |
| 00003114 |                   |          |          | 2752+        | DROP                                      | R5                                        |                                     |
| 00003114 | FFFFFFFF FFFFFFFF |          |          | 2753         | DC                                        | XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFF' | result                              |
| 0000311C | FFFFFFFF FFFFFFFF |          |          |              |                                           |                                           |                                     |
| 00003124 | 80000000 00000000 |          |          | 2754         | DC                                        | XL16'8000000000000000 0000000000000000'   | v2                                  |
| 0000312C | 00000000 00000000 |          |          |              |                                           |                                           |                                     |
|          |                   |          |          | 2755         |                                           |                                           |                                     |
|          |                   |          |          | 2756         | VRR_A                                     | VGEM, 4                                   |                                     |
| 00003138 |                   |          |          | 2757+        | DS                                        | 0FD                                       |                                     |
| 00003138 |                   | 00003138 |          | 2758+        | USING                                     | *, R5                                     | base for test data and test routine |
| 00003138 | 00003178          |          |          | 2759+T66     | DC                                        | A(X66)                                    | address of test routine             |
| 0000313C | 0042              |          |          | 2760+        | DC                                        | H'66'                                     | test number                         |
| 0000313E | 00                |          |          | 2761+        | DC                                        | X'00'                                     |                                     |
| 0000313F | 04                |          |          | 2762+        | DC                                        | HL1'4'                                    | M3                                  |
| 00003140 | E5C7C5D4 40404040 |          |          | 2763+        | DC                                        | CL8'VGEM'                                 | instruction name                    |
| 00003148 | 000031A4          |          |          | 2764+        | DC                                        | A(RE66+16)                                | address of v2 source                |
| 0000314C | 00000010          |          |          | 2765+        | DC                                        | A(16)                                     | result length                       |
| 00003150 | 00003194          |          |          | 2766+REA66   | DC                                        | A(RE66)                                   | result address                      |
| 00003158 | 00000000 00000000 |          |          | 2767+        | DS                                        | FD                                        | gap                                 |
| 00003160 | 00000000 00000000 |          |          | 2768+V1066   | DS                                        | XL16                                      | V1 output                           |
| 00003168 | 00000000 00000000 |          |          |              |                                           |                                           |                                     |
| 00003170 | 00000000 00000000 |          |          | 2769+        | DS                                        | FD                                        | gap                                 |
|          |                   |          |          | 2770+*       |                                           |                                           |                                     |
| 00003178 |                   |          |          | 2771+X66     | DS                                        | 0F                                        |                                     |
| 00003178 | E310 5010 0014    |          | 00000010 | 2772+        | LGF                                       | R1, V2ADDR                                | load v2 source                      |
| 0000317E | E721 0000 0006    |          | 00000000 | 2773+        | VL                                        | v2, 0(R1)                                 |                                     |
|          |                   |          |          |              |                                           |                                           |                                     |
| 00003184 |                   |          |          | 2776+        | DS                                        | 0H                                        | E754 VGEM                           |
| 00003184 | E7                |          |          | 2777+        | DC                                        | X'E7'                                     | - Vector Generate Element Masks     |
| 00003185 | 12                |          |          | 2778+        | DC                                        | X'12'                                     | v1, v2                              |
| 00003186 | 0000              |          |          | 2779+        | DC                                        | x'0000'                                   | reserved                            |
| 00003188 | 40                |          |          | 2780+        | DC                                        | HL1'64'                                   | M3, RXB                             |
| 00003189 | 54                |          |          | 2781+        | DC                                        | X'54'                                     |                                     |
|          |                   |          |          |              |                                           |                                           |                                     |
| 0000318A | E710 5028 000E    |          | 00003160 | 2783+        | VST                                       | V1, V1066                                 | save v1 output                      |
| 00003190 | 07FB              |          |          | 2784+        | BR                                        | R11                                       | return                              |
| 00003194 |                   |          |          | 2785+RE66    | DC                                        | 0F                                        | xl16 expected result                |
| 00003194 |                   |          |          | 2786+        | DROP                                      | R5                                        |                                     |
| 00003194 | FFFFFFFF FFFFFFFF |          |          | 2787         | DC                                        | XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFF' | result                              |
| 0000319C | FFFFFFFF FFFFFFFF |          |          |              |                                           |                                           |                                     |
| 000031A4 | F0000000 00000000 |          |          | 2788         | DC                                        | XL16'F000000000000000 0000000000000000'   | v2                                  |
| 000031AC | 00000000 00000000 |          |          |              |                                           |                                           |                                     |
|          |                   |          |          | 2789         |                                           |                                           |                                     |
|          |                   |          |          | 2790         |                                           |                                           |                                     |
| 000031B4 | 00000000          |          |          | 2791         | DC                                        | F'0'                                      | END OF TABLE                        |
| 000031B8 | 00000000          |          |          | 2792         | DC                                        | F'0'                                      |                                     |
|          |                   |          |          | 2793 *       |                                           |                                           |                                     |
|          |                   |          |          | 2794 *       | table of pointers to individual load test |                                           |                                     |
|          |                   |          |          | 2795 *       |                                           |                                           |                                     |
| 000031BC |                   |          |          | 2796 E7TESTS | DS                                        | 0F                                        |                                     |
|          |                   |          |          | 2797         | PTTABLE                                   |                                           |                                     |
| 000031BC |                   |          |          | 2798+TTABLE  | DS                                        | 0F                                        |                                     |
| 000031BC | 000010B8          |          |          | 2799+        | DC                                        | A(T1)                                     |                                     |
| 000031C0 | 00001138          |          |          | 2800+        | DC                                        | A(T2)                                     |                                     |
| 000031C4 | 000011B8          |          |          | 2801+        | DC                                        | A(T3)                                     |                                     |
| 000031C8 | 00001238          |          |          | 2802+        | DC                                        | A(T4)                                     |                                     |

| LOC      | OBJECT CODE | ADDR1 | ADDR2 | STMT  |    |        |
|----------|-------------|-------|-------|-------|----|--------|
| 000031CC | 000012B8    |       |       | 2803+ | DC | A(T5)  |
| 000031D0 | 00001338    |       |       | 2804+ | DC | A(T6)  |
| 000031D4 | 000013B8    |       |       | 2805+ | DC | A(T7)  |
| 000031D8 | 00001438    |       |       | 2806+ | DC | A(T8)  |
| 000031DC | 000014B8    |       |       | 2807+ | DC | A(T9)  |
| 000031E0 | 00001538    |       |       | 2808+ | DC | A(T10) |
| 000031E4 | 000015B8    |       |       | 2809+ | DC | A(T11) |
| 000031E8 | 00001638    |       |       | 2810+ | DC | A(T12) |
| 000031EC | 000016B8    |       |       | 2811+ | DC | A(T13) |
| 000031F0 | 00001738    |       |       | 2812+ | DC | A(T14) |
| 000031F4 | 000017B8    |       |       | 2813+ | DC | A(T15) |
| 000031F8 | 00001838    |       |       | 2814+ | DC | A(T16) |
| 000031FC | 000018B8    |       |       | 2815+ | DC | A(T17) |
| 00003200 | 00001938    |       |       | 2816+ | DC | A(T18) |
| 00003204 | 000019B8    |       |       | 2817+ | DC | A(T19) |
| 00003208 | 00001A38    |       |       | 2818+ | DC | A(T20) |
| 0000320C | 00001AB8    |       |       | 2819+ | DC | A(T21) |
| 00003210 | 00001B38    |       |       | 2820+ | DC | A(T22) |
| 00003214 | 00001BB8    |       |       | 2821+ | DC | A(T23) |
| 00003218 | 00001C38    |       |       | 2822+ | DC | A(T24) |
| 0000321C | 00001CB8    |       |       | 2823+ | DC | A(T25) |
| 00003220 | 00001D38    |       |       | 2824+ | DC | A(T26) |
| 00003224 | 00001DB8    |       |       | 2825+ | DC | A(T27) |
| 00003228 | 00001E38    |       |       | 2826+ | DC | A(T28) |
| 0000322C | 00001EB8    |       |       | 2827+ | DC | A(T29) |
| 00003230 | 00001F38    |       |       | 2828+ | DC | A(T30) |
| 00003234 | 00001FB8    |       |       | 2829+ | DC | A(T31) |
| 00003238 | 00002038    |       |       | 2830+ | DC | A(T32) |
| 0000323C | 000020B8    |       |       | 2831+ | DC | A(T33) |
| 00003240 | 00002138    |       |       | 2832+ | DC | A(T34) |
| 00003244 | 000021B8    |       |       | 2833+ | DC | A(T35) |
| 00003248 | 00002238    |       |       | 2834+ | DC | A(T36) |
| 0000324C | 000022B8    |       |       | 2835+ | DC | A(T37) |
| 00003250 | 00002338    |       |       | 2836+ | DC | A(T38) |
| 00003254 | 000023B8    |       |       | 2837+ | DC | A(T39) |
| 00003258 | 00002438    |       |       | 2838+ | DC | A(T40) |
| 0000325C | 000024B8    |       |       | 2839+ | DC | A(T41) |
| 00003260 | 00002538    |       |       | 2840+ | DC | A(T42) |
| 00003264 | 000025B8    |       |       | 2841+ | DC | A(T43) |
| 00003268 | 00002638    |       |       | 2842+ | DC | A(T44) |
| 0000326C | 000026B8    |       |       | 2843+ | DC | A(T45) |
| 00003270 | 00002738    |       |       | 2844+ | DC | A(T46) |
| 00003274 | 000027B8    |       |       | 2845+ | DC | A(T47) |
| 00003278 | 00002838    |       |       | 2846+ | DC | A(T48) |
| 0000327C | 000028B8    |       |       | 2847+ | DC | A(T49) |
| 00003280 | 00002938    |       |       | 2848+ | DC | A(T50) |
| 00003284 | 000029B8    |       |       | 2849+ | DC | A(T51) |
| 00003288 | 00002A38    |       |       | 2850+ | DC | A(T52) |
| 0000328C | 00002AB8    |       |       | 2851+ | DC | A(T53) |
| 00003290 | 00002B38    |       |       | 2852+ | DC | A(T54) |
| 00003294 | 00002BB8    |       |       | 2853+ | DC | A(T55) |
| 00003298 | 00002C38    |       |       | 2854+ | DC | A(T56) |
| 0000329C | 00002CB8    |       |       | 2855+ | DC | A(T57) |
| 000032A0 | 00002D38    |       |       | 2856+ | DC | A(T58) |
| 000032A4 | 00002DB8    |       |       | 2857+ | DC | A(T59) |
| 000032A8 | 00002E38    |       |       | 2858+ | DC | A(T60) |



| LOC | OBJECT CODE | ADDR1    | ADDR2    | STMT |                               |     |    |
|-----|-------------|----------|----------|------|-------------------------------|-----|----|
|     |             |          |          | 2872 | *****                         |     |    |
|     |             |          |          | 2873 | *            Register equates |     |    |
|     |             |          |          | 2874 | *****                         |     |    |
|     |             | 00000000 | 00000001 | 2876 | R0                            | EQU | 0  |
|     |             | 00000001 | 00000001 | 2877 | R1                            | EQU | 1  |
|     |             | 00000002 | 00000001 | 2878 | R2                            | EQU | 2  |
|     |             | 00000003 | 00000001 | 2879 | R3                            | EQU | 3  |
|     |             | 00000004 | 00000001 | 2880 | R4                            | EQU | 4  |
|     |             | 00000005 | 00000001 | 2881 | R5                            | EQU | 5  |
|     |             | 00000006 | 00000001 | 2882 | R6                            | EQU | 6  |
|     |             | 00000007 | 00000001 | 2883 | R7                            | EQU | 7  |
|     |             | 00000008 | 00000001 | 2884 | R8                            | EQU | 8  |
|     |             | 00000009 | 00000001 | 2885 | R9                            | EQU | 9  |
|     |             | 0000000A | 00000001 | 2886 | R10                           | EQU | 10 |
|     |             | 0000000B | 00000001 | 2887 | R11                           | EQU | 11 |
|     |             | 0000000C | 00000001 | 2888 | R12                           | EQU | 12 |
|     |             | 0000000D | 00000001 | 2889 | R13                           | EQU | 13 |
|     |             | 0000000E | 00000001 | 2890 | R14                           | EQU | 14 |
|     |             | 0000000F | 00000001 | 2891 | R15                           | EQU | 15 |
|     |             |          |          |      |                               |     |    |
|     |             |          |          |      |                               |     |    |
|     |             |          |          | 2893 | *****                         |     |    |
|     |             |          |          | 2894 | *            Register equates |     |    |
|     |             |          |          | 2895 | *****                         |     |    |
|     |             | 00000000 | 00000001 | 2897 | V0                            | EQU | 0  |
|     |             | 00000001 | 00000001 | 2898 | V1                            | EQU | 1  |
|     |             | 00000002 | 00000001 | 2899 | V2                            | EQU | 2  |
|     |             | 00000003 | 00000001 | 2900 | V3                            | EQU | 3  |
|     |             | 00000004 | 00000001 | 2901 | V4                            | EQU | 4  |
|     |             | 00000005 | 00000001 | 2902 | V5                            | EQU | 5  |
|     |             | 00000006 | 00000001 | 2903 | V6                            | EQU | 6  |
|     |             | 00000007 | 00000001 | 2904 | V7                            | EQU | 7  |
|     |             | 00000008 | 00000001 | 2905 | V8                            | EQU | 8  |
|     |             | 00000009 | 00000001 | 2906 | V9                            | EQU | 9  |
|     |             | 0000000A | 00000001 | 2907 | V10                           | EQU | 10 |
|     |             | 0000000B | 00000001 | 2908 | V11                           | EQU | 11 |
|     |             | 0000000C | 00000001 | 2909 | V12                           | EQU | 12 |
|     |             | 0000000D | 00000001 | 2910 | V13                           | EQU | 13 |
|     |             | 0000000E | 00000001 | 2911 | V14                           | EQU | 14 |
|     |             | 0000000F | 00000001 | 2912 | V15                           | EQU | 15 |
|     |             | 00000010 | 00000001 | 2913 | V16                           | EQU | 16 |
|     |             | 00000011 | 00000001 | 2914 | V17                           | EQU | 17 |
|     |             | 00000012 | 00000001 | 2915 | V18                           | EQU | 18 |
|     |             | 00000013 | 00000001 | 2916 | V19                           | EQU | 19 |
|     |             | 00000014 | 00000001 | 2917 | V20                           | EQU | 20 |
|     |             | 00000015 | 00000001 | 2918 | V21                           | EQU | 21 |









| SYMBOL | TYPE | VALUE    | LENGTH | DEFN | REFERENCES   |
|--------|------|----------|--------|------|--------------|
| RE4    | F    | 00001294 | 4      | 673  | 652    654   |
| RE40   | F    | 00002494 | 4      | 1898 | 1877    1879 |
| RE41   | F    | 00002514 | 4      | 1932 | 1911    1913 |
| RE42   | F    | 00002594 | 4      | 1966 | 1945    1947 |
| RE43   | F    | 00002614 | 4      | 2000 | 1979    1981 |
| RE44   | F    | 00002694 | 4      | 2034 | 2013    2015 |
| RE45   | F    | 00002714 | 4      | 2068 | 2047    2049 |
| RE46   | F    | 00002794 | 4      | 2103 | 2082    2084 |
| RE47   | F    | 00002814 | 4      | 2137 | 2116    2118 |
| RE48   | F    | 00002894 | 4      | 2171 | 2150    2152 |
| RE49   | F    | 00002914 | 4      | 2205 | 2184    2186 |
| RE5    | F    | 00001314 | 4      | 707  | 686    688   |
| RE50   | F    | 00002994 | 4      | 2239 | 2218    2220 |
| RE51   | F    | 00002A14 | 4      | 2273 | 2252    2254 |
| RE52   | F    | 00002A94 | 4      | 2307 | 2286    2288 |
| RE53   | F    | 00002B14 | 4      | 2341 | 2320    2322 |
| RE54   | F    | 00002B94 | 4      | 2375 | 2354    2356 |
| RE55   | F    | 00002C14 | 4      | 2409 | 2388    2390 |
| RE56   | F    | 00002C94 | 4      | 2443 | 2422    2424 |
| RE57   | F    | 00002D14 | 4      | 2478 | 2457    2459 |
| RE58   | F    | 00002D94 | 4      | 2512 | 2491    2493 |
| RE59   | F    | 00002E14 | 4      | 2546 | 2525    2527 |
| RE6    | F    | 00001394 | 4      | 741  | 720    722   |
| RE60   | F    | 00002E94 | 4      | 2580 | 2559    2561 |
| RE61   | F    | 00002F14 | 4      | 2614 | 2593    2595 |
| RE62   | F    | 00002F94 | 4      | 2648 | 2627    2629 |
| RE63   | F    | 00003014 | 4      | 2683 | 2662    2664 |
| RE64   | F    | 00003094 | 4      | 2717 | 2696    2698 |
| RE65   | F    | 00003114 | 4      | 2751 | 2730    2732 |
| RE66   | F    | 00003194 | 4      | 2785 | 2764    2766 |
| RE7    | F    | 00001414 | 4      | 775  | 754    756   |
| RE8    | F    | 00001494 | 4      | 809  | 788    790   |
| RE9    | F    | 00001514 | 4      | 843  | 822    824   |
| REA1   | A    | 000010D0 | 4      | 552  |              |
| REA10  | A    | 00001550 | 4      | 858  |              |
| REA11  | A    | 000015D0 | 4      | 892  |              |
| REA12  | A    | 00001650 | 4      | 926  |              |
| REA13  | A    | 000016D0 | 4      | 960  |              |
| REA14  | A    | 00001750 | 4      | 994  |              |
| REA15  | A    | 000017D0 | 4      | 1028 |              |
| REA16  | A    | 00001850 | 4      | 1062 |              |
| REA17  | A    | 000018D0 | 4      | 1096 |              |
| REA18  | A    | 00001950 | 4      | 1130 |              |
| REA19  | A    | 000019D0 | 4      | 1164 |              |
| REA2   | A    | 00001150 | 4      | 586  |              |
| REA20  | A    | 00001A50 | 4      | 1198 |              |
| REA21  | A    | 00001AD0 | 4      | 1232 |              |
| REA22  | A    | 00001B50 | 4      | 1266 |              |
| REA23  | A    | 00001BD0 | 4      | 1300 |              |
| REA24  | A    | 00001C50 | 4      | 1334 |              |
| REA25  | A    | 00001CD0 | 4      | 1368 |              |
| REA26  | A    | 00001D50 | 4      | 1402 |              |
| REA27  | A    | 00001DD0 | 4      | 1436 |              |
| REA28  | A    | 00001E50 | 4      | 1470 |              |
| REA29  | A    | 00001ED0 | 4      | 1504 |              |
| REA3   | A    | 000011D0 | 4      | 620  |              |

| SYMBOL   | TYPE | VALUE    | LENGTH | DEFN | REFERENCES |     |
|----------|------|----------|--------|------|------------|-----|
| REA30    | A    | 00001F50 | 4      | 1538 |            |     |
| REA31    | A    | 00001FD0 | 4      | 1573 |            |     |
| REA32    | A    | 00002050 | 4      | 1607 |            |     |
| REA33    | A    | 000020D0 | 4      | 1641 |            |     |
| REA34    | A    | 00002150 | 4      | 1675 |            |     |
| REA35    | A    | 000021D0 | 4      | 1709 |            |     |
| REA36    | A    | 00002250 | 4      | 1743 |            |     |
| REA37    | A    | 000022D0 | 4      | 1777 |            |     |
| REA38    | A    | 00002350 | 4      | 1811 |            |     |
| REA39    | A    | 000023D0 | 4      | 1845 |            |     |
| REA4     | A    | 00001250 | 4      | 654  |            |     |
| REA40    | A    | 00002450 | 4      | 1879 |            |     |
| REA41    | A    | 000024D0 | 4      | 1913 |            |     |
| REA42    | A    | 00002550 | 4      | 1947 |            |     |
| REA43    | A    | 000025D0 | 4      | 1981 |            |     |
| REA44    | A    | 00002650 | 4      | 2015 |            |     |
| REA45    | A    | 000026D0 | 4      | 2049 |            |     |
| REA46    | A    | 00002750 | 4      | 2084 |            |     |
| REA47    | A    | 000027D0 | 4      | 2118 |            |     |
| REA48    | A    | 00002850 | 4      | 2152 |            |     |
| REA49    | A    | 000028D0 | 4      | 2186 |            |     |
| REA5     | A    | 000012D0 | 4      | 688  |            |     |
| REA50    | A    | 00002950 | 4      | 2220 |            |     |
| REA51    | A    | 000029D0 | 4      | 2254 |            |     |
| REA52    | A    | 00002A50 | 4      | 2288 |            |     |
| REA53    | A    | 00002AD0 | 4      | 2322 |            |     |
| REA54    | A    | 00002B50 | 4      | 2356 |            |     |
| REA55    | A    | 00002BD0 | 4      | 2390 |            |     |
| REA56    | A    | 00002C50 | 4      | 2424 |            |     |
| REA57    | A    | 00002CD0 | 4      | 2459 |            |     |
| REA58    | A    | 00002D50 | 4      | 2493 |            |     |
| REA59    | A    | 00002DD0 | 4      | 2527 |            |     |
| REA6     | A    | 00001350 | 4      | 722  |            |     |
| REA60    | A    | 00002E50 | 4      | 2561 |            |     |
| REA61    | A    | 00002ED0 | 4      | 2595 |            |     |
| REA62    | A    | 00002F50 | 4      | 2629 |            |     |
| REA63    | A    | 00002FD0 | 4      | 2664 |            |     |
| REA64    | A    | 00003050 | 4      | 2698 |            |     |
| REA65    | A    | 000030D0 | 4      | 2732 |            |     |
| REA66    | A    | 00003150 | 4      | 2766 |            |     |
| REA7     | A    | 000013D0 | 4      | 756  |            |     |
| REA8     | A    | 00001450 | 4      | 790  |            |     |
| REA9     | A    | 000014D0 | 4      | 824  |            |     |
| READDR   | A    | 00000018 | 4      | 446  | 244        |     |
| REG2LOW  | U    | 000000DD | 1      | 390  |            |     |
| REG2PATT | U    | AABBCCDD | 1      | 389  |            |     |
| RELEN    | A    | 00000014 | 4      | 445  |            |     |
| RPTDWSAV | D    | 00000398 | 8      | 315  | 302        | 306 |
| RPTERROR | I    | 0000032C | 4      | 282  | 257        |     |
| RPTSAVE  | F    | 00000390 | 4      | 312  | 282        | 309 |
| RPTSVR5  | F    | 00000394 | 4      | 313  | 283        | 308 |
| SKL0001  | U    | 0000004E | 1      | 202  | 218        |     |
| SKT0001  | C    | 0000022A | 20     | 199  | 202        | 219 |
| SVOLDPSW | U    | 00000140 | 0      | 138  |            |     |
| T1       | A    | 000010B8 | 4      | 545  | 2799       |     |
| T10      | A    | 00001538 | 4      | 851  | 2808       |     |

| SYMBOL | TYPE | VALUE    | LENGTH | DEFN | REFERENCES |
|--------|------|----------|--------|------|------------|
| T11    | A    | 000015B8 | 4      | 885  | 2809       |
| T12    | A    | 00001638 | 4      | 919  | 2810       |
| T13    | A    | 000016B8 | 4      | 953  | 2811       |
| T14    | A    | 00001738 | 4      | 987  | 2812       |
| T15    | A    | 000017B8 | 4      | 1021 | 2813       |
| T16    | A    | 00001838 | 4      | 1055 | 2814       |
| T17    | A    | 000018B8 | 4      | 1089 | 2815       |
| T18    | A    | 00001938 | 4      | 1123 | 2816       |
| T19    | A    | 000019B8 | 4      | 1157 | 2817       |
| T2     | A    | 00001138 | 4      | 579  | 2800       |
| T20    | A    | 00001A38 | 4      | 1191 | 2818       |
| T21    | A    | 00001AB8 | 4      | 1225 | 2819       |
| T22    | A    | 00001B38 | 4      | 1259 | 2820       |
| T23    | A    | 00001BB8 | 4      | 1293 | 2821       |
| T24    | A    | 00001C38 | 4      | 1327 | 2822       |
| T25    | A    | 00001CB8 | 4      | 1361 | 2823       |
| T26    | A    | 00001D38 | 4      | 1395 | 2824       |
| T27    | A    | 00001DB8 | 4      | 1429 | 2825       |
| T28    | A    | 00001E38 | 4      | 1463 | 2826       |
| T29    | A    | 00001EB8 | 4      | 1497 | 2827       |
| T3     | A    | 000011B8 | 4      | 613  | 2801       |
| T30    | A    | 00001F38 | 4      | 1531 | 2828       |
| T31    | A    | 00001FB8 | 4      | 1566 | 2829       |
| T32    | A    | 00002038 | 4      | 1600 | 2830       |
| T33    | A    | 000020B8 | 4      | 1634 | 2831       |
| T34    | A    | 00002138 | 4      | 1668 | 2832       |
| T35    | A    | 000021B8 | 4      | 1702 | 2833       |
| T36    | A    | 00002238 | 4      | 1736 | 2834       |
| T37    | A    | 000022B8 | 4      | 1770 | 2835       |
| T38    | A    | 00002338 | 4      | 1804 | 2836       |
| T39    | A    | 000023B8 | 4      | 1838 | 2837       |
| T4     | A    | 00001238 | 4      | 647  | 2802       |
| T40    | A    | 00002438 | 4      | 1872 | 2838       |
| T41    | A    | 000024B8 | 4      | 1906 | 2839       |
| T42    | A    | 00002538 | 4      | 1940 | 2840       |
| T43    | A    | 000025B8 | 4      | 1974 | 2841       |
| T44    | A    | 00002638 | 4      | 2008 | 2842       |
| T45    | A    | 000026B8 | 4      | 2042 | 2843       |
| T46    | A    | 00002738 | 4      | 2077 | 2844       |
| T47    | A    | 000027B8 | 4      | 2111 | 2845       |
| T48    | A    | 00002838 | 4      | 2145 | 2846       |
| T49    | A    | 000028B8 | 4      | 2179 | 2847       |
| T5     | A    | 000012B8 | 4      | 681  | 2803       |
| T50    | A    | 00002938 | 4      | 2213 | 2848       |
| T51    | A    | 000029B8 | 4      | 2247 | 2849       |
| T52    | A    | 00002A38 | 4      | 2281 | 2850       |
| T53    | A    | 00002AB8 | 4      | 2315 | 2851       |
| T54    | A    | 00002B38 | 4      | 2349 | 2852       |
| T55    | A    | 00002BB8 | 4      | 2383 | 2853       |
| T56    | A    | 00002C38 | 4      | 2417 | 2854       |
| T57    | A    | 00002CB8 | 4      | 2452 | 2855       |
| T58    | A    | 00002D38 | 4      | 2486 | 2856       |
| T59    | A    | 00002DB8 | 4      | 2520 | 2857       |
| T6     | A    | 00001338 | 4      | 715  | 2804       |
| T60    | A    | 00002E38 | 4      | 2554 | 2858       |
| T61    | A    | 00002EB8 | 4      | 2588 | 2859       |













| DESC | SYMBOL | SIZE | POS | ADDR |
|------|--------|------|-----|------|
|------|--------|------|-----|------|

**Entry: 0**

|        |         |       |           |           |
|--------|---------|-------|-----------|-----------|
| Image  | IMAGE   | 13012 | 0000-32D3 | 0000-32D3 |
| Region |         | 13012 | 0000-32D3 | 0000-32D3 |
| CSECT  | ZVE7TST | 13012 | 0000-32D3 | 0000-32D3 |

| STMT | FILE NAME |
|------|-----------|
|------|-----------|

```
1 /devstor/sharedvfp/tests/zvector-e7-32-VGEM.asm
```

**\*\* NO ERRORS FOUND \*\***